

Debiased Inference for Dynamic Panels with Low-Rank Coefficient Heterogeneity

Abstract

We study dynamic panel models in which the coefficients on lagged outcomes and observed covariates vary across units and time. Every autoregressive coefficient matrix, every slope coefficient matrix, and the interactive-effects matrix has its own fixed low rank; the matrices need not share ranks, factors, or loadings. Rank selection uses a short weighted nuclear-norm path and one bounded-rank pilot to screen a small set of rank vectors, refits every retained vector without shrinkage, and applies an information criterion to the post-refits. Inference is based on one full-panel rank-constrained least-squares fit rather than repeated data splitting. For each target, a Riesz representer solves a local Riesz equation and produces observation-level Riesz weights. Individual coefficients and suitable fixed-time means and contrasts use the full-panel plug-in estimator. A fixed two-way split-panel correction removes the leading loading and factor estimation biases for averages taken over many units and many periods. The probability theory is conditional on common shocks, allowing pervasive common shocks while requiring the remaining unit-specific innovations to be conditionally mixing and the predetermined covariates to be conditionally near-epoch dependent. The theory allows dynamic feedback through lagged outcomes, spatial and cross-sectional dependence, conditional heteroskedasticity, separate ranks, and different growth rates for the two panel dimensions. A spatial HAC estimator consistently estimates the conditional variance when cross-sectional dependence is allowed. After rank selection, inference reuses the selected full-panel post-refit for local targets and adds four fixed-rank split-panel fits for two-way averages, with no random folds or implemented buffers.

Keywords: dynamic panel data; heterogeneous coefficients; low-rank matrices; interactive effects; full-panel inference; rank selection; spatial dependence; near-epoch dependence.

Contents

1	Introduction	4
2	Related Literature	5
3	Model and Parameters of Interest	7
3.1	Model specification	7
3.2	Matrix notation	8
3.3	Parameters of interest	9
4	Full-Panel Estimation, Rank Selection, and Variance Estimation	10
4.1	Full-panel fixed-rank estimation	10
4.2	Tangent spaces and Riesz representers	11
4.3	Point estimates and the split-panel correction	12
4.4	Variance estimation	13
4.5	Rank selection	15
5	Assumptions	17
6	Asymptotic Theory	22
7	Implementation	25
7.1	Practical implementation	25
7.2	Computation of the rank-constrained estimator	27
7.3	Tuning choices	28
8	Monte Carlo Experiments	29
8.1	Common dynamic design	29
8.2	Calibration, targets, and reported quantities	32
9	Empirical Application	34
9.1	Regional unemployment	34
9.2	House-price momentum	35
9.3	Spatial inference and computation	35
10	Conclusion	37
A	Proofs and Supporting Results	40
A.1	Conditional moments and local approximation	41
A.2	Target directions and the population Riesz-score limit	43
A.3	Full-panel concentration and fixed-rank recovery	48
A.4	Auxiliary local-deletion comparison	57

A.5	Full-panel target expansion and split-panel correction	59
A.6	Spatial HAC variance estimation	68
A.7	Rank selection	71
B	Additional Monte Carlo Results	72
B.1	Calibration and coefficient summaries	72
B.2	Entrywise and fixed-time targets	73
B.3	Two-way averages and bias correction	73
B.4	Rank, spatial-weight, and computation sensitivity	73
C	Empirical Appendix	74
C.1	Data construction	74
C.2	Rank selection and numerical diagnostics	74
C.3	Unemployment robustness	75
C.4	Housing robustness	75
C.5	Variance and dependence diagnostics	76

1 Introduction

Dynamic panel studies often ask whether persistence or covariate effects differ across units, across time periods, or between economically meaningful groups. A common-coefficient model cannot answer these questions, while unrestricted unit–time coefficients cannot be estimated from a single panel. This paper uses separate low-rank restrictions to make the full coefficient matrices estimable without imposing known groups, smooth time paths, or a common latent structure across regressors.

The model contains a separate coefficient matrix for every lagged outcome and observed covariate. An additional low-rank interactive-effects matrix absorbs additive and common conditional-mean heterogeneity. Each matrix has its own rank and latent spaces. This separation is important in applications in which persistence, covariate effects, and common movements need not be generated by the same factors.

Inference is difficult for three reasons. First, all coefficient matrices enter one outcome equation jointly, so population identification must prevent a change in one matrix from being offset by changes in the others. Second, a current disturbance enters future observations through lagged outcomes, while nearby units may have correlated shocks. Third, estimating the loading and factor rows from the same data creates biases of order T^{-1} and N^{-1} . These biases are negligible for an individual coefficient but can be first order for an average taken over many units and many periods.

The estimator follows a post-regularization architecture. A convex nuclear-norm path and a bounded-rank pilot locate plausible singular spaces, generate rank candidates, and provide warm starts. Every retained rank vector is then refitted by unpenalized joint fixed-rank least squares, so the reported coefficient estimates and standard errors do not inherit nuclear-norm shrinkage. For a supplied target, a Riesz representer solves the least-squares Gram equation on the estimated fixed-rank tangent space. This inverse-Gram adjustment translates estimation errors in all autoregressive, slope, and interactive-effect matrices into first-order error for the selected parameter. At an exact full-panel least-squares solution, the residual-weighted Riesz sum is zero, so the Riesz weights are not added as a separate point-estimate correction. They are instead used to derive the influence expansion and the standard error.

For individual coefficients, fixed unit–time lag sums, and suitable fixed-time means and contrasts, the $N^{-1} + T^{-1}$ estimation bias is smaller than the standard error. These targets use the full-panel plug-in estimate. For averages whose weights are spread over many units and many periods, the standard error can be of order $(NT)^{-1/2}$, so the same bias is not negligible. A fixed two-way split-panel correction removes the leading T^{-1} loading bias and N^{-1} factor bias. The correction uses two time-split fits and two unit-split fits, with the full-panel selected ranks held fixed.

The stochastic assumptions are stated conditionally on common shocks. This does not require removing all common shocks. Common shocks may be strong, persistent, and relevant for covariates, conditional variances, and spatial dependence. Common effects in the conditional mean are represented by the coefficient matrices and the interactive-effects matrix; conditional on the common shocks, the remaining unit-specific innovations are weakly dependent over space and time. Predetermined covariates are approximable by recent innovations from nearby units, while dynamic

stability controls the effect of current disturbances on distant future outcomes.

The main contribution is a joint inferential framework for several separately ranked coefficient matrices in a dynamic panel. The method combines four ideas that are familiar separately in low-rank analysis but require new arguments in this setting: a convex nuclear-norm pilot for screening and initialization; unpenalized joint least-squares refinement to remove shrinkage; explicit alignment and expansion of the estimated singular spaces; and a target-specific Riesz equation that adjusts for estimation spillovers across all coefficient matrices. The resulting theory provides full-panel inference without random folds for individual coefficients and suitable fixed-time means and contrasts, a fixed split-panel correction for broad two-way averages, and conditional spatial inference under common shocks. The present results assume exact fixed ranks and consistent selection of the separate rank vector; approximate-low-rank inference, growing-rank asymptotics, semiparametric efficiency, and inference robust to rank overestimation are not claimed.

The remainder is organized as follows. [Section 2](#) relates the paper to the existing literature. [Section 3](#) presents the model and parameters of interest. [Section 4](#) defines estimation, rank selection, and variance estimation. [Section 5](#) states the maintained assumptions. [Section 6](#) gives the main asymptotic results. [Section 7](#) describes computation. [Section 8](#) and [Section 9](#) contain the Monte Carlo and empirical designs, with numerical output placeholders in this draft. The appendix is organized in theorem order.

2 Related Literature

The paper relates to interactive-effects panels, dynamic panels, heterogeneous coefficients, low-rank estimation, and inference under spatial and temporal dependence.

Large-panel factor methods and interactive-effects regressions use latent factors to absorb unobserved common heterogeneity ([Bai and Ng, 2002](#); [Bai, 2003](#); [Pesaran, 2006](#); [Moon and Weidner, 2015](#)). We retain a low-rank interactive-effects matrix but allow every autoregressive and slope coefficient matrix to vary over both units and time. Each coefficient matrix has its own rank and latent spaces; no common factor or loading space is imposed across matrices.

The classical dynamic-panel literature uses instruments or transformations to address short panels with common coefficients ([Anderson and Hsiao, 1981, 1982](#); [Arellano and Bond, 1991](#)). Here both panel dimensions are large, the autoregressive coefficients are heterogeneous, and least squares relies on conditional sequential exogeneity. In dynamic interactive-effects models, predetermined regressors and heteroskedastic or correlated idiosyncratic errors generate separate incidental-parameter biases ([Moon and Weidner, 2017](#)). The loading-side and factor-side biases derived below have the same economic sources, but they arise for target functionals of several separately ranked coefficient matrices under conditional spatial dependence. If lagged outcomes remain endogenous after conditioning on current covariates, observed past information, and the common shocks, the least-squares criterion would need to be replaced by an instrumented one.

Random-coefficient, mean-group, and grouped-pattern methods impose different structures

on coefficient heterogeneity (Swamy, 1970; Pesaran and Smith, 1995; Bonhomme and Manresa, 2015). Low rank instead represents variation through latent unit and time dimensions. It supports individual coefficients, cross-sectional and time averages, group contrasts, and lag-sum persistence targets without requiring a known group assignment.

The closest low-rank inferential benchmark is Chernozhukov et al. (2023). They study linear functionals of a matrix that is well approximated by a spiked low-rank structure, allow slowly growing rank, initialize estimation by nuclear-norm regularization, and use unpenalized least-squares refinements together with an explicit rotation analysis of the estimated singular spaces. Under their model, the resulting procedure attains a semiparametric efficiency bound. We adopt the same post-regularization principle—a convex nuclear-norm pilot followed by unpenalized refinement—and the same broad emphasis on singular-space alignment, but their estimator does not transfer mechanically to the present model. Here several coefficient matrices enter the same regression through different stochastic regressors, one regressor is the lagged dependent variable, and the unit-specific disturbances may be spatially dependent conditional on common shocks. The fixed-rank refit and the local Gram equation must therefore be joint across the autoregressive, slope, and interactive-effect matrices.

The target-specific Riesz equation addresses the corresponding first-order inferential problem. It represents the derivative of a selected parameter under the joint conditional least-squares Gram form, and therefore accounts simultaneously for estimation errors in every coefficient matrix. This inverse-Gram adjustment is related in purpose to the rotation and least-squares refinements in Chernozhukov et al. (2023), but the present paper establishes validity rather than a semiparametric efficiency result. An efficiency analysis under conditional heteroskedasticity and spatial dependence would require an optimally covariance-weighted version of the joint Gram equation and is outside the scope of this paper.

Choi et al. (2024) obtain low-rank completion inference without implemented sample splitting by combining a nuclear-norm pilot with loading and factor least-squares refinements. Their proof uses an auxiliary leave-one-out estimator that is not computed. Our full-panel analysis follows the same broad proof principle, but the auxiliary object must delete a space–time neighborhood rather than a single observation: a current disturbance reappears in future lagged outcomes and may remain correlated with nearby units. For individual coefficients and suitable fixed-time parameters, no split is implemented. The four split-panel fits used for broad two-way averages serve a different purpose: they estimate and remove the leading T^{-1} loading bias and N^{-1} factor bias rather than creating independence.

Recent rank-robust low-rank inference provides a useful contrast. Choi et al. (2026) construct inference that remains valid when a prespecified rank is no smaller than the true rank and show that overestimating the low-rank space can itself generate a nonnegligible bias. Our current theory instead requires consistent selection of a separate rank for every coefficient matrix because the tangent spaces and joint Riesz equation are rank specific. Extending rank-robust inference to the present model would require controlling the bias from overestimating several interacting latent

spaces and its propagation through the dynamic regression. We therefore report rank-selection and neighboring-rank diagnostics but do not describe the resulting intervals as rank robust.

The comparison also clarifies the scope of the maintained low-rank assumptions. [Chernozhukov et al. \(2023\)](#) and [Choi et al. \(2024\)](#) permit forms of approximate low rank and slowly growing rank in their respective settings. We impose exact fixed ranks because approximation remainders in an autoregressive coefficient matrix can propagate through future outcomes, while approximation errors from all coefficient matrices enter the same local Gram and Riesz equations. The rank-screening stage is designed to retain strong singular directions while treating weaker directions as part of the screening uncertainty, but the inferential theorems are stated for exact fixed rank. Approximate-low-rank inference and growing-rank theory are left for future work.

Related work on matrix completion and causal panels includes [Athey et al. \(2021\)](#) and [Bai and Ng \(2021\)](#). Those settings motivate low-rank recovery but do not directly cover a dynamic equation with several stochastic-regressor coefficient matrices, predetermined covariates, and conditional spatial dependence.

For computation, [Moon and Weidner \(2026\)](#) show that nuclear-norm panel estimators solve convex problems and can provide effective starting values for subsequent least-squares refinements. [Gao and Absil \(2022\)](#) develop rank-adaptive Riemannian updates that combine fixed-rank optimization with rank-increase and rank-reduction steps. We use these ideas for screening and numerical optimization, while the model-specific rank-consistency argument relies on a post-refit information criterion and candidate coverage tailored to the present dynamic regression.

The conditional weak-dependence framework follows the spatial near-epoch-dependence approach of [Jenish and Prucha \(2012\)](#). Spatial HAC variance estimation follows the logic of [Conley \(1999\)](#). Entrywise low-rank perturbation results motivate the pointwise loading and factor analysis, while the dynamic and dependent setting requires additional control of future lagged outcomes and local spatial dependence.

3 Model and Parameters of Interest

3.1 Model specification

We observe a complete panel for units $i = 1, \dots, N$ and periods $t = 1, \dots, T$. The model is

$$y_{it} = \sum_{\ell=1}^P a_{0,it,\ell} y_{i,t-\ell} + \sum_{k=1}^K x_{it,k} \beta_{0,it,k} + h_{0,it} + u_{it}. \quad (3.1)$$

Here $y_{i,t-\ell}$ is the ℓ th lag of the dependent variable, $x_{it,k}$ is the k th observed covariate, and u_{it} is the remaining unit-specific innovation. The term $h_{0,it}$ is an unobserved interactive fixed effect that captures latent common conditional-mean variation with heterogeneous unit loadings and is treated as a nuisance parameter. The autoregressive coefficients $a_{0,it,\ell}$, for $\ell = 1, \dots, P$, and the slope coefficients $\beta_{0,it,k}$, for $k = 1, \dots, K$, may vary across both units and time. The initial values

$y_{i,0}, \dots, y_{i,1-P}$ are observed. Throughout, P and K are fixed and known.

Let $\mathcal{C}_t$ be the sigma-field generated by the common shocks at time t , and let $\mathcal{C} = \bigvee_{s=1}^T \mathcal{C}_s$ collect the common shocks over the sample. These shocks need not be observed by the researcher, and conditioning on $\mathcal{C}$ is used only in the probability theory. The common shocks may affect the observed covariates, the heterogeneous coefficient paths, and the distribution of the unit-specific innovations. Their effects on the conditional mean of y_{it} are represented by the complete fitted value $\mathcal{Y}_{it}(\boldsymbol{\Theta}_0)$: through the observed covariates, through the autoregressive and slope coefficient matrices when those coefficients vary with aggregate conditions, and through $\mathbf{H}_0$ for additive latent common mean effects that are not mediated by the observed regressors. Thus, $\mathbf{H}_0$ does not represent every effect of the common shocks. It represents the remaining additive common conditional-mean variation after the lagged outcomes, observed covariates, and heterogeneous coefficients have been included. The common shocks may additionally affect conditional variances and the strength of spatial dependence.

For each lag ℓ , collect the autoregressive coefficients in the $N \times T$ matrix $\mathbf{A}_0^{(\ell)} = (a_{0,it,\ell})_{i \leq N, t \leq T}$. For each covariate k , collect the slope coefficients in $\mathbf{B}_0^{(k)} = (\beta_{0,it,k})_{i \leq N, t \leq T}$, and write $\mathbf{H}_0 = (h_{0,it})_{i \leq N, t \leq T}$ for the interactive fixed-effect matrix. The maintained dimension-reduction restriction is that every one of these matrices has fixed low rank as $N, T \rightarrow \infty$:

$$r_{A,\ell} = \text{rank}\{\mathbf{A}_0^{(\ell)}\}, \quad r_{B,k} = \text{rank}\{\mathbf{B}_0^{(k)}\}, \quad r_H = \text{rank}(\mathbf{H}_0).$$

The rank vector is $\mathbf{r}_0 = (r_{A,1}, \dots, r_{A,P}, r_{B,1}, \dots, r_{B,K}, r_H)$, and any rank may equal zero. No common rank, factor, or loading structure is imposed across coefficient matrices.

The low-rank restriction allows flexible heterogeneity while reducing the effective number of parameters. For example, if $\mathbf{A}_0^{(1)}$ has rank one, then $a_{0,it,1} = \lambda_i f_t$ for unit loading λ_i and time factor f_t . A larger rank permits several such terms. A rank- r $N \times T$ matrix has $r(N + T - r)$ locally free parameters rather than NT . Thus, each coefficient matrix may vary over both dimensions while retaining an effective dimension of order $N + T$.

The model contains no unrestricted unit-time intercept outside $\mathbf{H}_0$, because such a matrix would not generally be identified separately from $\mathbf{H}_0$.

3.2 Matrix notation

Write

$$\boldsymbol{\Theta} = (\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(P)}, \mathbf{B}^{(1)}, \dots, \mathbf{B}^{(K)}, \mathbf{H}) \in (\mathbb{R}^{N \times T})^{P+K+1}.$$

For two such collections, define

$$\langle \mathbf{Q}, \boldsymbol{\Delta} \rangle = \sum_{\ell=1}^P \text{tr}\{(\mathbf{Q}_A^{(\ell)})' \boldsymbol{\Delta}_A^{(\ell)}\} + \sum_{k=1}^K \text{tr}\{(\mathbf{Q}_B^{(k)})' \boldsymbol{\Delta}_B^{(k)}\} + \text{tr}(\mathbf{Q}_H' \boldsymbol{\Delta}_H),$$

and $\|\mathbf{Q}\|^2 = \langle \mathbf{Q}, \mathbf{Q} \rangle$.

For each observation, define the scalar fitted-value map

$$\mathcal{Y}_{it} : (\mathbb{R}^{N \times T})^{P+K+1} \longrightarrow \mathbb{R}, \quad \mathcal{Y}_{it}(\boldsymbol{\Theta}) = \sum_{\ell=1}^P A_{it}^{(\ell)} y_{i,t-\ell} + \sum_{k=1}^K B_{it}^{(k)} x_{it,k} + H_{it}. \quad (3.2)$$

Thus, $\mathcal{Y}_{it}(\boldsymbol{\Theta})$ is one scalar, while $\mathcal{Y}(\boldsymbol{\Theta}) = \{\mathcal{Y}_{it}(\boldsymbol{\Theta})\}_{i \leq N, t \leq T} \in \mathbb{R}^{N \times T}$ collects the fitted values for the full panel. Conditional on the observed regressors and lagged outcomes, $\mathcal{Y}_{it}(\boldsymbol{\Theta})$ is linear in $\boldsymbol{\Theta}$, and the model can be written as $y_{it} = \mathcal{Y}_{it}(\boldsymbol{\Theta}_0) + u_{it}$.

3.3 Parameters of interest

We write the main formulas for one generic scalar parameter of interest. This avoids carrying an index through every equation; joint inference for any fixed finite collection is stated separately in the theorems. The parameter is a deterministic linear summary of the autoregressive and slope coefficient matrices:

$$\phi(\boldsymbol{\Theta}_0) = \langle \mathbf{D}, \boldsymbol{\Theta}_0 \rangle. \quad (3.3)$$

The direction assigned to $\mathbf{H}_0$ is zero. Every matrix in $\mathbf{D}$ is $N \times T$, and matrices not involved in the parameter are set equal to zero.

Let $\mathbf{e}_i \in \mathbb{R}^N$ and $\mathbf{e}_t \in \mathbb{R}^T$ denote coordinate vectors. For a unit group $G_N \subseteq \{1, \dots, N\}$ and a time set $R_T \subseteq \{1, \dots, T\}$, define

$$\mathbf{a}_{G_N} = \frac{1}{|G_N|} \sum_{i \in G_N} \mathbf{e}_i, \quad \mathbf{b}_{R_T} = \frac{1}{|R_T|} \sum_{t \in R_T} \mathbf{e}_t.$$

The directions for the main parameters are as follows.

- (i) For the individual autoregressive coefficient $a_{0,i_0 t_0, \ell}$, set $\mathbf{D}_A^{(\ell)} = \mathbf{e}_{i_0} \mathbf{e}_{t_0}'$, and set every other target matrix to zero. For $\beta_{0,i_0 t_0, k}$, use the same direction in $\mathbf{D}_B^{(k)}$.
- (ii) For the mean of a coefficient matrix $\mathbf{M}_0$ over $G_N \times R_T$, set $\mathbf{D}_M = \mathbf{a}_{G_N} \mathbf{b}_{R_T}'$. Then $\langle \mathbf{D}, \boldsymbol{\Theta}_0 \rangle = |G_N|^{-1} |R_T|^{-1} \sum_{i \in G_N} \sum_{t \in R_T} M_{0,it}$.
- (iii) For a cross-sectional mean at a fixed time t_0 , set $\mathbf{D}_M = \mathbf{a}_{G_N} \mathbf{e}_{t_0}'$. The full cross-sectional mean is obtained by taking $G_N = \{1, \dots, N\}$.
- (iv) For two unit groups $G_{1,N}$ and $G_{2,N}$, the time- t_0 contrast uses $\mathbf{D}_M = (\mathbf{a}_{G_{1,N}} - \mathbf{a}_{G_{2,N}}) \mathbf{e}_{t_0}'$. A contrast averaged over R_T replaces $\mathbf{e}_{t_0}$ by $\mathbf{b}_{R_T}$.
- (v) For the unit-time lag-sum persistence index $\sum_{\ell=1}^P a_{0,i_0 t_0, \ell}$, set $\mathbf{D}_A^{(\ell)} = \mathbf{e}_{i_0} \mathbf{e}_{t_0}'$ for every $\ell = 1, \dots, P$. Group, time, and group-time averages of the lag sum use the same averaging direction in every autoregressive coefficient matrix.

Empirical work often focuses on economically meaningful groups. For example, the mean of the k th slope coefficient for group G_N at time t is $\phi(\boldsymbol{\Theta}_0) = \frac{1}{|G_N|} \sum_{i \in G_N} \beta_{0,it,k}$, and the contrast

between groups $G_{1,N}$ and $G_{2,N}$ is

$$\phi(\Theta_0) = \frac{1}{|G_{1,N}|} \sum_{i \in G_{1,N}} \beta_{0,it,k} - \frac{1}{|G_{2,N}|} \sum_{i \in G_{2,N}} \beta_{0,it,k}.$$

Autoregressive coefficient means and contrasts are defined in the same way.

A target whose nonzero weights cover many units and many periods is called a *two-way average*. This expression refers to the support of the target weights, not to the rank of any coefficient matrix. The full-panel mean is the leading example. The lag-sum target is a linear summary of the autoregressive coefficients; it is not an impulse response or a long-run multiplier.

4 Full-Panel Estimation, Rank Selection, and Variance Estimation

The procedure has four steps. First, estimate all coefficient matrices jointly under supplied or selected ranks. Second, use the estimated singular spaces to construct a Riesz representer for each parameter of interest. Third, use the full-panel plug-in estimate for local parameters and a fixed split-panel correction for averages over many units and time periods. Fourth, estimate the variance from the residuals and Riesz weights, allowing for spatial dependence. Theorems 2–5 provide the formal justification for these steps.

4.1 Full-panel fixed-rank estimation

A rank vector records the rank assigned to every coefficient matrix:

$$\mathbf{r} = (r_{A,1}, \dots, r_{A,P}, r_{B,1}, \dots, r_{B,K}, r_H).$$

For a supplied rank vector $\mathbf{r}$, let $\mathcal{M}(\mathbf{r})$ contain all collections Θ for which $\text{rank}\{\mathbf{A}^{(\ell)}\} = r_{A,\ell}$, $\text{rank}\{\mathbf{B}^{(k)}\} = r_{B,k}$, and $\text{rank}(\mathbf{H}) = r_H$. Rank zero means that the corresponding matrix is identically zero.

Let $B < \infty$ be a fixed coefficient bound, and assume that the true matrices lie strictly inside this bound. Let $\|\Theta\|_{\max}$ denote the largest absolute entry among all coefficient matrices. For supplied ranks, define

$$\hat{\Theta}(\mathbf{r}) \in \arg \min_{\substack{\Theta \in \mathcal{M}(\mathbf{r}) \\ \|\Theta\|_{\max} \leq B}} \frac{1}{2NT} \sum_{i=1}^N \sum_{t=1}^T \{y_{it} - \mathcal{Y}_{it}(\Theta)\}^2. \quad (4.1)$$

Fix a supplied rank vector $\mathbf{r}$ and write $\hat{\Theta} = \hat{\Theta}(\mathbf{r})$. Set

$$\hat{u}_{it} = y_{it} - \mathcal{Y}_{it}(\hat{\Theta}), \quad \hat{\phi} = \langle \mathbf{D}, \hat{\Theta} \rangle.$$

Thus, the point estimate is obtained by inserting the estimated coefficient matrices into the parameter of interest. Theorem 2 proves consistency for supplied ranks. The implementation reports numerical stationarity and agreement across starting values.

4.2 Tangent spaces and Riesz representers

If a positive-rank matrix has compact singular-value decomposition $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}'$, its fixed-rank tangent space is

$$\mathcal{T}_M = \{\mathbf{U}\mathbf{R}' + \mathbf{L}\mathbf{V}' : \mathbf{R} \in \mathbb{R}^{T \times r_M}, \mathbf{L} \in \mathbb{R}^{N \times r_M}\}.$$

A rank-zero matrix has tangent space $\{0\}$. Let $\mathcal{T}_0$ and $\hat{\mathcal{T}}$ collect the tangent spaces at $\mathbf{\Theta}_0$ and $\hat{\mathbf{\Theta}}$, and let $\mathcal{P}_0$ and $\hat{\mathcal{P}}$ denote their orthogonal projections.

For a parameter $\phi(\mathbf{\Theta}) = \langle \mathbf{D}, \mathbf{\Theta} \rangle$, the map $\mathbf{\Delta} \mapsto \langle \mathbf{D}, \mathbf{\Delta} \rangle$ gives its first-order change along a tangent direction. Conditional on the common shocks $\mathcal{C}$, the least-squares Gram form on $\mathcal{T}_0$ is

$$(\mathbf{Q}, \mathbf{\Delta}) \mapsto \mathbb{E}_0 \left[\sum_{i,t} \mathcal{Y}_{it}(\mathbf{Q}) \mathcal{Y}_{it}(\mathbf{\Delta}) \middle| \mathcal{C} \right].$$

When this Gram form is nonsingular, the finite-dimensional Riesz representation theorem gives a unique $\mathbf{q}_0 \in \mathcal{T}_0$ satisfying

$$\mathbb{E}_0 \left[\sum_{i,t} \mathcal{Y}_{it}(\mathbf{\Delta}) \mathcal{Y}_{it}(\mathbf{q}_0) \middle| \mathcal{C} \right] = \langle \mathbf{D}, \mathbf{\Delta} \rangle \quad \text{for every } \mathbf{\Delta} \in \mathcal{T}_0. \quad (4.2)$$

The zero subscript indicates the population Riesz representer associated with the true coefficient matrices. It is the matrix analogue of a parameter derivative multiplied by the inverse of a population least-squares Gram matrix.

The empirical Riesz representer $\hat{\mathbf{q}} \in \hat{\mathcal{T}}$ solves

$$\sum_{i,t} \mathcal{Y}_{it}(\mathbf{\Delta}) \mathcal{Y}_{it}(\hat{\mathbf{q}}) = \langle \mathbf{D}, \mathbf{\Delta} \rangle \quad \text{for every } \mathbf{\Delta} \in \hat{\mathcal{T}}. \quad (4.3)$$

Thus, $\mathbf{q}_0$ and $\hat{\mathbf{q}}$ are the population and sample versions of the same object. The zero subscript denotes the population weight, while a hat denotes its sample estimate:

$$\Psi_{0,it} = \mathcal{Y}_{it}(\mathbf{q}_0), \quad \hat{\Psi}_{it} = \mathcal{Y}_{it}(\hat{\mathbf{q}}).$$

Define the conditional variance of the population first-order term by

$$s_{NT}^2 = \mathbb{E}_0 \left[\left(\sum_{i,t} \Psi_{0,it} u_{it} \right)^2 \middle| \mathcal{C} \right]. \quad (4.4)$$

[Theorem 2](#) shows that the empirical Riesz representer consistently estimates its population counterpart, and [Theorem 3](#) shows how the population Riesz weights enter the first-order distribution of $\hat{\phi}$.

A tangent direction is a feasible infinitesimal change in the fitted coefficient matrices. Formally,

$\Delta \in \widehat{\mathcal{T}}$ means that there is a differentiable curve $a \mapsto \Theta(a)$ within the fitted fixed-rank set such that $\Theta(0) = \widehat{\Theta}$ and $d\Theta(a)/da|_{a=0} = \Delta$. If the coefficient bound is inactive and $\widehat{\Theta}$ is a first-order stationary point, the least-squares objective cannot decrease to first order along any such curve. Since $\mathcal{Y}_{it}$ is linear in the coefficient matrices,

$$0 = \left. \frac{d}{da} \frac{1}{2NT} \sum_{i,t} \{y_{it} - \mathcal{Y}_{it}(\Theta(a))\}^2 \right|_{a=0} = -\frac{1}{NT} \sum_{i,t} \widehat{u}_{it} \mathcal{Y}_{it}(\Delta).$$

Taking $\Delta = \widehat{\mathbf{q}}$ gives

$$\sum_{i,t} \widehat{\Psi}_{it} \widehat{u}_{it} = 0. \quad (4.5)$$

This is the constrained least-squares normal equation, not an additional debiasing assumption. It explains why the Riesz-weighted residual is not added as a separate same-panel correction. The formal argument and the proof that the coefficient bound is inactive with probability approaching one are given in [Section A.3](#).

4.3 Point estimates and the split-panel correction

Use the full-panel plug-in $\widehat{\phi}$ for a regular individual coefficient and for a fixed unit-time lag sum. For a mean over $G_N \times R_T$, use the plug-in when

$$\sqrt{|G_N||R_T|} \left(\frac{1}{N} + \frac{1}{T} \right) \rightarrow 0.$$

For a contrast between disjoint groups $G_{1,N}$ and $G_{2,N}$ averaged over R_T , use the plug-in when

$$\sqrt{\frac{|R_T|}{|G_{1,N}|^{-1} + |G_{2,N}|^{-1}}} \left(\frac{1}{N} + \frac{1}{T} \right) \rightarrow 0.$$

[Proposition 1](#) and [Theorem 3](#) show that these conditions make the T^{-1} loading bias and the N^{-1} factor bias smaller than the corresponding standard error. The conditions hold automatically for a regular individual coefficient and a fixed unit-time lag sum.

For an average whose weights extend over many units and many time periods, the standard error can be as small as order $(NT)^{-1/2}$, so biases of order T^{-1} and N^{-1} may affect first-order inference. Draw a balanced random partition $\mathcal{I}_{T,1}, \mathcal{I}_{T,2}$ of $\{1, \dots, T\}$ and an independent balanced random partition $\mathcal{I}_{N,1}, \mathcal{I}_{N,2}$ of $\{1, \dots, N\}$. The unit partition is stratified by every unit group entering the parameter of interest. Let $T_h = |\mathcal{I}_{T,h}|$ and $N_g = |\mathcal{I}_{N,g}|$.

For a general two-way target, define $\mathbf{D}^{T,h}$ by retaining in $\mathbf{D}$ only the columns whose time indices belong to $\mathcal{I}_{T,h}$, and define $\mathbf{D}^{N,g}$ by retaining only the rows whose unit indices belong to $\mathcal{I}_{N,g}$. All other entries are set to zero. Then

$$\mathbf{D}^{T,1} + \mathbf{D}^{T,2} = \mathbf{D}, \quad \mathbf{D}^{N,1} + \mathbf{D}^{N,2} = \mathbf{D}.$$

Thus, this construction preserves any mean or contrast defined in [Section 3](#), including a target supported on proper subsets G_N and R_T . For a full mean, it is equivalent to averaging the two normalized split means with weights T_h/T or N_g/N .

For $h = 1, 2$, $\hat{\Theta}_{T,h}$ is the fixed-rank estimator using all units and only times in $\mathcal{I}_{T,h}$; every selected time retains its observed lagged outcomes even when a lag belongs to the other time partition. For $g = 1, 2$, $\hat{\Theta}_{N,g}$ uses all times and only units in $\mathcal{I}_{N,g}$. Define the split target contributions by

$$\hat{\phi}_{T,h} = \langle \mathbf{D}^{T,h}, \hat{\Theta}_{T,h} \rangle, \quad \hat{\phi}_{N,g} = \langle \mathbf{D}^{N,g}, \hat{\Theta}_{N,g} \rangle,$$

and set

$$\bar{\phi}_T = \hat{\phi}_{T,1} + \hat{\phi}_{T,2}, \quad \bar{\phi}_N = \hat{\phi}_{N,1} + \hat{\phi}_{N,2}.$$

The full-panel selected ranks are held fixed in all four fits. These splits are used only to estimate the leading incidental-parameter biases; they are not validation samples, and no observations are buffered.

Define the corrected estimator

$$\hat{\phi}^{\text{corr}} = 3\hat{\phi} - \bar{\phi}_T - \bar{\phi}_N. \quad (4.6)$$

The superscript corr denotes the split-panel bias-corrected estimate.

[Theorem 3](#) establishes that the full-panel bias has the form $B_T/T + B_N/N$. [Proposition 6](#) proves three facts used here: the restricted split targets reassemble the original parameter exactly; the random halves preserve the relevant ranks and Gram matrices; and the split bias constants equal the corresponding portions of the full-panel constants up to $o_p(1)$. Consequently, the aggregate time-split estimate has leading bias $2B_T/T + B_N/N$, while the aggregate unit-split estimate has leading bias $B_T/T + 2B_N/N$. Hence

$$3 \left(\frac{B_T}{T} + \frac{B_N}{N} \right) - \left(\frac{2B_T}{T} + \frac{B_N}{N} \right) - \left(\frac{B_T}{T} + \frac{2B_N}{N} \right) = 0.$$

Thus, (4.6) removes both leading bias terms. [Theorem 4](#) proves the resulting asymptotic normality and the validity of its standard error.

4.4 Variance estimation

For each observation (i, t) , $\Psi_{0,it}^T$ and $\Psi_{0,it}^N$ denote the population Riesz weights from the time split containing t and the unit split containing i . Their sample versions are $\hat{\Psi}_{it}^T$ and $\hat{\Psi}_{it}^N$, with residuals $\hat{u}_{it}^T$ and $\hat{u}_{it}^N$. The conditional variance of the corrected first-order score is

$$(s_{NT}^{\text{corr}})^2 = \text{Var}_0 \left(\sum_{i,t} u_{it} \left[3\Psi_{0,it} - \Psi_{0,it}^T - \Psi_{0,it}^N \right] \middle| \mathcal{C} \right). \quad (4.7)$$

This single variance includes every covariance among the full-panel and four split-panel estimates, which are based on overlapping observations. [Proposition 6](#) also shows that $(s_{NT}^{\text{corr}})^2/s_{NT}^2 \rightarrow_p 1$, so the correction does not collapse the first-order score.

For an uncorrected parameter, conditional cross-sectional independence gives

$$\hat{s}_{\text{diag}}^2 = \sum_{i,t} (\hat{\Psi}_{it} \hat{u}_{it})^2.$$

Although (4.5) makes the sum of the residual-weighted Riesz terms zero at an exact interior solution, the individual terms are not zero. Their squares and cross-products estimate the variance, just as observation-level residual-regressor products do in the usual heteroskedasticity-robust regression variance formula.

When unit-specific shocks may be spatially dependent, let $\mathbf{W}_N = (w_{ik,N})$ be a prespecified positive-semidefinite spatial weight matrix. It is symmetric, $w_{ii,N} = 1$, $|w_{ik,N}| \leq 1$, and $w_{ik,N} = 0$ whenever $d_N(i, k) > h_N$, where $h_N = \lceil c_h \log(NT) \rceil$ for a fixed sufficiently large c_h . The spatial HAC estimator is

$$\hat{s}_{\text{sp}}^2 = \sum_{t=1}^T \sum_{i=1}^N \sum_{k=1}^N w_{ik,N} \hat{\Psi}_{it} \hat{u}_{it} \hat{\Psi}_{kt} \hat{u}_{kt}. \quad (4.8)$$

The matrix $\mathbf{W}_N$ retains nearby cross-unit covariances and excludes sufficiently distant pairs. If units belong to known conditionally independent clusters, set $w_{ik,N} = 1$ within a cluster and zero across clusters. The diagonal estimator is the special case $\mathbf{W}_N = \mathbf{I}_N$. A general distance-based calculation uses a positive-semidefinite construction so that the reported variance is nonnegative.

For the corrected estimator, use the same spatial quadratic form with the full- and split-panel residual terms combined inside each observation:

$$\begin{aligned} \hat{s}_{\text{corr}}^2 = & \sum_{t=1}^T \sum_{i=1}^N \sum_{k=1}^N w_{ik,N} \left[3\hat{\Psi}_{it} \hat{u}_{it} - \hat{\Psi}_{it}^T \hat{u}_{it}^T - \hat{\Psi}_{it}^N \hat{u}_{it}^N \right] \\ & \times \left[3\hat{\Psi}_{kt} \hat{u}_{kt} - \hat{\Psi}_{kt}^T \hat{u}_{kt}^T - \hat{\Psi}_{kt}^N \hat{u}_{kt}^N \right]. \end{aligned} \quad (4.9)$$

This construction automatically includes the variances of the full- and split-panel estimates and every covariance among them. [Theorem 4](#) proves consistency of (4.8) and (4.9) for the corresponding population variances.

No temporal HAC terms are added because conditional sequential exogeneity makes the population time-level Riesz sums martingale differences. For $s < t$,

$$\mathbb{E}_0 \left[\left\{ \sum_i \Psi_{0,it} u_{it} \right\} \left\{ \sum_i \Psi_{0,is} u_{is} \right\} \middle| \mathcal{C} \right] = 0.$$

Thus, cross-time covariance does not enter the first-order variance, while same-time spatial covariance remains in (4.8) and (4.9).

4.5 Rank selection

Let

$$\mathbf{r} = (r_{A,1}, \dots, r_{A,P}, r_{B,1}, \dots, r_{B,K}, r_H)$$

denote a rank vector. If $\bar{r}_{A,\ell}$, $\bar{r}_{B,k}$, and $\bar{r}_H$ are researcher-specified rank caps, define

$$\mathcal{R}_{\max} = \prod_{\ell=1}^P \{0, 1, \dots, \bar{r}_{A,\ell}\} \times \prod_{k=1}^K \{0, 1, \dots, \bar{r}_{B,k}\} \times \{0, 1, \dots, \bar{r}_H\}.$$

The rank procedure has two distinct layers. The first is computational screening: preliminary low-rank estimates generate a small list of plausible rank vectors and starting values. The second is statistical selection: every retained vector is refitted by unpenalized joint fixed-rank least squares and the post-refits are compared by an information criterion. Only the second layer determines the reported ranks, coefficient estimates, Riesz representers, and standard errors. The penalized estimates are never used as final coefficient estimates.

The rank-consistency theorem is deliberately stated in terms of the accuracy of an included preliminary estimate rather than a particular screening grid. The baseline implementation uses two sources of preliminary estimates: a convex nuclear-norm path for efficient screening and warm starts, and one rank-cap pilot that provides the candidate-coverage safeguard. Other preliminary estimators can replace either source if they satisfy the operator-norm requirement in [Theorem 5](#). Thus, the theorem concerns pilot accuracy, singular-value separation, and post-refit information-criterion separation; the numerical resolution of the nuclear path is an implementation choice.

Write the normalized least-squares loss as

$$\mathcal{L}_{NT}(\boldsymbol{\Theta}) = \frac{1}{2NT} \sum_{i,t} \{y_{it} - \mathcal{Y}_{it}(\boldsymbol{\Theta})\}^2.$$

To put the penalties for the different coefficient matrices on comparable fitted-value scales, use

$$w_{A,\ell} = \left(\frac{1}{NT} \sum_{i,t} y_{i,t-\ell}^2 \right)^{1/2}, \quad w_{B,k} = \left(\frac{1}{NT} \sum_{i,t} x_{i,t,k}^2 \right)^{1/2}, \quad w_H = 1.$$

Equivalently, the lagged outcomes and covariates can be standardized before the screening step. The unpenalized post-refits and the reported coefficient matrices remain in the original units.

For $\lambda > 0$, compute

$$\tilde{\boldsymbol{\Theta}}(\lambda) \in \arg \min_{\|\boldsymbol{\Theta}\|_{\max} \leq B} [\mathcal{L}_{NT}(\boldsymbol{\Theta}) + \lambda \mathcal{J}_{NT}(\boldsymbol{\Theta})], \quad (4.10)$$

where

$$\mathcal{J}_{NT}(\boldsymbol{\Theta}) = \sum_{\ell=1}^P w_{A,\ell} \|\mathbf{A}^{(\ell)}\|_* + \sum_{k=1}^K w_{B,k} \|\mathbf{B}^{(k)}\|_* + w_H \|\mathbf{H}\|_*.$$

For a coefficient matrix $\mathbf{M}$, let $\nabla_M \mathcal{L}_{NT}(\mathbf{0})$ be the gradient of the loss with respect to that matrix

at the zero coefficient collection. Define

$$\lambda_{\max} = \max_M \frac{\|\nabla_M \mathcal{L}_{NT}(\mathbf{0})\|_{\text{op}}}{w_M}. \quad (4.11)$$

The zero coefficient collection satisfies the penalized first-order conditions for every $\lambda \geq \lambda_{\max}$. Thus, (4.11) supplies a reproducible upper endpoint rather than an arbitrary starting penalty.

Let $\Lambda_{NT} = \{\lambda_1 > \dots > \lambda_L > 0\}$ be a finite decreasing screening grid with $\lambda_1 = \lambda_{\max}$. The exact geometric grid and its denser sensitivity version are specified in Section 7; they are computational choices and do not enter the rank-consistency theorem.

In addition to the convex path, compute one rank-cap pilot

$$\hat{\Theta}^{\max} \in \arg \min_{\substack{\|\Theta\|_{\max} \leq B \\ \text{rank}(\mathbf{M}) \leq \bar{r}_M \text{ for every coefficient matrix } \mathbf{M}}} \mathcal{L}_{NT}(\Theta). \quad (4.12)$$

This is one bounded-rank fit, not a search over $\mathcal{R}_{\max}$. It may be computed by rank-adaptive fixed-rank updates. In the baseline procedure, $\hat{\Theta}^{\max}$ is the candidate-coverage pilot. More generally, the theory permits any included pilot $\hat{\Theta}^{\text{pil}}$ satisfying the operator-norm condition in Theorem 5.

Let $\tau_{NT} > 0$ be a singular-value threshold. For every path estimate and for the rank-cap pilot, record the number of singular values of each coefficient matrix exceeding τ_{NT} . Denote the resulting rank vectors by $\tilde{\mathbf{r}}(\lambda)$ and $\hat{\mathbf{r}}^{\max}$. The theory restricts only the rate of τ_{NT} ; the baseline numerical value and its sensitivity multipliers are specified in Section 7.

Form

$$\mathcal{R}_{\text{base}} = \{\hat{\mathbf{r}}^{\max}\} \cup \{\tilde{\mathbf{r}}(\lambda) : \lambda \in \Lambda_{NT}\}.$$

For each $\mathbf{r} \in \mathcal{R}_{\text{base}}$, retain $\mathbf{r}$ and every valid one-rank change $\mathbf{r} \pm \mathbf{e}_m$, where $\mathbf{e}_m$ changes the rank of only one coefficient matrix. The candidate set is

$$\mathcal{R}_{\text{cand}} = \bigcup_{\mathbf{r} \in \mathcal{R}_{\text{base}}} (\{\mathbf{r}\} \cup \{\mathbf{r} \pm \mathbf{e}_m : m = 1, \dots, P + K + 1\}) \cap \mathcal{R}_{\max}, \quad (4.13)$$

with invalid and duplicate vectors removed. Before duplicates are removed, the procedure generates at most

$$(|\Lambda_{NT}| + 1)\{1 + 2(P + K + 1)\}$$

candidates. This grows linearly in the number of coefficient matrices rather than exponentially as in a Cartesian search or a full radius-one neighborhood.

For every $\mathbf{r} \in \mathcal{R}_{\text{cand}}$, compute the unpenalized fixed-rank post-refit $\hat{\Theta}(\mathbf{r})$ in (4.1). Its residual mean square is

$$\hat{Q}_{NT}(\mathbf{r}) = \frac{1}{NT} \sum_{i,t} \{y_{it} - \mathcal{Y}_{it}(\hat{\Theta}(\mathbf{r}))\}^2. \quad (4.14)$$

For a rank- r , $N \times T$ matrix, $r(N + T - r)$ is the number of locally free parameters. Hence

$$d(\mathbf{r}) = \sum_{\ell=1}^P r_{A,\ell}(N + T - r_{A,\ell}) + \sum_{k=1}^K r_{B,k}(N + T - r_{B,k}) + r_H(N + T - r_H).$$

For a positive deterministic sequence κ_{NT} , select

$$\hat{\mathbf{r}} \in \arg \min_{\mathbf{r} \in \mathcal{R}_{\text{cand}}} \left[\log\{\hat{Q}_{NT}(\mathbf{r})\} + \kappa_{NT} \frac{d(\mathbf{r})}{NT} \right]. \quad (4.15)$$

Ties are resolved in favor of the smaller dimension. Before finalizing the selected vector, the implementation evaluates every valid one-coordinate neighbor of the provisional minimizer that has not already been post-refitted. If an added vector has a lower information criterion, it is included and the minimization is repeated until no omitted one-coordinate neighbor improves the criterion. This local completion adds at most a small number of fits and prevents a coarse screening path from stopping immediately next to a better rank vector. After selecting $\hat{\mathbf{r}}$, all preceding formulas are used with $\mathbf{r} = \hat{\mathbf{r}}$ and $\hat{\Theta} = \hat{\Theta}(\hat{\mathbf{r}})$. Rank selection is not repeated for parameters of interest or split panels. Nuclear shrinkage is therefore removed before residual fit is compared and before any inferential object is constructed.

The nuclear path implements the post-regularization idea of using a convex low-rank estimate to locate promising singular spaces and provide warm starts, while the unpenalized post-refits jointly update all autoregressive, slope, and interactive-effect matrices. Rank-adaptive Riemannian updates can be used for the rank-cap pilot and nearby post-refits. These computational ideas are related to [Chernozhukov et al. \(2023\)](#); [Moon and Weidner \(2026\)](#); [Gao and Absil \(2022\)](#), but the rank-consistency result below is model specific. It depends on pilot accuracy, threshold separation, and information-criterion separation, not on a particular nuclear grid. The implementation reports rank-stability diagnostics because the inferential theorems require consistent rank selection and do not provide robustness to rank overestimation.

5 Assumptions

Let $\mathcal{F}_{t-1} = \sigma(y_{is}, x_{is} : i \leq N, s \leq t-1)$, including the observed presample outcomes. The common-shock sigma-field $\mathcal{C}$ was defined in [Section 3](#). Conditional on $\mathcal{C}$, the true coefficient matrices and deterministic parameter weights are treated as fixed. The true rank vector $\mathbf{r}_0$ is nonrandom and does not depend on the realization of $\mathcal{C}$. A *regular common-shock realization* is a realization for which the conditional moment, dependence, covariance, signal, and Gram bounds stated below hold with their deterministic envelopes; the assumptions require this set of realizations to have probability one. A *regular individual coefficient* is an entry target satisfying the leverage and target-variance noncancellation requirements in [Assumption 9](#). All conditional statements below hold for almost every regular common-shock realization. Fix $\eta > 0$. Throughout this section, c and C denote fixed positive bounded constants that do not depend on N , T , i , t , or the realization of the common

shocks; their values may differ from line to line. Other constants introduced below are also fixed unless stated otherwise.

Assumption 1 (Dynamic stability). *Let $\mathbf{C}_{0,it}$ be the companion matrix formed from $a_{0,it,1}, \dots, a_{0,it,P}$. For some $\rho \in (0, 1)$,*

$$\sup_{i,h,t+h \leq T} \|\mathbf{C}_{0,i,t+h} \cdots \mathbf{C}_{0,i,t+1}\|_{\text{op}} \leq C\rho^h. \quad (5.1)$$

The observed initial outcome states satisfy $\sup_{i,s=1-P,\dots,0} \mathbb{E}_0(|y_{is}|^{8+\eta} \mid \mathcal{C}) \leq C$ almost surely.

This assumption permits time-varying autoregressive coefficients but rules out explosive propagation. The effect of a current disturbance through future lagged outcomes decreases geometrically with the time horizon, while the initial observations have the same conditional moment order used elsewhere in the paper.

Assumption 2 (Conditional sequential exogeneity). *For every i and t ,*

$$\mathbb{E}_0[u_{it} \mid x_{it}, \mathcal{F}_{t-1}, \mathcal{C}] = 0. \quad (5.2)$$

This is the conditional sequential-exogeneity formulation used in dynamic interactive-effects and dynamic spatial-panel models; see [Moon and Weidner \(2017\)](#) and [Kuersteiner and Prucha \(2020\)](#). It defines u_{it} as the new unit-specific innovation after the current covariates, past panel information, and common shocks are held fixed. Future unit-level covariates are not in the conditioning set and may respond to past outcome disturbances.

Assumption 3 (Spatial sampling geometry). *The unit locations lie in $\mathbb{R}^{d_s}$ for fixed d_s and are uniformly separated. If*

$$v_N(r) = \max_{i \leq N} |\{j \leq N : d_N(i, j) \leq r\}|,$$

then $v_N(r) \leq C(1+r)^{d_s}$ uniformly in N and $r \geq 0$.

This assumption limits how quickly the number of spatial neighbors can grow with distance. It allows irregular locations while making sums of local spatial covariances manageable.

Let ε_{it}^x collect the unit-specific innovations that generate the covariates and set $\varepsilon_{it} = ((\varepsilon_{it}^x)', u_{it})'$. For a set $\mathcal{A}$ of unit-time indices, write $\mathcal{G}(\mathcal{A}) = \sigma\{\varepsilon_{it} : (i, t) \in \mathcal{A}\}$. For two index sets $\mathcal{A}$ and $\mathcal{B}$, define

$$\text{dist}(\mathcal{A}, \mathcal{B}) = \min_{\substack{(i,t) \in \mathcal{A} \\ (j,s) \in \mathcal{B}}} \{d_N(i, j) + |t - s|\}.$$

Conditional on $\mathcal{C}$, the strong-mixing, or α -mixing, coefficient at separation a is

$$\alpha_{\mathcal{C}}(a) = \sup_{\text{dist}(\mathcal{A}, \mathcal{B}) \geq a} \sup_{\substack{E \in \mathcal{G}(\mathcal{A}) \\ F \in \mathcal{G}(\mathcal{B})}} |\mathbb{P}_0(E \cap F \mid \mathcal{C}) - \mathbb{P}_0(E \mid \mathcal{C})\mathbb{P}_0(F \mid \mathcal{C})|.$$

Let $\bar{\alpha}(a)$ be a deterministic, nonincreasing sequence, independent of N, T and the realization of $\mathcal{C}$,

that uniformly bounds $\alpha_{\mathcal{C}}(a)$. For $h, r \geq 0$, let

$$\mathcal{E}_{it}(h, r) = \sigma\left(\varepsilon_{js}^x : t - h \leq s \leq t, d_N(i, j) \leq r; u_{js} : t - h \leq s \leq t - 1, d_N(i, j) \leq r\right)$$

Assumption 4 (Conditional alpha mixing and near-epoch dependence). *Almost surely, $\alpha_{\mathcal{C}}(a) \leq \bar{\alpha}(a) \leq Ce^{-ca}$. Moreover, for every covariate,*

$$\sup_{i,t,k} \left\{ \mathbb{E}_0 \left[|x_{it,k} - \mathbb{E}_0[x_{it,k} \mid \mathcal{E}_{it}(h, r) \vee \mathcal{C}]|^{8+\eta} \mid \mathcal{C} \right] \right\}^{1/(8+\eta)} \leq C\{\psi_T(h) + \psi_S(r)\}, \quad (5.3)$$

where $\psi_T(h) \leq Ce^{-ch}$ and $\psi_S(r) \leq Ce^{-cr}$. Current covariate innovations are included in $\mathcal{E}_{it}(h, r)$, while the current outcome innovation is excluded.

Conditional on the common shocks, distant unit-specific innovations become nearly independent, while each covariate can be approximated by recent innovations from nearby units. The deterministic envelope $\bar{\alpha}(a)$ permits the realized conditional dependence to vary with the common shocks while imposing one uniform decay rate. The near-epoch-dependence formulation follows the spatial random-field framework of [Jenish and Prucha \(2012\)](#), adapted here to the combined unit–time index and to conditioning on common shocks. Excluding current u_{it} from the approximation preserves the predetermined timing of the covariates.

Assumption 5 (Conditional moments and weak cross-sectional covariance). *Let $\mathbf{u}_t = (u_{1t}, \dots, u_{Nt})'$, let $\mathbf{x}_t$ collect the current covariates of all units, and define $\boldsymbol{\Sigma}_{u,t} = \mathbb{E}_0[\mathbf{u}_t \mathbf{u}_t' \mid \mathbf{x}_t, \mathcal{F}_{t-1}, \mathcal{C}]$.*

(i) *For the fixed $\eta > 0$, uniformly in i, t, k*

$$\mathbb{E}_0[|u_{it}|^{8+\eta} \mid x_{it}, \mathcal{F}_{t-1}, \mathcal{C}] \leq C, \quad \mathbb{E}_0[|x_{it,k}|^{8+\eta} \mid \mathcal{C}] \leq C.$$

(ii) *Uniformly in t ,*

$$c \leq (\boldsymbol{\Sigma}_{u,t})_{ii} \leq C \quad \text{for every } i, \quad \max_{i \leq N} \sum_{j=1}^N |(\boldsymbol{\Sigma}_{u,t})_{ij}| \leq C.$$

The moment bounds control the products and quadratic forms used in estimation and variance calculation. The row-sum bound permits conditional spatial dependence but prevents the variance of a cross-sectional sum from growing because each unit is strongly related to too many other units. It implies $\|\boldsymbol{\Sigma}_{u,t}\|_{\text{op}} \leq C$. We do not require the smallest eigenvalue of the entire $N \times N$ covariance matrix to be uniformly positive. Instead, [Assumption 9](#) imposes nondegeneracy only in the Riesz directions used for inference. For comparison, [Bai et al. \(2021\)](#) impose the stronger requirement that each relevant covariance block have a smallest eigenvalue bounded away from zero and bounded absolute row sums.

Assumption 6 (Separate low-rank signals and incoherence). *For every positive-rank matrix*

$$\mathbf{M}_0 \in \{\mathbf{A}_0^{(1)}, \dots, \mathbf{A}_0^{(P)}, \mathbf{B}_0^{(1)}, \dots, \mathbf{B}_0^{(K)}, \mathbf{H}_0\},$$

let $\mathbf{M}_0 = \mathbf{U}_M \boldsymbol{\Sigma}_M \mathbf{V}_M'$ be its compact singular-value decomposition. Uniformly over these matrices,

$$c\sqrt{NT} \leq \sigma_{r_M}(\mathbf{M}_0) \leq \sigma_1(\mathbf{M}_0) \leq C\sqrt{NT},$$

and

$$\max_i \|\mathbf{e}_i' \mathbf{U}_M\|_2^2 \leq \frac{\mu_{r_M}}{N}, \quad \max_t \|\mathbf{e}_t' \mathbf{V}_M\|_2^2 \leq \frac{\mu_{r_M}}{T}.$$

Every true coefficient entry is bounded in absolute value by $B - c_B$ for some fixed interior margin $c_B \in (0, B)$. No singular-value condition is imposed on a rank-zero matrix.

The singular-value bounds make every nonzero low-rank matrix statistically visible, while incoherence spreads its variation across units and time periods. Thus, no single row or column carries the entire low-rank signal. The margin c_B need not be small; it only places the true coefficients a fixed positive distance from the boundary of the estimation set. Together with max-norm consistency, this makes the coefficient bound asymptotically inactive and permits the ordinary fixed-rank first-order conditions to be used.

Assumption 7 (Conditional prediction identification). *The lower bound*

$$\mathbb{E}_0 \left[\sum_{i,t} \mathcal{Y}_{it}(\boldsymbol{\Delta})^2 \middle| \mathcal{C} \right] \geq c \|\boldsymbol{\Delta}\|^2 \quad (5.4)$$

holds both (i) for every nonzero admissible difference $\boldsymbol{\Delta} = \boldsymbol{\Theta} - \boldsymbol{\Theta}_0$, with $\boldsymbol{\Theta}$ ranging over the ranks in $\mathcal{R}_{\max}$ and satisfying the coefficient bound, and (ii) for every nonzero tangent direction $\boldsymbol{\Delta} \in \mathcal{T}_0$.

This is an identification condition on the coefficient matrices, not a bound on u_{it} . The first part says that two distinct admissible low-rank coefficient collections cannot produce the same conditional fitted values. The second part is the local version needed for the Riesz equation and the loading-factor normal equations: it makes the conditional least-squares Gram form coercive on $\mathcal{T}_0$. A simple sufficient setting is one in which the lagged outcomes, covariates, and constant have conditionally nonsingular second moments after their low-rank directions are taken into account. Identification can fail, for example, if a covariate is exactly equal to a lagged outcome and the associated coefficient matrices can offset one another, or if a constant covariate duplicates the unrestricted role of $\mathbf{H}_0$. The corresponding upper bound follows from the fixed numbers of lags and covariates and the moment bounds in [Assumption 5](#).

For the next assumption, write every positive-rank matrix as

$$M_{0,it} = \boldsymbol{\lambda}_{M,0,i}' \mathbf{f}_{M,0,t}, \quad \frac{1}{N} \sum_{i=1}^N \boldsymbol{\lambda}_{M,0,i} \boldsymbol{\lambda}_{M,0,i}' = \mathbf{I}_{r_M}.$$

Let $r_+ = \sum_{\ell=1}^P r_{A,\ell} + \sum_{k=1}^K r_{B,k} + r_H$, omitting zero-rank matrices. Set $z_{it,A^{(\ell)}} = y_{i,t-\ell}$, $z_{it,B^{(k)}} = x_{it,k}$, and $z_{it,H} = 1$. Stack the r_+ loading-regression terms $z_{it,M} \mathbf{f}_{M,0,t}$ in $\mathbf{g}_{it}^\lambda \in \mathbb{R}^{r_+}$, and stack the factor-regression terms $z_{it,M} \boldsymbol{\lambda}_{M,0,i}$ in $\mathbf{g}_{it}^f \in \mathbb{R}^{r_+}$.

Assumption 8 (Loading and factor Gram matrices). *Let $\mathbf{I}$ below denote the $r_+ \times r_+$ identity matrix. Uniformly over i and t ,*

$$c\mathbf{I} \preceq \mathbb{E}_0 \left[\frac{1}{T} \sum_{s=1}^T \mathbf{g}_{is}^\lambda (\mathbf{g}_{is}^\lambda)' \middle| \mathcal{C} \right] \preceq C\mathbf{I},$$

and

$$c\mathbf{I} \preceq \mathbb{E}_0 \left[\frac{1}{N} \sum_{j=1}^N \mathbf{g}_{jt}^f (\mathbf{g}_{jt}^f)' \middle| \mathcal{C} \right] \preceq C\mathbf{I}.$$

These are ordinary least-squares second-moment matrices, not autoregressive or slope coefficient matrices. The assumption requires enough time-series variation to estimate each unit's loading rows and enough cross-sectional variation to estimate each time factor row. A close recent analogue is Assumption 5 of Wang et al. (2022), which imposes uniform eigenvalue bounds on the row- and column-wise regression matrices in a low-rank panel model. Peng et al. (2025) impose a related positive-definiteness condition on regressor moments after accounting for interactive effects.

Assumption 9 (Regularity of the parameters of interest). *If a unit or time averaging weight has growing support, its largest absolute value converges to zero. For a mean or contrast,*

$$\|\mathcal{P}_0 \mathbf{D}\| \geq c \|\mathbf{D}\|.$$

For an individual entry of a positive-rank matrix $\mathbf{M}_0 = \mathbf{U}_M \boldsymbol{\Sigma}_M \mathbf{V}_M'$, the selected unit and time satisfy

$$\|\mathbf{e}_i' \mathbf{U}_M\|_2^2 \geq c/N, \quad \|\mathbf{e}_t' \mathbf{V}_M\|_2^2 \geq c/T.$$

Let $\boldsymbol{\psi}_t = (\Psi_{0,1t}, \dots, \Psi_{0,Nt})'$. For every parameter of interest,

$$\sum_{t=1}^T \mathbb{E}_0[\boldsymbol{\psi}_t' \boldsymbol{\Sigma}_{u,t} \boldsymbol{\psi}_t \mid \mathcal{C}] \geq c \mathbb{E}_0 \left[\sum_{t=1}^T \|\boldsymbol{\psi}_t\|_2^2 \middle| \mathcal{C} \right].$$

For every fixed finite vector of parameters of interest, the limiting conditional correlation matrix of their variance-normalized first-order Riesz terms is positive definite.

The first restriction prevents one unit or one time from dominating a growing average. The projection and leverage restrictions ensure that the chosen parameter changes along statistically identified low-rank directions. The displayed variance restriction rules out near cancellation only for the Riesz-weighted combinations used in the reported inference; unlike a global lower eigenvalue bound on $\boldsymbol{\Sigma}_{u,t}$, it does not restrict irrelevant cross-sectional directions. The final restriction rules out redundant linear combinations in joint inference.

Let

$$b_{NT} = (NT)^{1/(8+\eta)} \log(NT). \quad (5.5)$$

The quantity b_{NT} is used only in the theoretical rates and reflects the finite $8 + \eta$ moments; it is not selected from the data. The rank-selection theorem states the admissible rate conditions for τ_{NT} and κ_{NT} , while [Section 7](#) gives the baseline numerical choices.

Assumption 10 (Panel growth). *As $N, T \rightarrow \infty$,*

$$(NT)^{4/(8+\eta)} \{\log(NT)\}^{d_s+6} \left(\frac{1}{N} + \frac{1}{T} \right) \rightarrow 0. \quad (5.6)$$

This condition allows N and T to grow at different rates but requires both dimensions to be large enough relative to the moment and spatial-complexity terms. It delivers the recovery rate used by the full-panel estimator and the rank-cap pilot. The additional threshold and penalty rates needed only for rank selection are stated directly in [Theorem 5](#).

The assumptions above contain the maintained restrictions on the data-generating process, parameters of interest, and panel growth. No separate high-level conditions are imposed. The appendix derives the uniform low-rank score and Gram bounds, the proof-only local-deletion rates, the preservation of ranks and Gram matrices by the balanced random splits, feasible spatial variance estimation, and rank consistency. Short numerical requirements for approximate optimization are stated with the results that use them.

6 Asymptotic Theory

All probability statements in this section are conditional on $\mathcal{C}$. The deterministic bounds in the assumptions make the conclusions uniform over regular realizations of the common shocks, so conditional validity implies unconditional validity. Let $\Phi(z)$ denote the cumulative distribution function of a standard normal random variable, and define the first-stage rate

$$\zeta_{NT} = b_{NT}^2 \{\log(NT)\}^{d_s+2} \left(\frac{1}{N} + \frac{1}{T} \right).$$

The logarithmic power reflects the fixed-dimensional unit-time blocking and the low-rank covering argument in [Lemma 4](#) and [Proposition 2](#). The panel-growth assumption is stronger than what is needed for $\zeta_{NT} \rightarrow 0$ and continues to imply all rank-threshold and split-panel rates below. Proofs are collected in the mathematical appendix: [Section A.2](#) proves [Theorem 1](#); [Sections A.3](#) and [A.4](#) prove [Theorem 2](#); [Section A.5](#) proves [Theorems 3](#) and [4](#); [Section A.6](#) proves the variance statements in [Theorem 4](#); and [Section A.7](#) proves [Theorem 5](#).

Theorem 1 (Conditional normality of the population Riesz term). *Under [Assumptions 1](#) to [10](#),*

$$\frac{\sum_{i,t} \Psi_{0,it} u_{it}}{s_{NT}} \xrightarrow{d} N(0, 1) \quad \text{conditionally on } \mathcal{C}.$$

Equivalently,

$$\sup_{z \in \mathbb{R}} \left| \mathbb{P}_0 \left[\frac{\sum_{i,t} \Psi_{0,it} u_{it}}{s_{NT}} \leq z \middle| \mathcal{C} \right] - \Phi(z) \right| \rightarrow_p 0. \quad (6.1)$$

The conclusion holds jointly for every fixed finite vector of parameters of interest after conditional covariance normalization.

Assumption 7 is a population restriction: it says that distinct admissible coefficient matrices imply distinct conditional fitted values. The next theorem is a sample result. It uses identification together with the moment and dependence assumptions to show that the least-squares estimator, its singular spaces, and the empirical Riesz representer converge to their population counterparts.

Theorem 2 (Consistency of the full-panel estimator and Riesz representer). *Suppose the true ranks are supplied and (4.1) is solved exactly. Under Assumptions 1 to 8 and 10,*

$$\frac{1}{NT} \|\hat{\Theta}(\mathbf{r}_0) - \Theta_0\|^2 + \frac{1}{NT} \sum_{i,t} \mathcal{Y}_{it} \{\hat{\Theta}(\mathbf{r}_0) - \Theta_0\}^2 = O_p(\zeta_{NT}),$$

$$\|\hat{\Theta}(\mathbf{r}_0) - \Theta_0\|_{\max} = o_p(1), \quad \|\hat{\mathcal{P}} - \mathcal{P}_0\|_{\text{op}} = O_p(\zeta_{NT}^{1/2}).$$

If Assumption 9 also holds for ϕ , then

$$\frac{\|\hat{\mathbf{q}} - \mathbf{q}_0\|}{\|\mathcal{P}_0 \mathbf{D}\|} = o_p(1), \quad \sum_{i,t} (\hat{\Psi}_{it} - \Psi_{0,it})^2 = o_p(\|\mathcal{P}_0 \mathbf{D}\|^2).$$

The first line establishes consistency of all separately ranked coefficient matrices in Frobenius and prediction norms. The second line gives entrywise and tangent-space recovery, and the last line justifies replacing the population Riesz weights by feasible weights. Root- T loading-row and root- N factor-row expansions used in the bias proof are given in Section A.3. The same conclusions hold for numerical solutions satisfying the objective, first-order, and Riesz-equation tolerances reported in Section 7.

Theorem 3 (Full-panel expansion and asymptotic normality). *Suppose the true ranks are supplied. Under Assumptions 1 to 10, there are bounded $\mathcal{C}$ -measurable terms B_T and B_N such that*

$$\hat{\phi} - \phi(\Theta_0) = \sum_{i,t} \Psi_{0,it} u_{it} + \frac{B_T}{T} + \frac{B_N}{N} + r_{NT}, \quad (6.2)$$

where $r_{NT} = o_p(s_{NT}) + o_p(N^{-1} + T^{-1})$. The T^{-1} term is generated by loading estimation and pre-terminated dynamic regressors; the N^{-1} term is generated by factor estimation and contemporaneous cross-sectional dependence.

For a regular individual coefficient or a fixed unit-time lag sum,

$$\frac{\hat{\phi} - \phi(\Theta_0)}{s_{NT}} \xrightarrow{d} N(0, 1) \quad \text{conditionally on } \mathcal{C}.$$

The same conclusion holds for a mean over $G_N \times R_T$ when

$$\sqrt{|G_N||R_T|} \left(\frac{1}{N} + \frac{1}{T} \right) \rightarrow 0,$$

and for a contrast between disjoint groups $G_{1,N}$ and $G_{2,N}$ averaged over R_T when

$$\sqrt{\frac{|R_T|}{|G_{1,N}|^{-1} + |G_{2,N}|^{-1}}} \left(\frac{1}{N} + \frac{1}{T} \right) \rightarrow 0.$$

Proposition 1 proves the variance rates used in this statement. In particular, the support-growth restriction is automatic for a regular individual coefficient and a fixed unit-time lag sum. A full-panel mean generally fails the displayed mean condition and is therefore covered by the split-panel correction in **Theorem 4**.

Theorem 4 (Split-panel correction and feasible asymptotic normality). *Suppose the true ranks are supplied. Use the independent balanced time and unit partitions defined in [Section 4.3](#), drawn independently of the panel, and hold the full-panel ranks fixed in all four split-panel fits. The conclusions hold conditionally on $\mathcal{C}$ for almost every partition sequence. Under [Assumptions 1 to 10](#),*

$$\hat{\phi}^{\text{corr}} - \phi(\Theta_0) = \sum_{i,t} u_{it} [3\Psi_{0,it} - \Psi_{0,it}^T - \Psi_{0,it}^N] + o_p(s_{NT}^{\text{corr}}).$$

Moreover,

$$\frac{(s_{NT}^{\text{corr}})^2}{s_{NT}^2} \rightarrow_p 1, \quad \frac{\hat{s}_{\text{corr}}^2}{(s_{NT}^{\text{corr}})^2} \rightarrow_p 1, \quad \frac{\hat{\phi}^{\text{corr}} - \phi(\Theta_0)}{\hat{s}_{\text{corr}}} \xrightarrow{d} N(0, 1) \quad \text{conditionally on } \mathcal{C}.$$

The analogous feasible normality result holds for an uncorrected parameter with [\(4.8\)](#).

Theorem 5 (Rank consistency and selected-rank inference). *Suppose [Assumptions 1 to 8](#) and [10](#) hold and $\mathbf{r}_0 \in \mathcal{R}_{\max}$. Suppose the candidate set contains the thresholded rank vector of at least one pilot $\hat{\Theta}^{\text{pil}}$ satisfying*

$$\max_M \|\widehat{\mathbf{M}}^{\text{pil}} - \mathbf{M}_0\|_{\text{op}} = o_p(\tau_{NT}), \quad \tau_{NT} = o(\sqrt{NT}). \quad (6.3)$$

Let the information-criterion penalty satisfy

$$\zeta_{NT} = o\left(\kappa_{NT} \frac{N+T}{NT}\right), \quad \kappa_{NT} \frac{N+T}{NT} \rightarrow 0. \quad (6.4)$$

Assume that the normalized objective errors of the candidate post-refits are

$$o_p\left(\kappa_{NT} \frac{N+T}{NT}\right).$$

Then the screened information criterion (4.15) satisfies

$$\mathbb{P}_0(\hat{\mathbf{r}} = \mathbf{r}_0 \mid \mathcal{C}) \rightarrow_p 1. \quad (6.5)$$

Consequently, Theorems 2 to 4 remain valid when the supplied ranks are replaced by the selected ranks. Rank selection is performed once on the full panel and is not repeated in the split panels.

Remark 1 (Baseline pilot and threshold). Under the maintained assumptions, the rank-cap pilot in (4.12) obeys

$$\max_M \|\widehat{\mathbf{M}}^{\max} - \mathbf{M}_0\|_{\text{op}} = O_p\{\sqrt{NT\zeta_{NT}}\}$$

when its normalized objective gap is $o_p(\zeta_{NT})$. It therefore satisfies (6.3) whenever

$$\sqrt{NT\zeta_{NT}} = o(\tau_{NT}) \quad \text{and} \quad \tau_{NT} = o(\sqrt{NT}).$$

The baseline threshold in Section 7 has these properties under Assumption 10. The nuclear path enriches the candidate set and supplies warm starts, but its grid is not needed for candidate coverage.

Remark 2 (Strong singular values under approximate low rank). The thresholding argument has a limited screening interpretation beyond exact low rank. If, for one coefficient matrix,

$$\mathbf{M}_0 = \mathbf{M}_{0,r} + \mathbf{R}_M,$$

where $\mathbf{M}_{0,r}$ has r singular values of order $\sqrt{NT}$ and $\|\mathbf{R}_M\|_{\text{op}} = o(\tau_{NT})$, then a pilot satisfying (6.3) identifies the number r of strong singular values by the same perturbation argument. This observation concerns screening only. All inferential theorems in the paper continue to assume exact fixed ranks because the approximation remainder can enter the parameter of interest, the dynamic recursion, and the joint Riesz equation.

7 Implementation

The implemented procedure follows the estimators in Section 4. It uses no validation folds and no forward or spatial buffers. The local deletions used in the proofs are never computed.

7.1 Practical implementation

The implementation separates screening, post-refitting, rank selection, and inference. This separation is important: the nuclear estimates are used only to propose ranks and initialize numerical optimization, whereas every reported coefficient and standard error is based on an unpenalized fixed-rank post-refit.

A practical implementation proceeds as follows.

1. *Preprocessing and admissible ranks.* Standardize the lagged outcomes and observed covariates used in the screening problems. Specify the matrix-specific rank caps, the coefficient bound

B , and the admissible set $\mathcal{R}_{\max}$. Store the transformations so that all final estimates can be returned to the original units.

2. *Convex screening and the rank-cap pilot.* Compute the penalty weights and $\lambda_{\max}$ in (4.11). Solve the prespecified warm-started nuclear path described in Section 7.3. Separately compute the single rank-cap pilot in (4.12). The path supplies additional rank proposals and warm starts; the rank-cap pilot supplies the baseline candidate-coverage safeguard.
3. *Candidate generation.* Threshold the singular values of every preliminary estimate, add all valid one-coordinate neighboring ranks, intersect with $\mathcal{R}_{\max}$, and remove duplicates. Record which pilot generated each candidate and whether the rank-cap pilot rank is retained.
4. *Unpenalized post-refitting and final rank choice.* For every candidate rank vector, compute the joint unpenalized fixed-rank least-squares post-refit. Compare only these post-refits using (4.15). The penalized objective values and penalized coefficient matrices do not enter the final information criterion. Retain the selected post-refit as $\hat{\Theta}$.
5. *Rank-stability assessment.* Fit any valid one-coordinate neighbor of the selected vector that has not already been evaluated. If one has a lower information criterion, add it to the candidate set and repeat the selection step. Report the final selected rank vector, the rank-cap pilot vector, the distinct nuclear-path rank vectors, the candidate count after duplicate removal and local completion, the smallest and second-smallest information criteria, and whether the selected vector changes under the denser path, threshold multipliers, penalty multipliers, or larger rank caps. For each main parameter, report the change in its plug-in estimate under the one-coordinate neighboring rank vectors with the smallest information criteria. These are diagnostics, not rank-robust confidence intervals.
6. *Parameters and Riesz systems.* For each parameter of interest, compute the plug-in estimate and the empirical Riesz representer from the selected unpenalized post-refit. Report the full-panel plug-in estimate for the parameters covered by Theorem 3. For a two-way average covered by Theorem 4, compute the two time-split and two unit-split fits, holding $\hat{\mathbf{r}}$ fixed, and form the corrected estimate in (4.6).
7. *Variance estimation and numerical reporting.* Compute the diagonal, spatial, or known-cluster variance estimator appropriate for the maintained cross-sectional dependence. For the corrected estimate, combine the full- and split-panel residual–Riesz products as in (4.9). Store objective values, first-order residuals, coefficient-bound activity, retained singular values, Gram-matrix eigenvalues, Riesz-equation residuals, split-panel group counts, spatial distance cutoffs, runtimes, and random seeds.

The full-panel estimate initializes all four split-panel fits. The unit split is balanced within every group entering the parameter of interest and, when useful, within broad geographic strata. The time split is balanced over the full sample. For a selected time t , its observed lagged outcomes remain

available as predetermined regressors even when a lagged time belongs to the other partition; the dynamic process is not restarted.

A selected rank at its imposed cap, disagreement between the baseline and denser screening paths, failure of the rank-cap pilot vector to enter the candidate set, a nearly tied information criterion, or a nearly singular Riesz system is reported as rank sensitivity. The paper does not describe such results as rank robust. A primary interval is not reported when the selected rank remains unresolved at its cap, the numerical fit fails the stated tolerances, the coefficient bound is active, a split loses a selected rank or target support, or the relevant Gram or Riesz system is numerically singular.

7.2 Computation of the rank-constrained estimator

For a supplied rank vector, write each positive-rank coefficient matrix as $\mathbf{M} = \mathbf{L}_M \mathbf{F}_M'$ and set every rank-zero matrix equal to zero. Conditional on the factor rows, the least-squares criterion separates into loading regressions across units. Conditional on the loading rows, it separates into factor regressions across time. The estimator can therefore be computed by warm-started alternating least squares or by a fixed-rank manifold optimization routine. The optimization is joint: every update is evaluated using residuals formed from all autoregressive, slope, and interactive-effect matrices. For the rank-cap pilot, rank-adaptive Riemannian updates may increase a rank when the objective gradient has a substantial direction outside the current tangent space and reduce a rank when a fitted singular value falls below the numerical threshold. After each update, the factors and loadings are renormalized without changing their product.

Each candidate post-refit uses two deterministic starting values. The first is the nuclear-path or rank-cap estimate whose thresholded rank vector has the smallest ℓ_1 distance from the candidate, truncated or augmented to the candidate ranks. The second is the rank-cap pilot truncated to the candidate ranks. The lower-objective solution is retained. A third randomized start is used only when the two deterministic solutions disagree by more than the reported objective tolerance. A fit is retained only when the objective is stable across the best starts and the first-order residual is below the reported tolerance. Candidate post-refits are independent after their starting values are formed and can be run in parallel. The selected full-panel solution supplies starting values for all split-panel fits. No coefficient estimate from the nuclear path is carried directly into a reported table or standard error.

After estimating the matrices, construct the tangent projection from their singular spaces. The empirical Riesz equation should be solved without forming a dense tangent-space basis. The required linear operator maps a proposed matrix collection $\mathbf{Q}$ to the fitted-value changes $\{\mathcal{Y}_{it}(\mathbf{Q})\}_{i,t}$ and maps an observation array back through the adjoint fitted-value map followed by the tangent projection. Conjugate gradients or LSQR can then solve the positive-definite system using matrix–vector products. The same operator and preconditioner are reused across parameters of interest; only the direction $\mathbf{D}$ changes. The weighted residual identity (4.5) and the residual of the Riesz equation are reported as numerical checks.

For the spatial variance calculation, neighbor lists are precomputed for pairs satisfying $d_N(i, k) \leq h_N$. The calculation then has cost proportional to $TNv_N(h_N)$ rather than TN^2 .

7.3 Tuning choices

The rank theorem and the numerical implementation use different levels of specification. Theorem 5 requires an accurate included pilot, a separating singular-value threshold, and an information-criterion penalty satisfying the stated rates. The exact nuclear grid, threshold multipliers, optimization tolerances, and sensitivity calculations below are prespecified numerical choices. They affect screening resolution and computation, not the form of the consistency theorem.

The rank caps should be modest but high enough that the selected ranks do not repeatedly reach their upper bounds. The baseline nuclear path is the geometric grid

$$\lambda_m = \lambda_{\max} \gamma^{m-1}, \quad m = 1, \dots, L, \quad L = 1 + \left\lceil \frac{\log(\epsilon)}{\log(\gamma)} \right\rceil. \quad (7.1)$$

The implementation fixes $\gamma = 0.8$ and $\epsilon = 0.01$, giving $L = 22$ points between $\lambda_{\max}$ and approximately $0.01\lambda_{\max}$. These values determine numerical resolution and warm-start spacing; they are not used in Theorem 5. The denser path with $\gamma_{\text{dense}} = \sqrt{0.8}$ inserts one logarithmic midpoint between adjacent baseline values and is a prespecified sensitivity calculation. The implementation reports $\lambda_{\max}$, $\lambda_L/\lambda_{\max}$, the number of distinct thresholded path ranks, the size of $\mathcal{R}_{\text{cand}}$, and whether the denser path changes the selected rank or the ordering of the two smallest information criteria.

The penalty weights are the sample root-mean-square scales of the corresponding lagged outcome or covariate, with $w_H = 1$. The baseline singular-value threshold is

$$\tau_{NT} = \frac{\sqrt{NT}}{\log(NT)},$$

with sensitivity to the constants $1/2$ and 2 multiplying this threshold. This choice lies above the rank-cap pilot error and below the order- $\sqrt{NT}$ genuine singular values under the maintained rates. The theory permits any threshold satisfying (6.3).

The theory-aligned baseline information-criterion penalty is

$$\kappa_{NT} = b_{NT}^2 \{\log(NT)\}^{d_s+3}. \quad (7.2)$$

Because

$$\kappa_{NT} \frac{N+T}{NT} = \log(NT) \zeta_{NT},$$

this sequence satisfies (6.4) under Assumption 10. Sensitivity to the constants $1/2$ and 2 multiplying (7.2) is reported. The exhaustive Cartesian search is used only as a Monte Carlo benchmark when the caps make it computationally feasible. No cross-validation is used to choose the nuclear penalty because the penalized estimator is only a screening and warm-start device; final rank selection is based on unpenalized post-refits and the information criterion.

Rank stability is assessed using quantities already generated by the procedure. The implementation must report: the gap between the smallest and second-smallest information criteria; whether the selected rank changes on the denser path; whether it changes under the threshold and penalty multipliers; whether any selected rank equals its cap; and how the main plug-in estimates change under the best one-coordinate neighboring ranks. When a neighboring rank has a nearly identical criterion, the corresponding point-estimate sensitivity is reported even though the primary theorem continues to condition on consistent rank selection. These checks do not create rank-robust inference; they reveal when the exact-rank approximation is empirically fragile.

The coefficient bound B is chosen conservatively above the economically plausible coefficient range. The margin c_B in [Assumption 6](#) is a theoretical interior margin and is not selected by the researcher. A fitted coefficient at the bound is treated as a failed interiority diagnostic.

For general spatial dependence, use a prespecified positive-semidefinite spatial weight matrix and a distance cutoff $h_N = \lceil c_h \log(NT) \rceil$. Results should be reported over a prespecified range of c_h . Conditional cross-sectional independence uses $\mathbf{W}_N = \mathbf{I}_N$, while known clusters use within-cluster weights.

Optimization tolerances are stated relative to the scale of the objective and the estimated standard error. The implementation reports the relative objective change, the fixed-rank first-order residual, and the Riesz-equation residual. The random balanced time and unit partitions and their seeds are stored. The primary interval is not reported when the selected rank remains unresolved at its cap, no numerical solution satisfies the tolerances, the coefficient bound is active, a Riesz system or split-panel Gram matrix is nearly singular, or the chosen spatial variance calculation fails its positive-semidefiniteness check.

8 Monte Carlo Experiments

We consider four data-generating processes arranged in increasing order of difficulty. DGP 1 introduces unit-level heteroskedasticity. DGP 2 adds spatial dependence. DGP 3 adds predetermined covariates. DGP 4 retains the spatial and predetermined structure of DGP 3 and adds systematic group heterogeneity in the autoregressive and slope coefficient matrices.

8.1 Common dynamic design

The baseline model has one lag and one observed covariate:

$$y_{it} = a_{it}y_{i,t-1} + \beta_{it}x_{it} + H_{0,it} + u_{it}, \quad i = 1, \dots, N. \quad (8.1)$$

The process is generated for $t = -49, \dots, T$, with $y_{i,-50} = 0$, and the first 50 periods are discarded. To keep the coefficient matrices uniformly bounded and compatible with the maintained incoherence restrictions, the loading perturbations and common-factor innovations below are independent standardized uniform variables on $[-\sqrt{3}, \sqrt{3}]$. Gaussian draws are retained for idiosyncratic

disturbances.

For $m \in \{a, b\}$,

$$g_{m,t} = \rho_g g_{m,t-1} + (1 - \rho_g^2)^{1/2} v_{m,t}, \quad f_{m,t} = \mu_{f,m} + \kappa_{f,m} g_{m,t},$$

where $v_{m,t} \stackrel{\text{iid}}{\sim} U[-\sqrt{3}, \sqrt{3}]$, $g_{m,-50} = 0$, and $\rho_g = 0.5$. For DGPs 1–4, set

$$\lambda_{a,i} = 1 + 0.1 z_{a,i}, \quad \lambda_{b,i} = 1 + 0.4 z_{b,i},$$

where $z_{a,i}$ and $z_{b,i}$ are independent standardized uniform variables. Use $\mu_{f,a} = 0.5$, $\kappa_{f,a} = 0.1$, $\mu_{f,b} = 0.6$, and $\kappa_{f,b} = 0.2$. The raw autoregressive and slope matrices are

$$a_{it}^{\text{raw}} = \lambda_{a,i} f_{a,t}, \quad \beta_{it} = \lambda_{b,i} f_{b,t}.$$

Dynamic stability is imposed by

$$a_{it} = c_a a_{it}^{\text{raw}}, \quad c_a = \min \left\{ 1, \frac{0.85}{\max_{i,t} |a_{it}^{\text{raw}}|} \right\}.$$

Let $\lambda_{h,i}$ and the innovations in $g_{h,t}$ be independent standardized uniform variables, and generate

$$g_{h,t} = \rho_g g_{h,t-1} + (1 - \rho_g^2)^{1/2} v_{h,t}, \quad H_{it}^{\text{raw}} = \lambda_{h,i} g_{h,t}.$$

For each DGP and panel size, define

$$H_{0,it} = c_\xi c_H H_{it}^{\text{raw}}, \quad u_{it} = c_\xi \tilde{u}_{it}.$$

Thus, (8.1) exactly matches the maintained model: $\mathbf{H}_0 = (H_{0,it})$ is the true interactive-effect matrix and u_{it} is the unit-specific disturbance. In the baseline design, $\mathbf{A}_0$, $\mathbf{B}_0$, and $\mathbf{H}_0$ all have rank one.

DGP 1: independent heteroskedastic disturbances. Let $\tilde{u}_{it} = \sigma_i \varepsilon_{it}$, where $\varepsilon_{it} \stackrel{\text{iid}}{\sim} N(0, 1)$ and $\sigma_i^2 \stackrel{\text{iid}}{\sim} U(0.5, 1.5)$.

DGP 2: spatially dependent heteroskedastic disturbances. Units lie on a one-dimensional lattice with $d_N(i, j) = |i - j|$. Draw $\varepsilon_{it} \stackrel{\text{iid}}{\sim} N(0, 1)$ and $\sigma_i^2 \stackrel{\text{iid}}{\sim} U(0.5, 1.5)$. For every time t , generate

$$z_{1t} = \varepsilon_{1t}, \quad z_{it} = \rho_s z_{i-1,t} + (1 - \rho_s^2)^{1/2} \varepsilon_{it}, \quad \rho_s = 0.5,$$

and set $\tilde{u}_{it} = \sigma_i z_{it}$. Conditional on the unit variances,

$$\text{Cov}(\tilde{u}_{it}, \tilde{u}_{jt}) = \sigma_i \sigma_j \rho_s^{|i-j|}.$$

The recursive construction has linear cost in N and avoids a dense covariance square root.

For DGPs 1 and 2, generate

$$x_{it} = \rho_x x_{i,t-1} + \delta_x \lambda_{x,i} f_{x,t} + (1 - \rho_x^2)^{1/2} e_{it}, \quad (8.2)$$

where $\rho_x = 0.5$, $\delta_x = 0.5$, $\lambda_{x,i} \stackrel{\text{iid}}{\sim} U[-\sqrt{3}, \sqrt{3}]$, $\sigma_{e,i}^2 \stackrel{\text{iid}}{\sim} U(0.5, 1.5)$, and $e_{it} \sim N(0, \sigma_{e,i}^2)$. The common covariate factor satisfies

$$f_{x,t} = \rho_{fx} f_{x,t-1} + (1 - \rho_{fx}^2)^{1/2} v_{x,t}, \quad \rho_{fx} = 0.5,$$

where $v_{x,t} \stackrel{\text{iid}}{\sim} U[-\sqrt{3}, \sqrt{3}]$.

DGP 3: predetermined covariates. DGP 3 retains the spatial disturbance from DGP 2 and generates

$$x_{it} = \rho_x x_{i,t-1} + \delta_x \lambda_{x,i} f_{x,t} + \eta_x \tilde{u}_{i,t-1} + (1 - \rho_x^2)^{1/2} e_{it}, \quad (8.3)$$

where $\eta_x = 0.3$ and the other terms follow (8.2). Thus, x_{it} responds to the lagged disturbance but not to the current disturbance, matching the maintained predeterminedness restriction.

DGP 4: predetermined covariates with group heterogeneity. DGP 4 retains the disturbance and covariate processes from DGP 3 and changes only the coefficient loadings. Divide the units into two equal groups,

$$G_{1,N} = \{1, \dots, N/2\}, \quad G_{2,N} = \{N/2 + 1, \dots, N\}.$$

For $m \in \{a, b\}$, set

$$\lambda_{m,i} = \mu_{\lambda,m,1} \mathbf{1}\{i \in G_{1,N}\} + \mu_{\lambda,m,2} \mathbf{1}\{i \in G_{2,N}\} + \sigma_{\lambda,m} z_{m,i},$$

where $z_{m,i} \stackrel{\text{iid}}{\sim} U[-\sqrt{3}, \sqrt{3}]$. The baseline values are

$$\mu_{\lambda,a,1} = 0.9, \quad \mu_{\lambda,a,2} = 1.1, \quad \sigma_{\lambda,a} = 0.08,$$

and

$$\mu_{\lambda,b,1} = 0.8, \quad \mu_{\lambda,b,2} = 1.2, \quad \sigma_{\lambda,b} = 0.25.$$

Construct $a_{it}^{\text{raw}} = \lambda_{a,i} f_{a,t}$, apply the same common stability scaling c_a , and set $\beta_{it} = \lambda_{b,i} f_{b,t}$. Because the loading vector remains multiplied by one time factor, all three coefficient matrices remain rank one. DGP 4 therefore adds systematic group heterogeneity without changing the rank vector or the spatial and predetermined structure inherited from DGP 3.

The common-shock sigma-field for all four DGPs is generated by the simulated paths $g_{a,t}$, $g_{b,t}$, $g_{h,t}$, and $f_{x,t}$. Conditional on these paths, the current disturbances have mean zero, and DGPs 2–4 satisfy the spatial weak-dependence design.

8.2 Calibration, targets, and reported quantities

A separate deterministic calibration experiment is used for every DGP and panel size. First, choose c_H so that the unscaled interactive effect accounts for $\pi_H = 0.3$ of the variance of $c_H H_{it}^{\text{raw}} + \tilde{u}_{it}$:

$$c_H^2 = \frac{\pi_H}{1 - \pi_H} \frac{\text{Var}(\tilde{u}_{it})}{\text{Var}(H_{it}^{\text{raw}})}.$$

Second, select c_ξ by deterministic root finding with common random numbers so that the average pooled coefficient of determination is 0.5, where the fitted value includes the lagged-outcome term, the observed-covariate term, and $H_{0,it}$. Calibration draws are not reused in the reported replications, and all selected constants and achieved moments are stored.

DGP 4 additionally reports the realized true group means and $G_{2,N}$ -minus- $G_{1,N}$ differences for both a_{it} and β_{it} , at the target time and after averaging over time. These truths are computed after applying the common autoregressive stability scaling c_a . Before production simulations, a deterministic pilot calibration checks that the post-scaling autoregressive group difference is not numerically negligible. If the baseline loading means fail a prespecified contrast floor, their separation is widened symmetrically around their common center, the pilot is rerun, and the final values are frozen and reported before any production replication is examined. No group-separation parameter is changed within or after the reported Monte Carlo replications.

The balanced-panel sequence uses $N = T \in \{50, 100, 200, 400\}$. Rectangular panels are included in the appendix to assess unequal growth of N and T . The production design uses 1,000 replications per cell when computationally feasible; any reduced count for the largest cells is fixed before results are examined and reported together with Monte Carlo standard errors.

DGP 1 uses the diagonal variance estimator. DGPs 2–4 use the positive-semidefinite Bartlett weights

$$w_{ik,N} = \left(1 - \frac{|i - k|}{h_N + 1}\right)_+, \quad h_N = \lceil c_{\text{sp}} \log(NT) \rceil,$$

with baseline $c_{\text{sp}} = 1$ and prespecified sensitivity values in the appendix.

Let $i_0 = \lfloor N/4 \rfloor$ and $t_0 = \lfloor T/2 \rfloor$. The main parameters of interest are:

- (i) the individual entries $a_{i_0 t_0}$ and $\beta_{i_0 t_0}$;
- (ii) the cross-sectional means of a_{it_0} and β_{it_0} ;
- (iii) the group-specific fixed-time means and the $G_{2,N}$ -minus- $G_{1,N}$ fixed-time contrasts;
- (iv) the full-panel means of $\mathbf{A}_0$ and $\mathbf{B}_0$;
- (v) the group-specific time-averaged means and the $G_{2,N}$ -minus- $G_{1,N}$ time-averaged contrasts.

The first three parameter classes use the full-panel plug-in estimator. The last two use the two-way split-panel correction. For DGPs 1–4, the deterministic groups provide zero-centered or weak-noise

contrast benchmarks. DGP 4 provides systematically nonzero group means and contrasts and therefore evaluates both confidence-interval coverage and power against zero.

For every parameter and panel cell, report the exact true value, mean estimate, bias, RMSE, Monte Carlo standard deviation, mean estimated standard error, the ratio of mean estimated standard error to Monte Carlo standard deviation, empirical size of the two-sided 5% test centered at the truth, power against zero when the truth is nonzero, 95% coverage, Monte Carlo standard errors for size and coverage, and the number and reasons for failed replications.

The main inference designs have true rank vector $(1, 1, 1)$. A separate rank-selection experiment uses the four disturbance and covariate designs together with rank vectors $(1, 1, 1)$, $(2, 1, 1)$, and $(1, 0, 2)$. Higher-rank matrices are generated by adding independent bounded loading-factor terms and are rescaled to preserve dynamic stability and coefficient bounds.

Table 1: Monte Carlo results for DGP 1

Placeholder. Panels will report individual entries, fixed-time means, full-panel means before and after the two-way correction, standard-error accuracy, size, and coverage.

Table 2: Monte Carlo results for DGP 2

Placeholder. The table will use the same parameter layout as [Table 1](#) and compare diagonal and spatial HAC standard errors.

Table 3: Monte Carlo results for DGP 3

Placeholder. The table will emphasize predetermined covariates, the two leading bias terms, and the performance of the two-way correction.

Table 4: Monte Carlo results for DGP 4

Placeholder. Panels will report the realized true group means and contrasts after stability scaling, their estimates, standard-error accuracy, coverage, and power against zero.

Table 5: Full-panel rank-selection accuracy

Placeholder. Rows will report candidate-set coverage, exact selected-rank recovery, matrix-specific underselection and overselection, results for separate and zero ranks, screened-versus-exhaustive agreement when feasible, distinct path ranks, candidate-set size before and after local completion, selected-to-cap margins, information-criterion gaps to the next-best and one-coordinate neighboring vectors, and agreement under the denser grid and threshold and penalty sensitivities for DGPs 1–4.

Table 6: Computation and numerical diagnostics

Placeholder. Rows will report runtime, fixed-rank fit counts, Riesz iterations, objective and first-order tolerances, coefficient-bound hits, split-panel failures, memory use, and the computational gain relative to the buffered-fold benchmark.

Placeholder for separate plots of individual entries, fixed-time means, group contrasts, and full-panel means before and after the two-way correction under DGPs 1–4.

Figure 1: Coverage and bias as the panel grows

9 Empirical Application

The empirical section contains two applications—regional unemployment and house-price momentum—and will recompute all estimates with the full-panel estimator.

9.1 Regional unemployment

The planned application uses monthly metropolitan unemployment rates, deseasonalized metro by metro. The working data build contains 315 metropolitan areas over 240 months in the main specification. The interactive-effects matrix absorbs the common cycle, while the autoregressive coefficient matrices describe remaining local persistence. Predetermined controls include lagged metropolitan population growth and real-GDP growth.

The main targets will be the full-panel persistence mean, high- and low-unemployment group means, their contrast, selected individual entries, and time paths of cross-sectional mean persistence. A time-invariant high/low split will be defined before estimation. Full-period means and contrasts will use the two-way correction; fixed-time targets and individual entries will use the full-panel plug-in when the target-rate condition holds.

Table 7: Idiosyncratic persistence in metropolitan unemployment

Placeholder. Placeholder for selected ranks, full and group means, the high-minus-low contrast, bias-corrected estimates, spatial standard errors, confidence intervals, and target-specific diagnostics.

9.2 House-price momentum

The planned housing application stacks the top and bottom house-price tiers for each metropolitan area as separate units. The working data build contains 610 metro-tier units over 315 months. The outcome is monthly log price growth. The main targets are the mean first- and second-lag coefficients, lag-sum persistence, top- and bottom-tier means, their contrast, selected individual entries, and time paths of mean momentum. Predetermined controls include lagged building-permit growth, population growth, and real-GDP growth.

Table 8: House-price dynamics and price-tier contrasts

Placeholder. Placeholder for selected ranks, lag means, lag-sum persistence, top and bottom tier means, the top-minus-bottom contrast, bias-corrected estimates, spatial standard errors, and confidence intervals.

9.3 Spatial inference and computation

When metropolitan coordinates are available, the primary dependence-robust calculation will use great-circle distance and several prespecified spatial distance cutoffs. The diagonal calculation will be reported as a benchmark. A known-cluster calculation may be reported when a credible cluster partition is available. Period clustering is not treated as a substitute for the spatial theorem unless additional conditions are verified.

Table 9: Geographic spatial HAC sensitivity

Placeholder. Placeholder for diagonal and spatial HAC standard errors at several geographic distance cutoffs for the main unemployment and housing targets.

Table 10: Rank, split-panel, and optimization diagnostics

Placeholder. Placeholder for rank-path singular values, selected-rank margins, full- and split-panel Gram-matrix eigenvalues, objective tolerances, Riesz iterations, coefficient-envelope checks, and spatial-weight diagnostics.

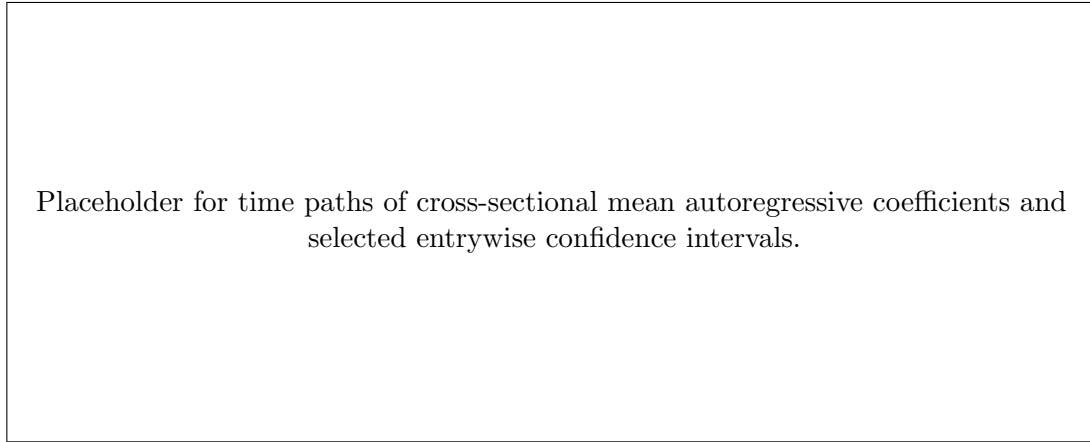


Figure 2: Estimated coefficient paths and selected entrywise estimates

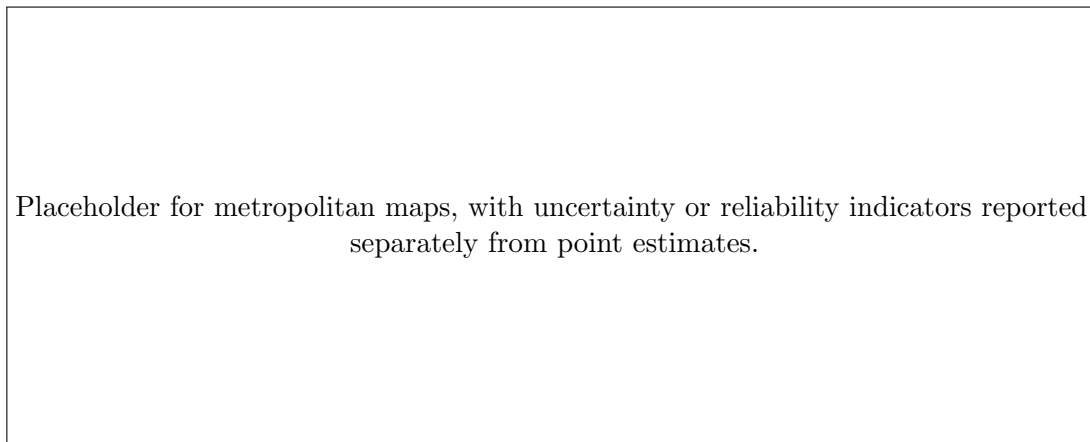


Figure 3: Geographic distribution of selected coefficient summaries

10 Conclusion

This paper develops full-panel estimation and inference for dynamic panels with several separately ranked coefficient matrices. A convex nuclear-norm pilot screens plausible low-rank spaces and supplies numerical starting values, while all reported coefficients are obtained from unpenalized joint fixed-rank least squares. The analysis aligns and expands the estimated singular spaces and uses a target-specific Riesz representer to translate estimation errors in the autoregressive, slope, and interactive-effect matrices into first-order uncertainty for a selected parameter. The framework allows predetermined covariates, conditioning on common shocks, dynamic feedback through lagged outcomes, and spatially dependent unit-specific innovations.

The inferential procedure depends on the parameter being reported. Individual coefficients, fixed unit-time lag sums, and suitable fixed-time means and contrasts use one full-panel fit because the leading T^{-1} loading bias and N^{-1} factor bias are smaller than their standard errors. Averages whose weights extend over many units and many time periods receive a fixed two-way split-panel correction that removes both leading bias terms. Rank selection is performed once on the full panel by combining pilot-based screening, unpenalized post-refits, and an information criterion. The nuclear path improves screening and computation, but the rank-consistency argument depends on pilot accuracy, singular-value separation, and post-refit information-criterion separation rather than on a particular numerical grid.

The scope of the theory is deliberately narrower than that of some single-matrix low-rank models. The inferential results assume exact fixed ranks and consistent selection of the separate rank vector. The screening argument can still identify strong singular directions when a smaller remainder lies below the threshold, but the confidence intervals are not presented as approximate-low-rank or rank-misspecification-robust procedures. Allowing growing ranks would change the dimensions of the loading and factor regressions, the uniform concentration rates, and the split-panel bias calculation. Rank-robust inference would require correcting the effects of overestimating several interacting latent spaces, including their propagation through the dynamic equation. Finally, the present Riesz equation establishes valid inverse-Gram adjustment under conditional heteroskedasticity and spatial dependence but does not characterize the semiparametric efficiency bound; an efficient extension would require covariance-optimal weighting of the joint Gram equation. These questions are natural directions for future work rather than consequences of the current results.

The Monte Carlo and empirical sections are designed to evaluate the finite-sample bias correction, spatial inference, rank-selection stability, and computational gains of the proposed full-panel procedure. Their numerical tables and figures will be populated after implementation is complete.

References

Anderson, T. W., and Hsiao, C. (1981). Estimation of dynamic models with error components. *Journal of the American Statistical Association*, 76(375), 598–606. [doi:](#)

[10.1080/01621459.1981.10477691](https://doi.org/10.1080/01621459.1981.10477691).

- Anderson, T. W., and Hsiao, C. (1982). Formulation and estimation of dynamic models using panel data. *Journal of Econometrics*, 18(1), 47–82. doi: [10.1016/0304-4076\(82\)90095-1](https://doi.org/10.1016/0304-4076(82)90095-1).
- Arellano, M., and Bond, S. (1991). Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations. *Review of Economic Studies*, 58(2), 277–297. doi: [10.2307/2297968](https://doi.org/10.2307/2297968).
- Athey, S., Bayati, M., Doudchenko, N., Imbens, G. W., and Khosravi, K. (2021). Matrix completion methods for causal panel data models. *Journal of the American Statistical Association*, 116(536), 1716–1730. doi: [10.1080/01621459.2021.1891924](https://doi.org/10.1080/01621459.2021.1891924).
- Bai, J. (2003). Inferential theory for factor models of large dimensions. *Econometrica*, 71(1), 135–171. doi: [10.1111/1468-0262.00392](https://doi.org/10.1111/1468-0262.00392).
- Bai, J., and Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1), 191–221. doi: [10.1111/1468-0262.00273](https://doi.org/10.1111/1468-0262.00273).
- Bai, J., and Ng, S. (2021). Matrix completion, counterfactuals, and factor analysis of missing data. *Journal of the American Statistical Association*, 116(536), 1746–1763. doi: [10.1080/01621459.2021.1967163](https://doi.org/10.1080/01621459.2021.1967163).
- Bai, J., Choi, S. H., and Liao, Y. (2021). Feasible generalized least squares for panel data with cross-sectional and serial correlations. *Empirical Economics*, 60(1), 309–326. doi: [10.1007/s00181-020-01977-2](https://doi.org/10.1007/s00181-020-01977-2).
- Bonhomme, S., and Manresa, E. (2015). Grouped patterns of heterogeneity in panel data. *Econometrica*, 83(3), 1147–1184. doi: [10.3982/ECTA11319](https://doi.org/10.3982/ECTA11319).
- Bradley, R. C. (1983). Approximation theorems for strongly mixing random variables. *Michigan Mathematical Journal*, 30(1), 69–81. doi: [10.1307/mmj/1029002789](https://doi.org/10.1307/mmj/1029002789).
- Chernozhukov, V., Hansen, C., Liao, Y., and Zhu, Y. (2023). Inference for low-rank models. *Annals of Statistics*, 51(3), 1309–1330. doi: [10.1214/23-AOS2293](https://doi.org/10.1214/23-AOS2293).
- Choi, J., Kwon, H., and Liao, Y. (2024). Inference for low-rank completion without sample splitting with application to treatment effect estimation. *Journal of Econometrics*, 240(1), Article 105682. doi: [10.1016/j.jeconom.2024.105682](https://doi.org/10.1016/j.jeconom.2024.105682).
- Choi, J., Kwon, H., and Liao, Y. (2026). Inference for low-rank models without estimating the rank. *Journal of the American Statistical Association*, 121(553), 655–666. doi: [10.1080/01621459.2025.2538272](https://doi.org/10.1080/01621459.2025.2538272).
- Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of Econometrics*, 92(1), 1–45. doi: [10.1016/S0304-4076\(98\)00084-0](https://doi.org/10.1016/S0304-4076(98)00084-0).
- Gao, B., and Absil, P.-A. (2022). A Riemannian rank-adaptive method for low-rank matrix completion. *Computational Optimization and Applications*, 81, 67–90. doi: [10.1007/s10589-021-00328-w](https://doi.org/10.1007/s10589-021-00328-w).
- Jenish, N., and Prucha, I. R. (2012). On spatial processes and asymptotic inference under near-epoch dependence. *Journal of Econometrics*, 170(1), 178–190. doi: [10.1016/j.jeconom.2012.05.022](https://doi.org/10.1016/j.jeconom.2012.05.022).

- Moon, H. R., and Weidner, M. (2015). Linear regression for panel with unknown number of factors as interactive fixed effects. *Econometrica*, 83(4), 1543–1579. doi: [10.3982/ECTA9382](https://doi.org/10.3982/ECTA9382).
- Moon, H. R., and Weidner, M. (2026). Nuclear norm regularized estimation of panel regression models. *Journal of Econometrics*, 255, Article 106246. doi: [10.1016/j.jeconom.2026.106246](https://doi.org/10.1016/j.jeconom.2026.106246).
- Moon, H. R., and Weidner, M. (2017). Dynamic linear panel regression models with interactive fixed effects. *Econometric Theory*, 33(1), 158–195. doi: [10.1017/S0266466615000328](https://doi.org/10.1017/S0266466615000328).
- Kuersteiner, G. M., and Prucha, I. R. (2020). Dynamic spatial panel models: Networks, common shocks, and sequential exogeneity. *Econometrica*, 88(5), 2109–2146. doi: [10.3982/ECTA13660](https://doi.org/10.3982/ECTA13660).
- Peng, B., Su, L., Westerlund, J., and Yang, Y. (2025). Interactive effects panel data models with general factors and regressors. *Econometric Theory*, 41(2), 472–488. doi: [10.1017/S0266466623000270](https://doi.org/10.1017/S0266466623000270).
- Pesaran, M. H. (2006). Estimation and inference in large heterogeneous panels with a multifactor error structure. *Econometrica*, 74(4), 967–1012. doi: [10.1111/j.1468-0262.2006.00692.x](https://doi.org/10.1111/j.1468-0262.2006.00692.x).
- Pesaran, M. H., and Smith, R. (1995). Estimating long-run relationships from dynamic heterogeneous panels. *Journal of Econometrics*, 68(1), 79–113. doi: [10.1016/0304-4076\(94\)01644-F](https://doi.org/10.1016/0304-4076(94)01644-F).
- Swamy, P. A. V. B. (1970). Efficient inference in a random coefficient regression model. *Econometrica*, 38(2), 311–323. doi: [10.2307/1913012](https://doi.org/10.2307/1913012).
- Tropp, J. A. (2012). User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics*, 12(4), 389–434. doi: [10.1007/s10208-011-9099-z](https://doi.org/10.1007/s10208-011-9099-z).
- Wang, Y., Su, L., and Zhang, Y. (2022). Low-rank panel quantile regression: Estimation and inference. *arXiv preprint* doi: [10.48550/arXiv.2210.11062](https://doi.org/10.48550/arXiv.2210.11062).
- Wedin, P.-Å. (1972). Perturbation bounds in connection with singular value decomposition. *BIT Numerical Mathematics*, 12, 99–111. doi: [10.1007/BF01932678](https://doi.org/10.1007/BF01932678).
- Yuan, Z., and Spindler, M. (2025). Bernstein-type inequalities and nonparametric estimation under near-epoch dependence. *Journal of Econometrics*, 251, Article 106054. doi: [10.1016/j.jeconom.2025.106054](https://doi.org/10.1016/j.jeconom.2025.106054).

A Proofs and Supporting Results

All proofs are carried out under the conditional law given $\mathcal{C}$. Conditional O_p and o_p statements are uniform over the regular common-shock realizations covered by the deterministic constants in the assumptions. Integrating a conditional conclusion over $\mathcal{C}$ gives its unconditional counterpart.

Lemma 1 (Conditional use of random-field limit results). *Fix a regular realization $\mathcal{C} = c$ and let $\mathbb{P}_0^c$ denote a regular conditional probability given $\mathcal{C} = c$.*

- (i) *If a cited theorem is an asymptotic weak-limit result and its assumptions hold under $\mathbb{P}_0^c$, then it may be applied under that conditional law for almost every regular c . If the resulting conditional distribution functions converge pointwise to a continuous limit, Pólya’s theorem gives uniform convergence in the argument for that fixed c , and bounded convergence over c yields convergence in probability of the conditional distribution distance and the corresponding unconditional weak limit.*
- (ii) *If a cited theorem is a finite-sample probability inequality whose constants depend only on deterministic moment, sampling-geometry, NED, and mixing envelopes, then the same inequality holds under $\mathbb{P}_0^c$ with constants uniform over regular c . Integrating the conditional probability bound gives the corresponding unconditional bound.*

Proof. For almost every c , the regular conditional law $\mathbb{P}_0^c$ is an ordinary probability measure. Once c is fixed, the arrays of unit-specific innovations, localized covariates, and scores satisfy the same deterministic geometry, moment, NED, and mixing inequalities as in the maintained assumptions. The random-field central limit theorem of [Jenish and Prucha \(2012\)](#) is therefore applied separately under each such $\mathbb{P}_0^c$; no uniform Berry–Esseen bound is invoked. Because the Gaussian limit is continuous, conditional weak convergence for almost every c implies uniform convergence of the corresponding conditional distribution functions for that fixed c , and the distance is bounded by one. Bounded convergence over the distribution of $\mathcal{C}$ then gives convergence in probability of the conditional distance and the unconditional conclusion.

For finite-sample concentration inequalities the transfer is more literal: if an external inequality is applied after conditioning on $\mathcal{C} = c$, its own sampling, weighting, moment, NED, and mixing hypotheses must first be verified under $\mathbb{P}_0^c$, and any constants that are to be uniform in c must depend only on the deterministic envelopes in the maintained assumptions. The Bernstein inequality of [Yuan and Spindler \(2025\)](#) is a useful irregular-spacing NED benchmark, but its published Theorem 1 concerns a scalar average under its own domain-expanding-infill geometry. The concentration argument needed here is instead proved directly in [Lemma 4](#) from the stronger setwise conditional mixing coefficient in [Assumption 4](#), exponential local approximation, and the fixed-dimensional unit–time geometry. Thus, no unverified weighted extension of the Yuan–Spindler theorem is used below. $\square$

For compactness in the appendix only, write

$$Z_{it} = \Psi_{0,it} u_{it}, \quad \widehat{Z}_{it} = \widehat{\Psi}_{it} \widehat{u}_{it}.$$

For the split-panel correction, write

$$Z_{it}^{\text{corr}} = u_{it} [3\Psi_{0,it} - \Psi_{0,it}^T - \Psi_{0,it}^N],$$

Equivalently, Z_{it}^{corr} is the bracketed Riesz weight multiplied by u_{it} . The feasible analogue combines the full- and split-panel residual scores as in (4.9). These abbreviations are used only to keep the proofs readable.

A.1 Conditional moments and local approximation

This subsection establishes the moment and localization results used in every later proof. The dynamic equation makes a current fitted value depend on lagged outcomes, and the covariates may depend on a long history of nearby unit-specific innovations. We first show that stability keeps the required moments bounded. We then replace each fitted value or score by a version that depends only on innovations from a finite time window and a finite spatial neighborhood. Conditional alpha mixing can be applied after this replacement because the innovation sets associated with sufficiently separated observations are then disjoint.

For $p \geq 1$ and a random variable X , define its conditional L_p norm by

$$\|X\|_{p,\mathcal{C}} = \{\mathbb{E}_0(|X|^p \mid \mathcal{C})\}^{1/p}.$$

Thus, $\|X\|_{p,\mathcal{C}}$ measures the size of X under the conditional probability law obtained after the common shocks are held fixed. All bounds below are uniform over the regular common-shock realizations covered by the assumptions.

Recall from [Assumption 4](#) that $\mathcal{E}_{it}(h, r)$ contains unit-specific innovations from the previous h time periods and from units within distance r of unit i , with current covariate innovations included and current outcome innovations excluded. The approximation error has three sources: propagation through the dynamic equation, dependence of the covariates on remote past innovations, and dependence of the covariates on spatially distant innovations. For $h \geq 3P$ and $r \geq 0$, collect these three errors in

$$\chi(h, r) = \underbrace{\rho^{\lfloor h/3 \rfloor}}_{\text{dynamic propagation}} + \underbrace{\psi_T(\lfloor h/3 \rfloor)}_{\text{remote time innovations}} + \underbrace{\psi_S(\lfloor r/3 \rfloor)}_{\text{remote spatial innovations}}.$$

Dividing both localization radii by three is only a proof device: it leaves separate margins for the outcome lags, the dynamic recursion, the covariate approximation, and the mixing separation. Any fixed division with sufficiently large margins would give the same asymptotic conclusion. By stability and the NED assumption, $\chi(h, r) \rightarrow 0$ as $h, r \rightarrow \infty$.

The field $\mathcal{E}_{it}(h, r)$ is appropriate for approximating predetermined covariates because it excludes the current outcome innovation. A score such as $u_{it}\mathcal{Y}_{it}(\Delta)$ contains the current innovation explicitly. For score approximations only, define the enlarged local information set

$$\mathcal{E}_{it}^+(h, r) = \mathcal{E}_{it}(h, r) \vee \sigma\{u_{jt} : d_N(i, j) \leq r\}.$$

The added sigma-field contains current outcome innovations for unit i and its spatial neighbors. This enlargement does not alter the timing assumption for the covariates; it is used only because the score itself contains u_{it} .

Lemma 2 (Moments and local approximations). *Under [Assumptions 1, 4](#) and [5](#), outcomes and their fixed number of lags have uniformly bounded conditional $8 + \eta$ moments. For every $\ell = 1, \dots, P$, $y_{i,t-\ell}$ has an approximation measurable with respect to $\mathcal{E}_{it}(h, r) \vee \mathcal{C}$ whose conditional $8 + \eta$ error is bounded by $C\chi(h, r)$. Consequently, for every fixed Δ ,*

$$\left\| \mathcal{Y}_{it}(\Delta) - \mathcal{Y}_{it}^{[h,r]}(\Delta) \right\|_{8+\eta, \mathcal{C}} \leq C\|\Delta\|\chi(h, r),$$

where $\mathcal{Y}_{it}^{[h,r]}(\Delta)$ is measurable with respect to $\mathcal{E}_{it}(h, r) \vee \mathcal{C}$. The product

$$u_{it}\mathcal{Y}_{it}(\Delta)$$

has an approximation measurable with respect to $\mathcal{E}_{it}^+(h, r) \vee \mathcal{C}$ with conditional $4 + \eta/2$ error bounded by $C\|\Delta\|\chi(h, r)$. Products used in the Gram-matrix and variance calculations satisfy the corresponding $2 + \eta/4$ bounds.

Proof. Let

$$\mathbf{y}_{it} = (y_{it}, y_{i,t-1}, \dots, y_{i,t-P+1})'$$

and let $\mathbf{e}_1$ be the first coordinate vector in $\mathbb{R}^P$. The dynamic model can be written as

$$\mathbf{y}_{it} = \mathbf{C}_{0,it}\mathbf{y}_{i,t-1} + \mathbf{e}_1 \left\{ \sum_{k=1}^K x_{it,k}\beta_{0,it,k} + h_{0,it} + u_{it} \right\}.$$

For $j \geq 1$, define

$$\Phi_{i,t}(j) = \mathbf{C}_{0,it}\mathbf{C}_{0,i,t-1} \cdots \mathbf{C}_{0,i,t-j+1}, \quad \Phi_{i,t}(0) = \mathbf{I}_P.$$

Iterating backward m periods gives

$$\begin{aligned} \mathbf{y}_{it} &= \Phi_{i,t}(m)\mathbf{y}_{i,t-m} \\ &\quad + \sum_{j=0}^{m-1} \Phi_{i,t}(j)\mathbf{e}_1 \left\{ \sum_{k=1}^K x_{i,t-j,k}\beta_{0,i,t-j,k} + h_{0,i,t-j} + u_{i,t-j} \right\}. \end{aligned} \tag{A.1}$$

By [\(5.1\)](#), $\|\Phi_{i,t}(j)\|_{\text{op}} \leq C\rho^j$. The coefficient entries are bounded, P and K are fixed, and [Assump-](#)

tion 5 bounds the moments of $x_{it,k}$ and u_{it} . Conditional Minkowski applied to (A.1), followed by $m \rightarrow \infty$, yields

$$\sup_{i,t} \|\mathbf{y}_{it}\|_{8+\eta, \mathcal{C}} \leq C.$$

Take $m = \lfloor h/3 \rfloor$. Replace every retained covariate in (A.1) by its conditional expectation given the corresponding local innovation sigma-field. The omitted initial term is bounded by $C\rho^m$. The accumulated temporal and spatial covariate approximation errors are bounded by

$$C \sum_{j=0}^{m-1} \rho^j \{\psi_T(m) + \psi_S(r)\} \leq C\{\psi_T(m) + \psi_S(r)\}.$$

For $\ell \geq 1$, every disturbance entering $y_{i,t-\ell}$ is dated no later than $t-1$, so the approximation is measurable with respect to $\mathcal{E}_{it}(h, r) \vee \mathcal{C}$. This proves the lagged-outcome statement.

The fitted-value map is a finite sum of bounded coefficient entries multiplied by the approximated lagged outcomes and covariates. Conditional Minkowski therefore gives the stated bound for $\mathcal{Y}_{it}(\Delta)$. For products, write

$$AB - A^{[h,r]}B^{[h,r]} = (A - A^{[h,r]})B + A^{[h,r]}(B - B^{[h,r]}).$$

Conditional Hölder with the $8 + \eta$ moment bound yields a $4 + \eta/2$ product bound. Applying the same argument once more gives the $2 + \eta/4$ bounds for quadratic products. The score approximation includes u_{it} itself and is therefore measurable with respect to $\mathcal{E}_{it}^+(h, r) \vee \mathcal{C}$. $\square$

For two locally measurable variables whose innovation neighborhoods are separated by distance a , the conditional mixing covariance inequality and Assumption 4 imply

$$|\text{Cov}_0(X, Y \mid \mathcal{C})| \leq C \|X\|_{p, \mathcal{C}} \|Y\|_{p, \mathcal{C}} \bar{\alpha}(a)^{\eta/(8+\eta)} \quad (\text{A.2})$$

for the values of p used below. Approximation errors add terms proportional to $\chi(h, r)$. Exponential decay and the polynomial neighborhood bound in Assumption 3 make the resulting covariance sums finite.

A.2 Target directions and the population Riesz-score limit

This subsection has three steps. First, it relates the size of the population Riesz representer to the direction $\mathbf{D}$ of the parameter of interest. Second, it derives the variance rates and verifies that no single observation dominates the Riesz score. Third, it applies the conditional NED central limit theorem to prove Theorem 1.

Let

$$\Gamma_0(\Delta, \mathbf{Q}) = \mathbb{E}_0 \left[\sum_{i,t} \mathcal{Y}_{it}(\Delta) \mathcal{Y}_{it}(\mathbf{Q}) \middle| \mathcal{C} \right].$$

Because Γ_0 is coercive on $\mathcal{T}_0$, the Riesz equation has a unique solution. Taking $\Delta = \mathbf{q}_0$ in (4.2) gives

$$\Gamma_0(\mathbf{q}_0, \mathbf{q}_0) = \langle \mathcal{P}_0 \mathbf{D}, \mathbf{q}_0 \rangle. \quad (\text{A.3})$$

The upper and lower Gram-matrix bounds therefore imply

$$c \|\mathcal{P}_0 \mathbf{D}\| \leq \|\mathbf{q}_0\| \leq C \|\mathcal{P}_0 \mathbf{D}\|. \quad (\text{A.4})$$

For use in both the score and recovery proofs, introduce one common set of local loading and factor coordinates. If $\mathbf{M}_0 = \mathbf{U}_M \Sigma_M \mathbf{V}_M'$ is a positive-rank coefficient matrix, take

$$\Lambda_{M,0} = \sqrt{N} \mathbf{U}_M, \quad \mathbf{F}_{M,0} = N^{-1/2} \mathbf{V}_M \Sigma_M.$$

Then $\mathbf{M}_0 = \Lambda_{M,0} \mathbf{F}_{M,0}'$, $N^{-1} \Lambda_{M,0}' \Lambda_{M,0} = \mathbf{I}_{r_M}$, and the eigenvalues of $T^{-1} \mathbf{F}_{M,0}' \mathbf{F}_{M,0} = \Sigma_M^2 / (NT)$ are bounded above and away from zero by [Assumption 6](#). This gives the explicit map between the singular-value and loading-factor parameterizations used in the paper.

Let $z_{it,M}$ be the regressor multiplying matrix M :

$$z_{it,A^{(\ell)}} = y_{i,t-\ell}, \quad z_{it,B^{(k)}} = x_{it,k}, \quad z_{it,H} = 1.$$

Stack the loading rows over the positive-rank matrices in $\lambda_{0,i}$, stack their factor rows in $\mathbf{f}_{0,t}$, and set

$$\mathbb{Z}_{it} = \text{diag}\{z_{it,M} \mathbf{I}_{r_M} : r_M > 0\}.$$

After imposing the usual normalization restrictions that remove loading-factor rotations, a local direction is written as $\mathbf{h} = (\{\mathbf{h}_i^\lambda\}_i, \{\mathbf{h}_t^f\}_t)$ and its fitted-value derivative is

$$\dot{\mathbf{y}}_{it}(\mathbf{h}) = \frac{(\mathbb{Z}_{it} \mathbf{f}_{0,t})' \mathbf{h}_i^\lambda}{\sqrt{T}} + \frac{(\mathbb{Z}_{it}' \lambda_{0,i})' \mathbf{h}_t^f}{\sqrt{N}}. \quad (\text{A.5})$$

Define the conditional local Gram operator by

$$\mathbb{G}_0(\mathbf{h}, \mathbf{q}) = \mathbb{E}_0 \left[\sum_{i,t} \dot{\mathbf{y}}_{it}(\mathbf{h}) \dot{\mathbf{y}}_{it}(\mathbf{q}) \middle| \mathcal{C} \right].$$

Lemma 3 (Rows of the inverse local Gram operator). *Under [Assumptions 5 to 8](#), $\mathbb{G}_0$ is invertible on the normalized local coordinate space. Uniformly for a fixed target unit i and target time t ,*

$$\begin{aligned} \|\mathbb{G}_0^{-1}\|_{\lambda_i, \lambda_i} \text{op} &\leq C, & \sum_{j \neq i} \|\mathbb{G}_0^{-1}\|_{\lambda_i, \lambda_j}^2 &\leq C/N, & \sum_s \|\mathbb{G}_0^{-1}\|_{\lambda_i, f_s}^2 &\leq C/N, \\ \|\mathbb{G}_0^{-1}\|_{f_t, f_t} \text{op} &\leq C, & \sum_{s \neq t} \|\mathbb{G}_0^{-1}\|_{f_t, f_s}^2 &\leq C/T, & \sum_j \|\mathbb{G}_0^{-1}\|_{f_t, \lambda_j}^2 &\leq C/T. \end{aligned} \quad (\text{A.6})$$

Proof. Partition $\mathbb{G}_0$ into loading and factor coordinates,

$$\mathbb{G}_0 = \begin{pmatrix} G_{\lambda\lambda} & G_{\lambda f} \\ G_{f\lambda} & G_{ff} \end{pmatrix}.$$

The diagonal block associated with loading row i is exactly the conditional time-series Gram matrix in [Assumption 8](#); the diagonal block associated with factor row t is the corresponding conditional cross-sectional Gram matrix. Hence these diagonal blocks have eigenvalues between c and C . The local part of [Assumption 7](#) makes the full operator coercive, and therefore bounds both Schur complements away from zero.

The (i, t) loading–factor cross block contains the factor $1/\sqrt{NT}$ from [\(A.5\)](#). Conditional Cauchy–Schwarz, the moment bounds, and incoherence give its Frobenius norm at most $C/\sqrt{NT}$. Consequently, the squared norm of all cross blocks in a fixed loading row is at most $T C/(NT) = C/N$, and the analogous sum for a fixed factor row is at most C/T .

Write $A = G_{\lambda\lambda}$, $D = G_{ff}$, and $B = G_{\lambda f}$. The matrices A and D are block diagonal across loading rows and factor rows, respectively, with uniformly bounded inverses. The block inverse identities

$$\begin{pmatrix} A & B \\ B' & D \end{pmatrix}_{11}^{-1} = (A - BD^{-1}B')^{-1} = A^{-1} + A^{-1}B(D - B'A^{-1}B)^{-1}B'A^{-1},$$

and

$$\begin{pmatrix} A & B \\ B' & D \end{pmatrix}_{12}^{-1} = -(A - BD^{-1}B')^{-1}BD^{-1},$$

together with their lower-block analogues, give the stated rates. In particular, an off-diagonal loading–loading block of the inverse factors through one loading–factor row of B , the bounded inverse factor Schur complement, and one factor–loading column of B' . The squared Frobenius norm of a fixed loading row of this correction is therefore bounded by C/N after summing over $j \neq i$. The factor–factor correction is symmetric and has squared row norm bounded by C/T . The same factorization gives the loading–factor and factor–loading row bounds. This proves [\(A.6\)](#). $\square$

Proposition 1 (Riesz-score regularity and variance rates). *Under [Assumptions 1 to 9](#), let $\delta = \eta/4$. For every parameter of interest defined in [Section 3](#),*

$$\frac{\sum_{i,t} \mathbb{E}_0(|Z_{it}|^{2+\delta} \mid \mathcal{C})}{s_{NT}^{2+\delta}} \longrightarrow 0, \quad \frac{\max_{i,t} \mathbb{E}_0(Z_{it}^2 \mid \mathcal{C})}{s_{NT}^2} \longrightarrow 0. \quad (\text{A.7})$$

Moreover, Z_{it} is conditionally $L_{2+\delta}$ -NED on the innovation field, with an approximation coefficient bounded by $C\chi(h, r)$ after normalization by its parameter-specific scale. The conditional variance satisfies:

(a) for a regular individual coefficient or a fixed unit-time lag sum,

$$s_{NT}^2 \asymp N^{-1} + T^{-1};$$

(b) for a mean over $G_N \times R_T$,

$$s_{NT}^2 \asymp (|G_N||R_T|)^{-1};$$

(c) for a contrast between disjoint groups $G_{1,N}$ and $G_{2,N}$ averaged over R_T ,

$$s_{NT}^2 \asymp \frac{1}{|R_T|} \left(\frac{1}{|G_{1,N}|} + \frac{1}{|G_{2,N}|} \right).$$

Proof. Let $\boldsymbol{\psi}_t = (\Psi_{0,1t}, \dots, \Psi_{0,Nt})'$. Conditional sequential exogeneity makes the time-level Riesz sums martingale differences, so cross-time covariances vanish. Because $\boldsymbol{\psi}_t$ is measurable with respect to current covariates, past panel information, and the common shocks,

$$s_{NT}^2 = \sum_{t=1}^T \mathbb{E}_0 [\boldsymbol{\psi}_t' \boldsymbol{\Sigma}_{u,t} \boldsymbol{\psi}_t | \mathcal{C}].$$

The absolute row-sum bound in [Assumption 5](#) gives

$$s_{NT}^2 \leq C \mathbb{E}_0 \left[\sum_{i,t} \Psi_{0,it}^2 \middle| \mathcal{C} \right],$$

while the target-specific noncancellation restriction in [Assumption 9](#) gives the reverse inequality. Therefore,

$$s_{NT}^2 \asymp \mathbb{E}_0 \left[\sum_{i,t} \Psi_{0,it}^2 \middle| \mathcal{C} \right] = \Gamma_0(\mathbf{q}_0, \mathbf{q}_0).$$

The Gram-matrix bounds, [\(A.3\)](#), and [\(A.4\)](#) therefore imply

$$s_{NT}^2 \asymp \|\mathcal{P}_0 \mathbf{D}\|^2.$$

For an individual entry, the explicit tangent projection

$$\mathcal{P}_{T_M}(D) = \mathbf{U}_M \mathbf{U}_M' D + D \mathbf{V}_M \mathbf{V}_M' - \mathbf{U}_M \mathbf{U}_M' D \mathbf{V}_M \mathbf{V}_M'$$

and the leverage bounds give

$$\|\mathcal{P}_0 \mathbf{D}\|^2 \asymp \frac{1}{N} + \frac{1}{T}.$$

For a fixed unit-time lag sum, the direction contains only the fixed number P of entry directions. The positive-definiteness requirement in [Assumption 9](#) rules out cancellation among this fixed collection, so its variance has the same order $N^{-1} + T^{-1}$.

For a mean over $G_N \times R_T$, orthogonal projection is contractive and [Assumption 9](#) gives

$$\|\mathcal{P}_0 \mathbf{D}\|^2 \asymp \|\mathbf{D}\|^2 = \|\mathbf{a}_{G_N}\|_2^2 \|\mathbf{b}_{R_T}\|_2^2 = \frac{1}{|G_N||R_T|}.$$

For a contrast between disjoint groups,

$$\|\mathbf{a}_{G_{1,N}} - \mathbf{a}_{G_{2,N}}\|_2^2 = |G_{1,N}|^{-1} + |G_{2,N}|^{-1},$$

which gives the stated contrast variance after multiplication by $\|\mathbf{b}_{R_T}\|_2^2 = |R_T|^{-1}$.

The block-inverse calculation in [\(A.6\)](#), which follows from [Assumptions 6 to 8](#), implies that the time-series Riesz-score contributions associated with the target unit have conditional $2 + \delta$ norms bounded by CT^{-1} and the cross-sectional contributions associated with the target time are bounded by CN^{-1} . Hence

$$\sum_{i,t} \mathbb{E}_0(|Z_{it}|^{2+\delta} \mid \mathcal{C}) \leq C\{T \cdot T^{-(2+\delta)} + N \cdot N^{-(2+\delta)}\},$$

$$s_{NT}^{2+\delta} \asymp (N^{-1} + T^{-1})^{1+\delta/2},$$

and their ratio converges to zero. The largest variance contribution is bounded by $C(N^{-2} + T^{-2})$, which is negligible relative to $N^{-1} + T^{-1}$.

For a mean over $G_N \times R_T$, the target direction is $\mathbf{a}_{G_N} \mathbf{b}'_{R_T}$. The largest unit and time weights are $|G_N|^{-1}$ and $|R_T|^{-1}$. The same block-inverse bounds transfer these maximum weights to the corresponding Riesz-score contributions. Because the maximum target weight converges to zero whenever a support grows, the same Lyapunov calculation gives [\(A.7\)](#). Fixed-time group means and contrasts use $R_T = \{t_0\}$; the cross-sectional averaging weights still make every individual unit contribution negligible. Lag sums contain only the fixed number P of autoregressive coefficient matrices, so the same bounds hold after multiplication by a constant.

Finally, [Lemma 2](#) applied to $\mathcal{Y}_{it}(\mathbf{q}_0)u_{it}$ gives the conditional NED approximation. The norm equivalence in [\(A.4\)](#) provides the target-specific normalization. $\square$

Proof of [Theorem 1](#). Conditional sequential exogeneity and measurability of $\Psi_{0,it}$ with respect to current covariates, lagged outcomes, and $\mathcal{C}$ imply

$$\mathbb{E}_0(Z_{it} \mid x_{it}, \mathcal{F}_{t-1}, \mathcal{C}) = \Psi_{0,it} \mathbb{E}_0(u_{it} \mid x_{it}, \mathcal{F}_{t-1}, \mathcal{C}) = 0.$$

Hence $\mathbb{E}_0(Z_{it} \mid \mathcal{C}) = 0$.

Treat (i, t) as a location in $\mathbb{R}^{d_s+1}$ with distance $d_N(i, k) + |t - s|$. Uniform separation in space and integer spacing in time satisfy the sampling requirement of the NED random-field central limit theorem in [Jenish and Prucha \(2012\)](#). By [Lemma 1](#), that theorem may be applied under the regular conditional law given $\mathcal{C}$. By [Lemma 2](#) and [Proposition 1](#), the normalized score array is conditionally $L_{2+\delta}$ -NED, has a summable approximation rate, satisfies the conditional Lyapunov condition, and

has no dominating observation. Exponential mixing and [Assumption 3](#) imply the required mixing summability:

$$\sum_{a=1}^{\infty} (1+a)^{d_s} \bar{\alpha}(a)^{\delta/(2+\delta)} < \infty.$$

Let $\boldsymbol{\psi}_t = (\Psi_{0,1t}, \dots, \Psi_{0,Nt})'$. Conditional sequential exogeneity removes cross-time covariance, so

$$s_{NT}^2 = \sum_t \mathbb{E}_0[\boldsymbol{\psi}_t' \boldsymbol{\Sigma}_{u,t} \boldsymbol{\psi}_t \mid \mathcal{C}].$$

The row-sum bound in [Assumption 5](#) gives the upper comparison

$$s_{NT}^2 \leq C \mathbb{E}_0 \left[\sum_t \|\boldsymbol{\psi}_t\|_2^2 \mid \mathcal{C} \right],$$

while the Riesz-direction noncancellation restriction in [Assumption 9](#) gives the matching lower comparison. The Riesz equation and the Gram bounds then make this squared-weight sum comparable to $\|\mathcal{P}_0 \mathbf{D}\|^2$. The conditional variance is therefore nondegenerate at the target-specific rate established in [Proposition 1](#). The cited theorem yields (6.1). Applying the same argument to every conditionally variance-normalized linear combination of a fixed target vector and then using Cramér–Wold proves the joint statement. $\square$

A.3 Full-panel concentration and fixed-rank recovery

This subsection proves [Theorem 2](#). The argument first controls the residual score and the empirical prediction criterion uniformly over the admissible low-rank differences. The least-squares basic inequality then gives recovery of all coefficient matrices. Strong singular values convert matrix recovery into singular-space and tangent-space recovery, after which local loading and factor equations yield max-norm and Riesz-representer consistency.

Define the empirical Gram form

$$\hat{\Gamma}(\boldsymbol{\Delta}, \mathbf{Q}) = \sum_{i,t} \mathcal{Y}_{it}(\boldsymbol{\Delta}) \mathcal{Y}_{it}(\mathbf{Q}).$$

Let

$$\mathcal{D}_{\max} = \left\{ \boldsymbol{\Theta} - \boldsymbol{\Theta}_0 : \boldsymbol{\Theta} \in \bigcup_{\mathbf{r} \in \mathcal{R}_{\max}} \mathcal{M}(\mathbf{r}), \|\boldsymbol{\Theta}\|_{\max} \leq B \right\}$$

denote the set of admissible coefficient differences. For every $\boldsymbol{\Delta} \in \mathcal{D}_{\max}$ and every coefficient matrix M , $\text{rank}(\boldsymbol{\Delta}_M) \leq \bar{r}_M + r_{M,0}$ and $\|\boldsymbol{\Delta}_M\|_{\max} \leq 2B$.

Lemma 4 (Weighted concentration on the unit-time index). *For this lemma only, write $n = NT$ and $d = d_s + 1$. Let X_a , $a = (i, t)$, be a scalar array satisfying, conditionally on $\mathcal{C}$,*

$$\mathbb{E}_0(X_a \mid \mathcal{C}) = 0, \quad \sup_a \|X_a\|_{2+\delta, \mathcal{C}} \leq C$$

for some fixed $\delta > 0$. Suppose that for every integer $q \geq 1$ there is a bounded local approximation $X_a^{(q)}$, measurable with respect to the unit-specific innovations in the ball of radius $q/3$ around a together with $\mathcal{C}$, such that

$$|X_a^{(q)}| \leq M_n, \quad \sup_a \|X_a - X_a^{(q)}\|_{2+\delta, \mathcal{C}} \leq CM_n e^{-cq},$$

where $M_n \leq n^C$. Let $\mathcal{W}_n$ be a finite deterministic collection of weights $w = (w_a)$ satisfying

$$\sum_a |w_a| \leq 1, \quad \sum_a w_a^2 \leq v_n^2, \quad \max_a |w_a| \leq \omega_n.$$

Then, for every fixed $A < \infty$, one may choose a constant K_A such that, with $q_n = \lceil K_A \log n \rceil$, for every $x_n \in (0, 1]$ that is bounded below by a fixed negative power of n ,

$$\mathbb{P}_0 \left(\max_{w \in \mathcal{W}_n} \left| \sum_a w_a X_a \right| > x_n \middle| \mathcal{C} \right) \leq 2|\mathcal{W}_n| \exp \left[-\frac{cx_n^2}{v_n^2 + M_n \omega_n \{\log n\}^d x_n} \right] + Cn^{-A} \quad (\text{A.8})$$

almost surely for all sufficiently large n .

The same blocking argument has the following matrix consequence. If $X = (X_{it})$ is an $N \times T$ centered array satisfying the same bounded local-approximation conditions, then

$$\|X\|_{\text{op}} = O_p \left(\sqrt{(N+T) \log n} + M_n \{\log n\}^{(d_s+3)/2} \right). \quad (\text{A.9})$$

The conclusion continues to hold for arrays of fixed-size matrix blocks, with only the constants changed.

Proof. The combined unit-time index has polynomial ball growth of dimension $d = d_s + 1$: by [Assumption 3](#),

$$\sup_{i,t} |\{(j, s) : d_N(i, j) + |t - s| \leq r\}| \leq C(1+r)^{d_s+1}.$$

Fix $q = q_n$. Using the spatial locations from [Assumption 3](#) together with the integer time coordinate, partition the product space into blocks of side length proportional to q . A fixed number, depending only on d , of residue classes is enough to color the blocks so that two distinct blocks with the same color are separated by at least q . Each block contains at most Cq^d observed unit-time points.

First replace X_a by $X_a^{(q)}$. Since the weights have ℓ_1 norm at most one, the replacement is uniform over $\mathcal{W}_n$. Conditional Markov's inequality and a union bound over the n observations give, for any fixed polynomially small x_n ,

$$\mathbb{P}_0 \left(\max_a |X_a - X_a^{(q)}| > x_n/4 \middle| \mathcal{C} \right) \leq Cn \left(\frac{M_n e^{-cq}}{x_n} \right)^{2+\delta}.$$

Choosing K_A sufficiently large makes this probability $O(n^{-A})$.

For one color, collect the localized variables in each block into a single finite-dimensional random

vector. The corresponding block sigma-fields are separated by at least a constant multiple of q . The definition of $\alpha_{\mathcal{C}}$ in [Assumption 4](#) already takes the supremum over arbitrary separated index sets, so there is no block-cardinality multiplier in the mixing coefficient. Iterating the coupling theorem of [Bradley \(1983\)](#) therefore constructs independent copies of the localized block vectors, preserving each block distribution, on an extension of the conditional probability space. Exponential mixing and $q = K_A \log n$ make the probability that any coupled block differs by more than the amount used in the Bernstein comparison $O(n^{-A})$. The number of colors is fixed, so the same bound holds for all colors simultaneously.

On the coupling event, for a fixed $w \in \mathcal{W}_n$ the sum within a block is bounded by

$$CM_n q^d \omega_n.$$

Moreover, the conditional covariance inequality in [\(A.2\)](#), exponential NED approximation, and polynomial shell growth imply an absolute covariance-row-sum bound for the original centered field. The localized covariance matrix differs from it by $O(ne^{-cq})$ in row sum. Hence, uniformly over the blocks,

$$\sum_{\text{blocks}} \text{Var}_0 \left(\sum_{a \text{ in block}} w_a X_a^{(q)} \middle| \mathcal{C} \right) \leq C \sum_a w_a^2 \leq C v_n^2.$$

Scalar Bernstein's inequality applied to the independent coupled block sums, followed by the fixed-color union bound and then the union bound over $\mathcal{W}_n$, gives [\(A.8\)](#) after constants are enlarged.

For the matrix statement, couple the entire localized block matrices rather than scalar block sums. A block matrix has at most Cq^d nonzero entries, so its operator norm is at most $CM_n q^{d/2}$. The same covariance-row-sum calculation gives the rectangular matrix variance bounds

$$\left\| \sum_{\text{blocks}} \mathbb{E}_0(BB' \mid \mathcal{C}) \right\|_{\text{op}} \leq CT, \quad \left\| \sum_{\text{blocks}} \mathbb{E}_0(B'B \mid \mathcal{C}) \right\|_{\text{op}} \leq CN.$$

The rectangular matrix Bernstein inequality of [Tropp \(2012\)](#), applied separately to the fixed number of color classes, therefore gives

$$\|X^{(q)}\|_{\text{op}} = O_p \left[\sqrt{(N+T) \log n} + M_n q^{d/2} \log n \right].$$

The localization and coupling errors are $o_p(1)$ after increasing K_A , and $q \asymp \log n$, which proves [\(A.9\)](#). Fixed block size only multiplies the displayed bounds by a constant. $\square$

Proposition 2 (Uniform full-panel concentration). *Under [Assumptions 1 to 8](#) and [10](#),*

$$\sup_{\Delta \in \mathcal{D}_{\max} \setminus \{0\}} \frac{|\sum_{i,t} u_{it} \mathcal{Y}_{it}(\Delta)|}{\|\Delta\|} = O_p \{ \sqrt{NT \zeta_{NT}} \}, \quad (\text{A.10})$$

and, uniformly over $\mathcal{D}_{\max}$,

$$\sum_{i,t} \mathcal{Y}_{it}(\Delta)^2 \geq \frac{c}{2} \|\Delta\|^2 - CNT\zeta_{NT}. \quad (\text{A.11})$$

On $\mathcal{T}_0$, the empirical least-squares Gram form converges uniformly to its conditional population counterpart.

Proof. Let $z_{it,M}$ denote the regressor multiplying coefficient matrix M . By Lemma 2, the arrays $u_{it}z_{it,M}$ and $z_{it,M}z_{it,M'}$ have the conditional moments and exponential local-approximation rates required below. Put $b = b_{NT}$ and let $\mathcal{E}_b$ be the event on which every $|u_{it}|$ and every $|z_{it,M}|$ is at most b . Stability and the conditional $8 + \eta$ moment bounds imply

$$\mathbb{P}_0(\mathcal{E}_b^c \mid \mathcal{C}) \leq CNTb^{-(8+\eta)} \leq C\{\log(NT)\}^{-(8+\eta)} \longrightarrow 0.$$

Thus all truncations used below agree simultaneously with the original panel on an event whose conditional probability approaches one.

Step 1: the score term. For each M , clip u_{it} and $z_{it,M}$ at b , multiply them, and subtract the conditional mean given $\mathcal{C}$. Let $\mathbf{S}_M^b$ denote the resulting centered $N \times T$ matrix. Its entries are bounded by Cb^2 , have uniformly bounded conditional $2 + \delta$ moments for some fixed $\delta > 0$, and have exponential local approximations by Lemma 2. The matrix conclusion of Lemma 4 gives

$$\max_M \|\mathbf{S}_M^b\|_{\text{op}} = O_p \left[\sqrt{(N+T)\log(NT)} + b^2 \{\log(NT)\}^{(d_s+3)/2} \right]. \quad (\text{A.12})$$

Conditional sequential exogeneity gives $\mathbb{E}_0(u_{it}z_{it,M} \mid \mathcal{C}) = 0$. Since clipping changes a score entry only on $\mathcal{E}_b^c$, conditional Hölder and the $8 + \eta$ moments imply that the matrix of clipping means has Frobenius norm $o(1)$. Therefore (A.12) also holds for the original score matrix $\mathbf{S}_M$. Under Assumption 10,

$$\sqrt{(N+T)\log(NT)} + b^2 \{\log(NT)\}^{(d_s+3)/2} = O\left(\sqrt{NT\zeta_{NT}}\right).$$

Nuclear/operator-norm duality and the fixed rank caps now give, uniformly over $\mathcal{D}_{\max}$,

$$\left| \sum_{i,t} u_{it} \mathcal{Y}_{it}(\Delta) \right| \leq C \max_M \|\mathbf{S}_M\|_{\text{op}} \|\Delta\|,$$

which proves (A.10).

Step 2: uniform prediction curvature on the rank-capped set. Write $n = NT$, $m = N + T$, and $d = d_s + 1$ within this step. Expanding the centered empirical prediction norm gives

$$\sum_{i,t} [\mathcal{Y}_{it}(\Delta)^2 - \mathbb{E}_0\{\mathcal{Y}_{it}(\Delta)^2 \mid \mathcal{C}\}] = \sum_{M,M'} \sum_{i,t} X_{it}^{MM'} \Delta_{M,it} \Delta_{M',it}, \quad (\text{A.13})$$

where

$$X_{it}^{MM'} = z_{it,M} z_{it,M'} - \mathbb{E}_0(z_{it,M} z_{it,M'} \mid \mathcal{C}).$$

On $\mathcal{E}_b$, replace $X_{it}^{MM'}$ by its clipped version; this is bounded by Cb^2 and satisfies the hypotheses of [Lemma 4](#).

Let

$$r_0^2 = C_0 n \zeta_{NT} = C_0 b^2 m \{\log n\}^{d+1},$$

where C_0 is a sufficiently large fixed constant. If $\|\Delta\| < r_0$, the lower bound ([A.11](#)) is immediate after the constant multiplying $n \zeta_{NT}$ is enlarged, because the empirical prediction norm is nonnegative. It remains to consider $\|\Delta\| \geq r_0$.

Peel this set into dyadic norm ranges $r \leq \|\Delta\| < 2r$. There are only $O(\log n)$ nonempty ranges because the coefficient bound implies $\|\Delta\| \leq C\sqrt{n}$. In each range choose an ϵ_n -net, in Frobenius norm, of the normalized rank-capped collections $\Delta/\|\Delta\|$, with

$$\epsilon_n = c_0/b^2.$$

The ranks and number of coefficient matrices are fixed, so the usual singular-vector parameterization yields

$$\log |\mathcal{N}_r| \leq C m \log(Cb) \leq C m \log n.$$

The net centers may be chosen from the normalized admissible set itself. Hence, for a center δ from a range with lower endpoint r ,

$$\max_{i,t,M} |\delta_{M,it}| \leq \frac{2B}{r}.$$

For every fixed pair (M, M') , the weights

$$w_{it}^{MM'} = \delta_{M,it} \delta_{M',it}$$

satisfy

$$\sum_{i,t} |w_{it}^{MM'}| \leq 1, \quad \sum_{i,t} (w_{it}^{MM'})^2 \leq \omega_r, \quad \omega_r \leq C/r^2.$$

Apply ([A.8](#)) simultaneously to the finite collection of all net points and all matrix pairs, with a fixed small threshold $x > 0$. Since $M_n = Cb^2$ and $q_n^d \asymp \{\log n\}^d$, the exponent is bounded below by

$$\frac{cr^2}{b^2 \{\log n\}^d} \geq cC_0 m \log n.$$

Choosing C_0 large enough makes this dominate the net entropy, the finite number of matrix pairs, and the $O(\log n)$ norm ranges. Thus, with conditional probability approaching one, the absolute value of ([A.13](#)) is at most $c\|\Delta\|^2/4$ at every net point.

On $\mathcal{E}_b$, multiplication by the regressors maps the coefficient collection into fitted values with operator norm at most Cb . Consequently, both the empirical and conditional-population quadratic

forms are Cb^2 -Lipschitz on the unit Frobenius sphere. The choice $\epsilon_n = c_0/b^2$ therefore extends the preceding bound from each net to its entire dyadic range after c_0 is chosen sufficiently small. Conditional prediction identification then gives, uniformly for $\|\Delta\| \geq r_0$,

$$\sum_{i,t} \mathcal{Y}_{it}(\Delta)^2 \geq \frac{c}{2} \|\Delta\|^2.$$

Combining the large- and small-norm ranges proves (A.11).

Step 3: the empirical Gram operator on the true tangent space. Use the normalized loading and factor coordinates introduced before Lemma 3. The loading–loading diagonal block for unit i is a T -average of fixed-dimensional products, and the factor–factor diagonal block for time t is an N -average. Apply (A.8) entry by entry with weights $1/T$ or $1/N$ and take a union bound over the $N + T$ blocks. Under Assumption 10,

$$\max_i \|\widehat{G}_{\lambda\lambda,i} - G_{\lambda\lambda,i}\|_{\text{op}} + \max_t \|\widehat{G}_{ff,t} - G_{ff,t}\|_{\text{op}} = o_p(1).$$

The loading–factor part is a rectangular block matrix whose (i, t) block is scaled by $(NT)^{-1/2}$. Apply the matrix conclusion (A.9) to the centered, clipped fixed-size block array and then divide by $\sqrt{NT}$ to obtain

$$\|\widehat{G}_{\lambda f} - G_{\lambda f}\|_{\text{op}} = O_p \left[\sqrt{\frac{(N+T) \log(NT)}{NT}} + \frac{b^2 \{\log(NT)\}^{(d_s+3)/2}}{\sqrt{NT}} \right] = o_p(1).$$

The transpose block has the same bound. Hence

$$\|\widehat{\mathbb{G}} - \mathbb{G}_0\|_{\text{op}} = o_p(1)$$

on the normalized tangent coordinates, which is equivalent to the stated uniform convergence of the empirical least-squares Gram form on $\mathcal{T}_0$. $\square$

Proof of Theorem 2. Let

$$\widehat{\Delta} = \widehat{\Theta}(\mathbf{r}_0) - \Theta_0.$$

Optimality of $\widehat{\Theta}(\mathbf{r}_0)$ and feasibility of Θ_0 imply

$$\begin{aligned} 0 &\geq \frac{1}{2NT} \sum_{i,t} \left[\{u_{it} - \mathcal{Y}_{it}(\widehat{\Delta})\}^2 - u_{it}^2 \right] \\ &= \frac{1}{2NT} \widehat{\Gamma}(\widehat{\Delta}, \widehat{\Delta}) - \frac{1}{NT} \sum_{i,t} u_{it} \mathcal{Y}_{it}(\widehat{\Delta}). \end{aligned}$$

Multiplying by NT and applying Proposition 2 and Assumption 10 gives

$$\frac{c}{4} \|\widehat{\Delta}\|^2 \leq C \sqrt{NT \zeta_{NT}} \|\widehat{\Delta}\| + CNT \zeta_{NT}.$$

If $x = \|\widehat{\Delta}\|$, the inequality $cx^2/4 \leq ax + b$ implies

$$x \leq \frac{4a}{c} + 2\sqrt{\frac{b}{c}}.$$

Therefore,

$$\|\widehat{\Delta}\| = O_p\{\sqrt{NT\zeta_{NT}}\}. \quad (\text{A.14})$$

Substituting (A.14) into the basic inequality yields

$$\sum_{i,t} \mathcal{Y}_{it}(\widehat{\Delta})^2 = O_p(NT\zeta_{NT}).$$

For a positive-rank coefficient matrix M ,

$$\|\widehat{M} - M_0\|_{\text{op}} \leq \|\widehat{M} - M_0\|_F = O_p\{\sqrt{NT\zeta_{NT}}\}.$$

Because $\sigma_{r_M}(M_0) \geq c\sqrt{NT}$, the Wedin $\sin \Theta$ perturbation bound (Wedin, 1972) gives

$$\|\widehat{U}_M \widehat{U}_M' - U_M U_M'\|_{\text{op}} + \|\widehat{V}_M \widehat{V}_M' - V_M V_M'\|_{\text{op}} = O_p(\zeta_{NT}^{1/2}).$$

The formula for the tangent projection then implies

$$\|\widehat{\mathcal{P}} - \mathcal{P}_0\|_{\text{op}} = O_p(\zeta_{NT}^{1/2}). \quad (\text{A.15})$$

A supplied rank-zero matrix is exactly zero.

We next derive the pointwise rates using the loading-factor normalization, the matrix $\mathbb{Z}_{it}$, and the local derivative introduced before Lemma 3. Align the estimated factors and loadings with that normalization and set

$$\mathbf{h}_i^\lambda = \sqrt{T}(\widehat{\lambda}_i - \lambda_{0,i}), \quad \mathbf{h}_t^f = \sqrt{N}(\widehat{\mathbf{f}}_t - \mathbf{f}_{0,t}).$$

the fitted-value change is exactly

$$\mathcal{Y}_{it}(\widehat{\Theta}) - \mathcal{Y}_{it}(\Theta_0) = \dot{\mathcal{Y}}_{it}(\mathbf{h}) + R_{it}(\mathbf{h}), \quad (\text{A.16})$$

where $\dot{\mathcal{Y}}_{it}(\mathbf{h})$ is defined in (A.5) and

$$R_{it}(\mathbf{h}) = \frac{(\mathbf{h}_i^\lambda)' \mathbb{Z}_{it} \mathbf{h}_t^f}{\sqrt{NT}}. \quad (\text{A.17})$$

For a direction $\mathbf{q}$, the derivative of (A.16) at $\mathbf{h}$ is

$$\dot{\mathcal{Y}}_{it}(\mathbf{q}) + C_{it}(\mathbf{h}, \mathbf{q}),$$

where

$$C_{it}(\mathbf{h}, \mathbf{q}) = \frac{(\mathbf{h}_i^\lambda)' \mathbb{Z}_{it} \mathbf{q}_t^f + (\mathbf{q}_i^\lambda)' \mathbb{Z}_{it} \mathbf{h}_t^f}{\sqrt{NT}}. \quad (\text{A.18})$$

Define the normalized score blocks

$$\begin{aligned} \mathbf{s}_i^\lambda &= \frac{1}{\sqrt{T}} \sum_{t=1}^T (\mathbb{Z}_{it} \mathbf{f}_{0,t}) u_{it}, \\ \mathbf{s}_t^f &= \frac{1}{\sqrt{N}} \sum_{i=1}^N (\mathbb{Z}_{it}' \boldsymbol{\lambda}_{0,i}) u_{it}. \end{aligned}$$

The empirical first-order conditions and (A.16)–(A.18) imply

$$\widehat{\mathbb{G}}(\widehat{\mathbf{h}}, \mathbf{q}) = \mathbf{s}(\mathbf{q}) + \mathcal{N}(\widehat{\mathbf{h}}, \mathbf{q}) \quad \text{for every } \mathbf{q}, \quad (\text{A.19})$$

where

$$\begin{aligned} \mathcal{N}(\mathbf{h}, \mathbf{q}) &= \sum_{i,t} u_{it} C_{it}(\mathbf{h}, \mathbf{q}) - \sum_{i,t} \dot{\mathbf{y}}_{it}(\mathbf{h}) C_{it}(\mathbf{h}, \mathbf{q}) \\ &\quad - \sum_{i,t} R_{it}(\mathbf{h}) \dot{\mathbf{y}}_{it}(\mathbf{q}) - \sum_{i,t} R_{it}(\mathbf{h}) C_{it}(\mathbf{h}, \mathbf{q}). \end{aligned} \quad (\text{A.20})$$

The tangent-space Gram part of Proposition 2 gives

$$\|\widehat{\mathbb{G}} - \mathbb{G}_0\|_{\text{op}} = o_p(1).$$

By Lemma 3, the inverse local Gram operator has bounded diagonal rows and square-summable loading–factor cross rows at the rates in (A.6). The bounds in (A.6), the conditional moment bounds for $\mathbf{s}_i^\lambda$ and $\mathbf{s}_t^f$, and (5.6) imply

$$\|\mathcal{N}(\widehat{\mathbf{h}}, \cdot)\| = o_p(1)$$

for every fixed target loading or factor row. Inverting (A.19) therefore gives

$$\mathbf{h}_i^\lambda = [\mathbb{G}_0^{-1} \mathbf{s}]_i^\lambda + o_p(1), \quad \mathbf{h}_t^f = [\mathbb{G}_0^{-1} \mathbf{s}]_t^f + o_p(1). \quad (\text{A.21})$$

This proves root- T loading and root- N factor recovery for fixed target rows. For the uniform version, truncate every coordinate of $\mathbf{s}_i^\lambda$ and $\mathbf{s}_t^f$ at the same level b_{NT}^2 used in Proposition 2. Applying Lemma 4 to the localized row scores with the equal time or unit weights, followed by a union bound over the fixed-dimensional coordinates of all $N + T$ rows, gives

$$\max_i \|\mathbf{s}_i^\lambda\| + \max_t \|\mathbf{s}_t^f\| = O_p\{b_{NT} \sqrt{\log(NT)}\}.$$

The same-side and cross-side inverse-row bounds in (A.6) imply the identical order for the rows of

$\mathbb{G}_0^{-1}\mathbf{s}$. On the ball with this radius, (A.17)–(A.20), conditional moments, and fixed ranks give

$$\max_i \|\mathcal{N}(\hat{\mathbf{h}}, \cdot)_{\lambda_i}\| + \max_t \|\mathcal{N}(\hat{\mathbf{h}}, \cdot)_{f_t}\| \leq C b_{NT}^2 \log(NT) \left(N^{-1/2} + T^{-1/2} \right) + o_p(1).$$

The right side is $o_p(1)$ by [Assumption 10](#). Applying the row bounds of $\mathbb{G}_0^{-1}$ once more to (A.19) therefore yields

$$\max_i \|\mathbf{h}_i^\lambda\| = O_p\{b_{NT}\sqrt{\log(NT)}\}, \quad \max_t \|\mathbf{h}_t^f\| = O_p\{b_{NT}\sqrt{\log(NT)}\}.$$

Because the true loading and factor rows are bounded by incoherence and the signal normalization, expanding each fitted entry yields

$$\|\hat{\Theta} - \Theta_0\|_{\max} = O_p \left[b_{NT} \sqrt{\frac{\log(NT)}{T}} + b_{NT} \sqrt{\frac{\log(NT)}{N}} + \frac{b_{NT}^2 \log(NT)}{\sqrt{NT}} \right] = O_p(\zeta_{NT}^{1/2}) = o_p(1). \quad (\text{A.22})$$

This proves the max-norm statement in [Theorem 2](#) and, together with the strict margin in [Assumption 6](#), makes the coefficient bound inactive with probability approaching one.

Let $\mathbf{h}^*$ denote the local coordinate of the population Riesz representer. It is the unique solution of

$$\mathbb{G}_0(\mathbf{q}, \mathbf{h}^*) = \dot{\phi}(\mathbf{q}) \quad \text{for every } \mathbf{q}, \quad (\text{A.23})$$

where $\dot{\phi}$ is the derivative of the parameter. The empirical Riesz coordinate satisfies the corresponding equation with $\hat{\mathbb{G}}$ and the estimated tangent space. For a mean or contrast, [Assumptions 8](#) and [9](#) and (A.15) give

$$\|\hat{\mathbf{q}} - \mathbf{q}_0\| \leq C \left[\|\hat{\mathbb{G}} - \mathbb{G}_0\|_{\text{op}} + \|\hat{\mathcal{P}} - \mathcal{P}_0\|_{\text{op}} \right] \|\mathcal{P}_0 \mathbf{D}\| = o_p(\|\mathcal{P}_0 \mathbf{D}\|).$$

For an individual entry, the global tangent bound must be evaluated at the target row. The explicit tangent projection and (A.21) give

$$\|(\hat{\mathcal{P}} - \mathcal{P}_0) \mathbf{D}\| = o_p \left\{ \left(\frac{1}{N} + \frac{1}{T} \right)^{1/2} \right\} = o_p(\|\mathcal{P}_0 \mathbf{D}\|).$$

The same inverse-Gram argument then yields the Riesz rate. Finally, conditional identification gives

$$\sum_{i,t} (\hat{\Psi}_{it} - \Psi_{0,it})^2 \leq C \|\hat{\mathbf{q}} - \mathbf{q}_0\|^2 = o_p(\|\mathcal{P}_0 \mathbf{D}\|^2).$$

Weighted residual identity. Let

$$L_{NT}(\Theta) = \frac{1}{2NT} \sum_{i,t} \{y_{it} - \mathcal{Y}_{it}(\Theta)\}^2.$$

On the event that the coefficient bound is inactive, every $\Delta \in \widehat{\mathcal{T}}$ is the derivative at zero of a differentiable path $a \mapsto \Theta(a)$ in the selected fixed-rank set with $\Theta(0) = \widehat{\Theta}$. First-order stationarity gives

$$0 = \left. \frac{d}{da} L_{NT}\{\Theta(a)\} \right|_{a=0} = -\frac{1}{NT} \sum_{i,t} \widehat{u}_{it} \mathcal{Y}_{it}(\Delta).$$

Since $\widehat{\mathbf{q}} \in \widehat{\mathcal{T}}$, choosing $\Delta = \widehat{\mathbf{q}}$ proves (4.5). The max-norm conclusion of [Theorem 2](#) and the strict interior margin in [Assumption 6](#) set the coefficient-bound multipliers equal to zero with probability approaching one. Numerical objective, first-order, and Riesz equation residuals enter additively and are negligible under the stated tolerances. $\square$

A.4 Auxiliary local-deletion comparison

The empirical Riesz weights and u_{it} are calculated from the same panel, so their product cannot be centered by treating the weights as fixed. This subsection introduces an auxiliary estimator that removes a small space–time neighborhood around (i, t) . The auxiliary estimator is used only in the proof. Conditional NED and alpha mixing separate it from the current score, while stability shows that deleting the neighborhood changes the full-panel estimator by a negligible amount.

Choose proof-only radii

$$q_T = \lceil c_q \log(NT) \rceil, \quad r_N = \lceil c_r \log(NT) \rceil,$$

with fixed constants c_q, c_r sufficiently large. Exponential dynamic, NED, and mixing decay, together with [Assumptions 3](#) and [10](#), gives

$$\begin{aligned} b_{NT}^2 \left\{ \frac{2q_T + 1}{T} + \frac{v_N(r_N)}{N} \right\} &\longrightarrow 0, \\ \sqrt{NT} \left[\rho^{q_T - P} + \psi_T(\lfloor q_T/3 \rfloor) + \psi_S(\lfloor r_N/3 \rfloor) + \sum_{a > \lfloor \min(q_T, r_N)/3 \rfloor} (1+a)^{d_s} \bar{\alpha}(a)^{\eta/(8+\eta)} \right] &\longrightarrow 0. \end{aligned} \quad (\text{A.24})$$

These radii are used only in the proof and are not estimator tuning parameters.

For observation $a = (i, t)$, define the proof-only neighborhood

$$\mathcal{B}_a = \{(k, s) : |s - t| \leq q_T, \ d_N(i, k) \leq r_N\}.$$

Let $\widehat{\Theta}^{(-a)}$ be the supplied-rank estimator after deleting $\mathcal{B}_a$, and let $\widehat{\mathbf{q}}^{(-a)}$ be its Riesz representer. These quantities are not computed by the estimator.

Proposition 3 (Local-deletion stability and comparison). *Under the assumptions of [Theorem 2](#) and [Proposition 2](#),*

$$\frac{1}{NT} \sum_a \|\widehat{\mathbf{h}} - \widehat{\mathbf{h}}^{(-a)}\|^2 = o_p(1), \quad (\text{A.25})$$

and

$$\frac{1}{NT} \sum_a \frac{\|\hat{\mathbf{q}} - \hat{\mathbf{q}}^{(-a)}\|^2}{\|\mathcal{P}_0 \mathbf{D}\|^2} = o_p(1). \quad (\text{A.26})$$

Proof. Work in the aligned local coordinates used in the recovery proof. Let $\hat{\mathbb{G}}^{(-a)}$ and $\hat{\mathbf{s}}^{(-a)}$ denote the Gram matrix and score after deleting $\mathcal{B}_a$. Deleting the neighborhood removes at most $2q_T + 1$ observations from a loading-row time series and at most $v_N(r_N)$ observations from a factor-row cross-section. Therefore,

$$\|\hat{\mathbb{G}} - \hat{\mathbb{G}}^{(-a)}\|_{\text{op}} \leq C b_{NT}^2 \left\{ \frac{2q_T + 1}{T} + \frac{v_N(r_N)}{N} \right\} = o_p(1). \quad (\text{A.27})$$

The conditional second moment of the deleted score satisfies

$$\frac{1}{NT} \sum_a \|\hat{\mathbf{s}} - \hat{\mathbf{s}}^{(-a)}\|^2 \leq C(2q_T + 1)v_N(r_N) \left(\frac{1}{N} + \frac{1}{T} \right) + o_p(1). \quad (\text{A.28})$$

Subtracting the full and deleted normal equations and using uniform nonsingularity gives

$$\begin{aligned} \|\hat{\mathbf{h}} - \hat{\mathbf{h}}^{(-a)}\| &\leq C \left[\|\hat{\mathbf{s}} - \hat{\mathbf{s}}^{(-a)}\| + \|\hat{\mathbb{G}} - \hat{\mathbb{G}}^{(-a)}\|_{\text{op}} \|\hat{\mathbf{h}}\| \right. \\ &\quad \left. + \|\mathcal{N}(\hat{\mathbf{h}}, \cdot) - \mathcal{N}^{(-a)}(\hat{\mathbf{h}}^{(-a)}, \cdot)\| \right]. \end{aligned} \quad (\text{A.29})$$

The nonlinear difference is of smaller order by recovery and (5.6). Averaging the square of (A.29) and applying (A.27)–(A.28) gives (A.25).

The Riesz equations satisfy, in aligned coordinates,

$$\begin{aligned} \hat{\mathbf{h}}^* - \hat{\mathbf{h}}^{*(-a)} &= (\hat{\mathbb{G}}^{(-a)})^{-1} \left[(\hat{\phi} - \hat{\phi}^{(-a)}) \right. \\ &\quad \left. - (\hat{\mathbb{G}} - \hat{\mathbb{G}}^{(-a)}) \hat{\mathbf{h}}^* \right]. \end{aligned}$$

The first difference is controlled by the tangent-space stability just obtained, and the second by (A.27). Averaging the squared norm proves (A.26).

The same deletion construction will also be used below as a decorrelation device. After replacing an anchored regressor or score contribution by its local innovation approximation, that contribution is measurable inside $\mathcal{B}_a$, whereas the corresponding deleted coordinate is measurable outside $\mathcal{B}_a$. The conditional NED covariance inequality therefore bounds their conditional covariance by the dependence-tail quantity in (A.24) times their conditional moments. For centered sums, splitting pairs according to the distance between their local innovation neighborhoods, then using exponential mixing and polynomial neighborhood growth, makes the covariance bound summable. This is the specific local-deletion property invoked in Proposition 4; no feasible-score decomposition is needed here. $\square$

A.5 Full-panel target expansion and split-panel correction

This subsection proves [Theorems 3](#) and [4](#). The local coordinates and score equation were constructed in [Section A.3](#). We now separate their first-order solution from the remaining loading–factor products. Substitution into the parameter of interest produces the Riesz score, a loading-side bias of order T^{-1} , a factor-side bias of order N^{-1} , and smaller terms. The final part shows algebraically why the two split-panel comparisons remove the two leading biases.

Let $\hat{\mathbf{h}}$ be the local coordinate of the full-panel estimate defined in [Section A.3](#), and define the first-order coordinate $\mathbf{h}^{(1)}$ by

$$\mathbb{G}_0(\mathbf{h}^{(1)}, \mathbf{q}) = \mathbf{s}(\mathbf{q}) \quad \text{for every } \mathbf{q}. \quad (\text{A.30})$$

The target has the exact local expansion

$$\phi\{\boldsymbol{\Theta}(\mathbf{h})\} - \phi(\boldsymbol{\Theta}_0) = \dot{\phi}(\mathbf{h}) + \mathcal{Q}(\mathbf{h}), \quad (\text{A.31})$$

where

$$\mathcal{Q}(\mathbf{h}) = \sum_{M:r_M>0} \frac{\left\langle \mathbf{D}_M, \mathbf{H}_M^\lambda (\mathbf{H}_M^f)' \right\rangle}{\sqrt{NT}}. \quad (\text{A.32})$$

Here $\mathbf{H}_M^\lambda$ and $\mathbf{H}_M^f$ collect the loading and factor coordinates for matrix M . By [\(A.23\)](#) and [\(A.30\)](#),

$$\dot{\phi}(\mathbf{h}^{(1)}) = \mathbb{G}_0(\mathbf{h}^{(1)}, \mathbf{h}^*) = \mathbf{s}(\mathbf{h}^*) = \sum_{i,t} \Psi_{0,it} u_{it}. \quad (\text{A.33})$$

Lemma 5 (Exact local target decomposition). *For an exact interior minimizer,*

$$\begin{aligned} \hat{\phi} - \phi(\boldsymbol{\Theta}_0) &= \sum_{i,t} \Psi_{0,it} u_{it} + \{\mathbb{G}_0 - \hat{\mathbb{G}}\}(\hat{\mathbf{h}}, \mathbf{h}^*) \\ &\quad + \mathcal{N}(\hat{\mathbf{h}}, \mathbf{h}^*) + \mathcal{Q}(\hat{\mathbf{h}}). \end{aligned} \quad (\text{A.34})$$

For a computed approximate stationary point, an additional numerical remainder is bounded by the reported first-order and Riesz-equation tolerances.

Proof. Subtract the first-order equation [\(A.30\)](#) from the exact local normal equation [\(A.19\)](#). For every direction $\mathbf{q}$,

$$\mathbb{G}_0(\hat{\mathbf{h}} - \mathbf{h}^{(1)}, \mathbf{q}) = \{\mathbb{G}_0 - \hat{\mathbb{G}}\}(\hat{\mathbf{h}}, \mathbf{q}) + \mathcal{N}(\hat{\mathbf{h}}, \mathbf{q}).$$

Set $\mathbf{q} = \mathbf{h}^*$ and use the local Riesz equation [\(A.23\)](#) to obtain

$$\dot{\phi}(\hat{\mathbf{h}} - \mathbf{h}^{(1)}) = \{\mathbb{G}_0 - \hat{\mathbb{G}}\}(\hat{\mathbf{h}}, \mathbf{h}^*) + \mathcal{N}(\hat{\mathbf{h}}, \mathbf{h}^*).$$

Combining this identity with the exact target expansion [\(A.31\)](#) and the first-order influence identity [\(A.33\)](#) gives [\(A.34\)](#). $\square$

Define the leading second-order expression

$$\mathcal{B}_{NT} = \{\mathbb{G}_0 - \widehat{\mathbb{G}}\}(\mathbf{h}^{(1)}, \mathbf{h}^*) + \mathcal{N}(\mathbf{h}^{(1)}, \mathbf{h}^*) + \mathcal{Q}(\mathbf{h}^{(1)}). \quad (\text{A.35})$$

Proposition 4 (Decorrelation of the bilinear remainders). *Under the assumptions of [Theorem 3](#), in the aligned local coordinates of [Section A.3](#),*

$$\begin{aligned} \{\mathbb{G}_0 - \widehat{\mathbb{G}}\}(\widehat{\mathbf{h}}, \mathbf{h}^*) - \{\mathbb{G}_0 - \widehat{\mathbb{G}}\}(\mathbf{h}^{(1)}, \mathbf{h}^*) \\ = o_p(s_{NT}) + o_p(N^{-1} + T^{-1}), \end{aligned} \quad (\text{A.36})$$

$$\mathcal{N}(\widehat{\mathbf{h}}, \mathbf{h}^*) - \mathcal{N}(\mathbf{h}^{(1)}, \mathbf{h}^*) = o_p(s_{NT}) + o_p(N^{-1} + T^{-1}), \quad (\text{A.37})$$

$$\mathcal{Q}(\widehat{\mathbf{h}}) - \mathcal{Q}(\mathbf{h}^{(1)}) = o_p(s_{NT}) + o_p(N^{-1} + T^{-1}). \quad (\text{A.38})$$

For the mixed loading-factor part of [\(A.35\)](#),

$$\mathbb{E}_0(\mathcal{B}_{NT}^{\lambda f} \mid \mathcal{C}) = o(N^{-1} + T^{-1}), \quad \text{Var}_0(\mathcal{B}_{NT}^{\lambda f} \mid \mathcal{C}) = o(s_{NT}^2). \quad (\text{A.39})$$

Consequently,

$$\mathcal{B}_{NT}^{\lambda f} = o_p(s_{NT}) + o_p(N^{-1} + T^{-1}).$$

Proof. Let δ_{NT} denote the bracketed dependence-tail expression in [\(A.24\)](#). By construction of the proof-only radii, $\sqrt{NT} \delta_{NT} \rightarrow 0$. Define

$$\epsilon_{NT} = b_{NT}^2 \log(NT) \left(N^{-1/2} + T^{-1/2} \right) + \sqrt{NT} \delta_{NT}.$$

The panel-growth condition and [\(A.24\)](#) imply $\epsilon_{NT} \rightarrow 0$.

We first prove the three replacement statements. Expand the empirical Gram difference and the nonlinear term block by block in loading and factor coordinates. Each summand is a product of a local regressor or score contribution and one row of $\widehat{\mathbf{h}} - \mathbf{h}^{(1)}$, multiplied by a row of the fixed target coordinate $\mathbf{h}^*$. For the summand anchored at $a = (i, t)$, add and subtract the corresponding coordinate from the estimator that deletes $\mathcal{B}_a$. The average squared difference between the full and deleted estimator is $o_p(1)$ by [\(A.25\)](#); the uniform row bounds in [Section A.3](#) and Cauchy-Schwarz therefore make the contribution of this replacement bounded by the first term in ϵ_{NT} times $s_{NT} + N^{-1} + T^{-1}$.

After deletion, localize the regressor and score contribution to radii $\lfloor q_T/3 \rfloor$ and $\lfloor r_N/3 \rfloor$. The localized contribution attached to a is measurable inside $\mathcal{B}_a$, while the localized deleted coordinate is measurable outside that neighborhood. The conditional NED covariance inequality used in the proof of [Proposition 3](#) therefore bounds the conditional mean by $C\delta_{NT}$ times the relevant conditional second moments. The centered covariances are split according to the distance between the two local innovation neighborhoods. Exponential mixing, polynomial spatial neighborhood growth, the same-side and cross-side inverse-Gram row bounds in [\(A.6\)](#), and $\|\mathbf{h}^*\| \asymp \|\mathcal{P}_0 \mathbf{D}\|$ then make the sum of the centered covariances $O(\epsilon_{NT}^2 s_{NT}^2)$. The localization errors satisfy the same order. This proves

(A.36) and (A.37). Finally, $\mathcal{Q}$ is bilinear in one loading and one factor coordinate. Expanding $\mathcal{Q}(\hat{\mathbf{h}}) - \mathcal{Q}(\mathbf{h}^{(1)})$ and using the uniform row recovery rates with the preceding deletion bound gives (A.38).

It remains to verify that the mixed loading–factor expectation is smaller than the two one-sided biases. The primitive normalized score blocks satisfy a sharper cross-side covariance bound than their same-side variances. For a loading score at unit i and a factor score at time t , sequential exogeneity eliminates every cross-time term in their product. Indeed, if $s < t$, the time- t disturbance has conditional mean zero given the time- s disturbance and the intervening past; if $s > t$, the time- s disturbance has conditional mean zero given the time- t disturbance and the intervening past. Hence only $s = t$ remains, and

$$\begin{aligned} & \left\| \mathbb{E}_0 \left[\mathbf{s}_i^\lambda (\mathbf{s}_t^f)' \middle| \mathcal{C} \right] \right\| \\ & \leq \frac{C}{\sqrt{NT}} \sum_j |\mathbb{E}_0(u_{it}u_{jt} \mid x_t, \mathcal{F}_{t-1}, \mathcal{C})| \leq \frac{C}{\sqrt{NT}}, \end{aligned} \quad (\text{A.40})$$

where the last inequality uses the conditional covariance row-sum bound and the moment bounds for the regressors and factor/loadings.

Write $\mathbb{G}_0^{-1}$ in loading–loading, loading–factor, factor–loading, and factor–factor blocks and substitute $\mathbf{h}^{(1)} = \mathbb{G}_0^{-1} \mathbf{s}$. The diagonal loading and factor blocks are bounded, while every off-diagonal same-side row and every cross-side row has squared norm $O(N^{-1})$ or $O(T^{-1})$ by (A.6). Combining these row bounds with (A.40) and conditional covariance summability gives, uniformly in i, t ,

$$\left\| \mathbb{E}_0 \left[h_i^{(1),\lambda} (h_t^{(1),f})' \middle| \mathcal{C} \right] \right\| \leq C \left[(NT)^{-1/2} + N^{-1/2} + T^{-1/2} \right].$$

Every mixed term in (A.35) has one additional factor $(NT)^{-1/2}$. The entry, mean, contrast, and fixed lag-sum directions in Section 3 have uniformly bounded total absolute weight. Therefore,

$$\left| \mathbb{E}_0(\mathcal{B}_{NT}^{\lambda f} \mid \mathcal{C}) \right| \leq C \left[\frac{1}{NT} + \frac{1}{N\sqrt{T}} + \frac{1}{\sqrt{N}T} \right] = o(N^{-1} + T^{-1}).$$

For the centered mixed term, substitute the four inverse-Gram block expansions just used, localize each score product, and again delete the space–time neighborhood around one anchor. The covariance of two localized products is bounded by the summable conditional mixing envelope; the squared coefficients are summable at the rates in (A.6). The resulting bound is

$$\text{Var}_0 \left(\mathcal{B}_{NT}^{\lambda f} \middle| \mathcal{C} \right) \leq C \epsilon_{NT}^2 s_{NT}^2 = o(s_{NT}^2).$$

This proves (A.39) and the proposition. $\square$

Proposition 5 (Orders and local form of the second-order terms). *Collect the terms in (A.35) according to whether the two first-order score coordinates are both loading coordinates, both factor*

coordinates, or one of each:

$$\mathcal{B}_{NT} = \mathcal{B}_{NT}^{\lambda\lambda} + \mathcal{B}_{NT}^{ff} + \mathcal{B}_{NT}^{\lambda f}.$$

Uniformly over the parameters of interest in Assumption 9,

$$\begin{aligned} \mathbb{E}_0(\mathcal{B}_{NT}^{\lambda\lambda} \mid \mathcal{C}) &= B_T/T, & |B_T| &\leq C, \\ \mathbb{E}_0(\mathcal{B}_{NT}^{ff} \mid \mathcal{C}) &= B_N/N, & |B_N| &\leq C, \\ \mathbb{E}_0(\mathcal{B}_{NT}^{\lambda f} \mid \mathcal{C}) &= o(N^{-1} + T^{-1}), & \text{Var}_0(\mathcal{B}_{NT}^{\lambda f} \mid \mathcal{C}) &= o(s_{NT}^2). \end{aligned}$$

The centered loading–loading and factor–factor terms have conditional variances $o(s_{NT}^2)$. Hence the mixed term is $o_p(s_{NT}) + o_p(N^{-1} + T^{-1})$.

The two leading constants are linear in the target direction. To make the local contributions explicit, for the coordinate direction $\mathbf{E}_{M,it}$ that has a one in entry (i, t) of coefficient matrix M and zeros elsewhere, define

$$b_{M,it}^T = T \mathbb{E}_0 \left\{ \mathcal{B}_{NT}^{\lambda\lambda}(\mathbf{E}_{M,it}) \mid \mathcal{C} \right\}, \quad b_{M,it}^N = N \mathbb{E}_0 \left\{ \mathcal{B}_{NT}^{ff}(\mathbf{E}_{M,it}) \mid \mathcal{C} \right\}.$$

Linearity of $\mathbf{h}^*$ and of the target curvature in $\mathbf{D}$ then gives the exact representation

$$B_T = \sum_M \sum_{i,t} D_{M,it} b_{M,it}^T, \quad B_N = \sum_M \sum_{i,t} D_{M,it} b_{M,it}^N. \quad (\text{A.41})$$

The inverse-Gram row bounds and the conditional moment assumptions imply $\sup_{M,i,t} \{|b_{M,it}^T| + |b_{M,it}^N|\} \leq C$ for the target classes in Section 3. For a target restricted to a set of rows or columns, the corresponding leading bias constant is obtained by restricting the same sums in (A.41).

Proof. Every first-order loading coordinate is a row of $\mathbb{G}_0^{-1}$ applied to a score normalized by $T^{-1/2}$; every first-order factor coordinate is the analogous object normalized by $N^{-1/2}$. The inverse-row bounds in (A.6) and the conditional covariance summability in (A.2) imply

$$\sup_i \mathbb{E}_0 \|h_i^{(1),\lambda}\|^2 \leq C, \quad \sup_t \mathbb{E}_0 \|h_t^{(1),f}\|^2 \leq C.$$

Every term with two loading coordinates carries the explicit factor T^{-1} from the local parameterization; every term with two factor coordinates carries N^{-1} . Conditional Hölder and the $8 + \eta$ moment bounds therefore give

$$\left| \mathbb{E}_0(\mathcal{B}_{NT}^{\lambda\lambda} \mid \mathcal{C}) \right| \leq C/T, \quad \left| \mathbb{E}_0(\mathcal{B}_{NT}^{ff} \mid \mathcal{C}) \right| \leq C/N,$$

which defines bounded B_T and B_N .

The mixed loading–factor term is handled by Proposition 4. The key point is not the explicit prefactor $(NT)^{-1/2}$ by itself, but the additional cross-side decorrelation in (A.40) together with the same-side and cross-side inverse-Gram row decay. These yield an expectation $o(N^{-1} + T^{-1})$ and a

centered variance $o(s_{NT}^2)$.

For a centered same-side term, replace its first-order scores by local approximations. The covariance of two products is bounded by a summable function of the distance between their space–time neighborhoods. Summing these covariance bounds and using the Riesz-weight rates in [Proposition 1](#) gives conditional variance $o(s_{NT}^2)$.

It remains to record the structure needed by the split-panel proof. The local Riesz coordinate $\mathbf{h}^*$ is linear in $\mathbf{D}$, and every same-side conditional expectation in [\(A.35\)](#) is therefore a bounded linear functional of $\mathbf{D}$. Evaluating these two functionals on the coordinate directions $\mathbf{E}_{M,it}$ gives the coefficients $b_{M,it}^T$ and $b_{M,it}^N$ defined above and proves [\(A.41\)](#). Because the functionals are finite-degree combinations of bounded conditional moments and blocks of $\mathbb{G}_0^{-1}$, the inverse-Gram bounds, conditional $8 + \eta$ moments, and fixed ranks make the coefficients uniformly bounded. Restricting the target direction to selected rows or columns simply removes the corresponding terms from the exact linear sums. $\square$

Proof of Theorem 3. Lemma 5 gives the exact expansion. The rate-level replacements [\(A.36\)](#)–[\(A.38\)](#) in [Proposition 4](#) show, for the specific bilinear objects appearing in that decomposition, that replacing $\hat{\mathbf{h}}$ by $\mathbf{h}^{(1)}$ changes the last three terms by

$$o_p(s_{NT}) + o_p(N^{-1} + T^{-1}).$$

Propositions 4 and 5 then yield

$$\hat{\phi} - \phi(\boldsymbol{\Theta}_0) = \sum_{i,t} \Psi_{0,it} u_{it} + \frac{B_T}{T} + \frac{B_N}{N} + r_{NT},$$

with the remainder stated in the theorem. For an individual coefficient and a fixed unit–time lag sum, the variance rates in [Proposition 1](#) make the two bias terms $o_p(s_{NT})$. For means and contrasts, the support-growth conditions stated in [Theorem 3](#) give the same conclusion. The conditional population limit and Slutsky yield the stated normal limits. $\square$

Proposition 6 (Balanced split-panel preservation). *Consider a two-way mean or contrast covered by [Theorem 4](#). Under the assumptions of [Theorem 3](#), the independent balanced time and unit partitions in [Section 4.3](#) have the following properties, conditionally on $\mathcal{C}$, with probability approaching one over the random partitions.*

(i) *The restricted target directions reassemble the full target exactly:*

$$\sum_{h=1}^2 \mathbf{D}^{T,h} = \mathbf{D}, \quad \sum_{g=1}^2 \mathbf{D}^{N,g} = \mathbf{D}.$$

(ii) *Every positive rank and every lower Gram-matrix eigenvalue bound is preserved in the corresponding split panel.*

(iii) Let $\pi_{T,h} = T_h/T$ and $\pi_{N,g} = N_g/N$. The leading constants for the restricted time targets satisfy

$$B_T^{T,h} = \pi_{T,h} B_T + o_p(1), \quad B_N^{T,h} = \pi_{T,h} B_N + o_p(1),$$

and the restricted unit targets satisfy

$$B_T^{N,g} = \pi_{N,g} B_T + o_p(1), \quad B_N^{N,g} = \pi_{N,g} B_N + o_p(1).$$

(iv) If $\Psi_{0,it}^T$ and $\Psi_{0,it}^N$ are the population split Riesz weights defined in [Section 4.4](#), then

$$\begin{aligned} & \sum_{h=1}^2 \sum_{t \in \mathcal{I}_{T,h}} \sum_i (\Psi_{0,it}^T - \Psi_{0,it})^2 + \sum_{g=1}^2 \sum_{i \in \mathcal{I}_{N,g}} \sum_t (\Psi_{0,it}^N - \Psi_{0,it})^2 \\ &= o_p \left(\mathbb{E}_0 \left[\sum_{i,t} \Psi_{0,it}^2 \middle| \mathcal{C} \right] \right). \end{aligned} \quad (\text{A.42})$$

Consequently,

$$\frac{(s_{NT}^{\text{corr}})^2}{s_{NT}^2} \rightarrow_p 1. \quad (\text{A.43})$$

Proof. We condition throughout on the realized common shocks and on the full panel, and take probability with respect to the independent split assignments. The conclusions are then integrated over the panel and the splits.

Step 1: the split targets reassemble the original parameter. By construction, each entry of $\mathbf{D}$ is retained in exactly one time-restricted direction and exactly one unit-restricted direction. Therefore,

$$\sum_h \langle \mathbf{D}^{T,h}, \boldsymbol{\Theta}_0 \rangle = \langle \mathbf{D}, \boldsymbol{\Theta}_0 \rangle, \quad \sum_g \langle \mathbf{D}^{N,g}, \boldsymbol{\Theta}_0 \rangle = \langle \mathbf{D}, \boldsymbol{\Theta}_0 \rangle.$$

This argument applies without change to a proper time subset R_T , a proper unit subset G_N , and a signed contrast, because the original target weights and signs are retained rather than renormalized inside a split.

Step 2: ranks and Gram matrices are preserved. For a positive-rank matrix $\mathbf{M}_0 = \mathbf{U}_M \boldsymbol{\Sigma}_M \mathbf{V}_M'$, let $v'_{M,t}$ be row t of $\mathbf{V}_M$. Conditional on $\mathbf{V}_M$, sampling a balanced random time half without replacement gives

$$\mathbb{E}_{\text{split}} \left(\sum_{t \in \mathcal{I}_{T,h}} v_{M,t} v'_{M,t} \right) = \pi_{T,h} \mathbf{I}.$$

Incoherence bounds every summand by C/T . The finite-population variance formula therefore gives, entry by entry,

$$\text{Var}_{\text{split}} \left(\sum_{t \in \mathcal{I}_{T,h}} v_{M,t,a} v_{M,t,b} \right) \leq C/T.$$

Since r_M is fixed,

$$\mathbf{V}'_{M,h} \mathbf{V}_{M,h} = \pi_{T,h} \mathbf{I} + o_p(1).$$

Hence the restricted matrix retains rank r_M and its nonzero singular values remain of order $\sqrt{NT_h}$. The same calculation with the rows of $\mathbf{U}_M$ establishes rank preservation in a random unit half.

The loading and factor Gram matrices are finite-dimensional averages of terms with uniformly bounded second moments. Applying the same sampling-without-replacement variance calculation to each entry gives

$$\max_h \|G_{\lambda,i}^{T,h} - G_{\lambda,i}\|_{\text{op}} = o_p(1), \quad \max_g \|G_{f,t}^{N,g} - G_{f,t}\|_{\text{op}} = o_p(1),$$

with the corresponding statements for the other Gram blocks. The lower eigenvalue bounds in [Assumptions 7](#) and [8](#) are therefore preserved.

Step 3: the leading bias constants are preserved. By [\(A.41\)](#), the loading- and factor-side constants are weighted sums of the explicitly defined local coefficients $b_{M,it}^T$ and $b_{M,it}^N$. These coefficients are finite-degree functions of: (a) the relevant blocks of the inverse local Gram operator; (b) the aligned loading and factor rows; and (c) a fixed-dimensional vector of conditional moments of the regressors and disturbances that enters the two same-side expectations in [Proposition 5](#). On the set where the Gram eigenvalues are bounded below by c , matrix inversion is Lipschitz. Consequently, if $b_{M,it}^{j,T,h}$ denotes the corresponding coefficient computed from time split h , there is a random envelope $\xi_{M,it}^j$ with $\sup_{M,i,t} \mathbb{E}_0[(\xi_{M,it}^j)^2 \mid \mathcal{C}] \leq C$ such that

$$\left| b_{M,it}^{j,T,h} - b_{M,it}^j \right| \leq C \Delta_{T,h} \xi_{M,it}^j, \quad j \in \{T, N\}, \quad (\text{A.44})$$

where $\Delta_{T,h}$ is the maximum of the operator-norm differences between the split and full aligned Gram blocks and the Euclidean differences between the corresponding fixed-dimensional conditional-moment vectors. Step 2 and the sampling-without-replacement variance bound give $\max_h \Delta_{T,h} = o_p(1)$. The unit-split coefficients satisfy the analogous bound with $\Delta_{N,g} = o_p(1)$. Thus the phrase “preservation of the local bias contributions” is a direct Lipschitz consequence of the objects entering their definition rather than an appeal to continuity alone.

For the time split, write

$$a_t^T = \sum_M \sum_i D_{M,it} b_{M,it}^T, \quad a_t^N = \sum_M \sum_i D_{M,it} b_{M,it}^N.$$

Then $B_T = \sum_t a_t^T$ and $B_N = \sum_t a_t^N$. For a two-way target, the largest time weight tends to zero; the uniform bounds in [Proposition 5](#) therefore imply

$$\sum_t (a_t^T)^2 + \sum_t (a_t^N)^2 = o_p(1).$$

The finite-population variance formula gives, for $j \in \{T, N\}$,

$$\mathbb{E}_{\text{split}} \left[\left(\sum_{t \in \mathcal{I}_{T,h}} a_t^j - \pi_{T,h} \sum_t a_t^j \right)^2 \middle| \mathcal{C}, \text{panel} \right] \leq C \sum_t (a_t^j)^2 = o_p(1).$$

The Lipschitz bound (A.44) and the bounded total absolute target weight also give

$$B_j^{T,h} = \sum_{t \in \mathcal{I}_{T,h}} \sum_{M,i} D_{M,it} b_{M,it}^{j,T,h} = \sum_{t \in \mathcal{I}_{T,h}} a_t^j + o_p(1) = \pi_{T,h} B_j + o_p(1), \quad j \in \{T, N\}.$$

This proves the two time-split constant relations. For the unit split, set $c_i^j = \sum_M \sum_t D_{M,it} b_{M,it}^j$ and repeat the argument. Stratification within every group entering a contrast preserves the signed group weights, and the vanishing maximum unit weight gives $\sum_i (c_i^j)^2 = o_p(1)$. The unit-split analogue of (A.44) then yields $B_j^{N,g} = \pi_{N,g} B_j + o_p(1)$.

For a time split, the loading-side bias of subtarget h is $B_T^{T,h}/T_h$. Since $\pi_{T,h}/T_h = 1/T$, summing over the two halves gives $2B_T/T + o_p(T^{-1})$. Its factor-side biases sum to $B_N/N + o_p(N^{-1})$. The unit-split calculation gives $B_T/T + 2B_N/N + o_p(N^{-1} + T^{-1})$.

Step 4: split Riesz weights and the corrected-score scale. Let $\Gamma_{T,h}$ be the population Gram form on time split h , and let $\mathbf{q}_0^{T,h}$ solve its Riesz equation for $\mathbf{D}^{T,h}$. After aligning the split and full singular spaces, let $\mathbf{q}_0^{[T,h]}$ denote the restriction of the full-panel Riesz coordinates to the loading rows and factor rows used by time split h . Define $\Gamma_{N,g}$, $\mathbf{q}_0^{N,g}$, and $\mathbf{q}_0^{[N,g]}$ analogously.

We now make the factor-of- $\pi_{T,h}$ bookkeeping explicit. Let $\dot{\phi}(\Delta) = \langle \mathbf{D}, \Delta \rangle$ and $\dot{\phi}_{T,h}(\Delta) = \langle \mathbf{D}^{T,h}, \Delta \rangle$. The same sampling-without-replacement calculation as in Step 2, now applied to every fixed-dimensional Gram block and to the target derivative, yields

$$\sup_{\|\Delta\|=1} \left| \pi_{T,h}^{-1} \Gamma_{T,h}(\mathbf{q}_0^{[T,h]}, \Delta) - \Gamma_0(\mathbf{q}_0, \Delta) \right| = o_p(\|\mathcal{P}_0 \mathbf{D}\|), \quad (\text{A.45})$$

$$\sup_{\|\Delta\|=1} \left| \pi_{T,h}^{-1} \dot{\phi}_{T,h}(\Delta) - \dot{\phi}(\Delta) \right| = o_p(\|\mathcal{P}_0 \mathbf{D}\|). \quad (\text{A.46})$$

For means and contrasts, the second display follows from the vanishing maximum time weight; for a full mean it reduces to the ordinary finite-population variance formula. Since the full Riesz equation gives $\Gamma_0(\mathbf{q}_0, \Delta) = \dot{\phi}(\Delta)$, subtracting (A.46) from (A.45) and multiplying by $\pi_{T,h}$ gives

$$\sup_{\|\Delta\|=1} \left| \Gamma_{T,h}(\mathbf{q}_0^{[T,h]}, \Delta) - \langle \mathbf{D}^{T,h}, \Delta \rangle \right| = o_p(\|\mathcal{P}_0 \mathbf{D}\|).$$

The same argument with $\pi_{N,g}$ proves the unit-split approximate Riesz equation. Set $\mathbf{d}_{T,h} = \mathbf{q}_0^{T,h} - \mathbf{q}_0^{[T,h]}$. Subtracting the two Riesz equations gives

$$\Gamma_{T,h}(\mathbf{d}_{T,h}, \Delta) = - \left\{ \Gamma_{T,h}(\mathbf{q}_0^{[T,h]}, \Delta) - \langle \mathbf{D}^{T,h}, \Delta \rangle \right\}$$

for every split tangent direction Δ . Taking $\Delta = \mathbf{d}_{T,h}$ and using split Gram coercivity yields

$$\Gamma_{T,h}(\mathbf{d}_{T,h}, \mathbf{d}_{T,h}) \leq C \sup_{\|\Delta\|=1} \left| \Gamma_{T,h}(\mathbf{q}_0^{[T,h]}, \Delta) - \langle \mathbf{D}^{T,h}, \Delta \rangle \right|^2 = o_p(\|\mathcal{P}_0 \mathbf{D}\|^2).$$

Since this Gram norm is the squared prediction norm of the split Riesz-weight difference, summing over h gives the first term in (A.42). Repeating the same argument with $\mathbf{d}_{N,g} = \mathbf{q}_0^{N,g} - \mathbf{q}_0^{[N,g]}$ gives the second term.

Let

$$e_{it}^T = \Psi_{0,it}^T - \Psi_{0,it}, \quad e_{it}^N = \Psi_{0,it}^N - \Psi_{0,it}.$$

The corrected population weight equals

$$3\Psi_{0,it} - \Psi_{0,it}^T - \Psi_{0,it}^N = \Psi_{0,it} - e_{it}^T - e_{it}^N.$$

The conditional covariance row-sum bound in Assumption 5 turns (A.42) into an $o_p(s_{NT}^2)$ bound for the variance of the score difference. The target-specific lower variance bound in Assumption 9 gives $s_{NT}^2 \asymp \mathbb{E}_0[\sum_{i,t} \Psi_{0,it}^2 \mid \mathcal{C}]$. Expanding the variance of the last display and applying conditional Cauchy–Schwarz proves (A.43). $\square$

For the split-panel estimator, define the population subtargets

$$\phi_{T,h}(\Theta_0) = \langle \mathbf{D}^{T,h}, \Theta_0 \rangle, \quad \phi_{N,g}(\Theta_0) = \langle \mathbf{D}^{N,g}, \Theta_0 \rangle.$$

Applying (6.2) to each split fit gives

$$\begin{aligned} \hat{\phi}_{T,h} - \phi_{T,h}(\Theta_0) &= S_{T,h} + \frac{B_T^{T,h}}{T_h} + \frac{B_N^{T,h}}{N} + r_{T,h}, \\ \hat{\phi}_{N,g} - \phi_{N,g}(\Theta_0) &= S_{N,g} + \frac{B_T^{N,g}}{T} + \frac{B_N^{N,g}}{N_g} + r_{N,g}. \end{aligned}$$

Here the S terms are the corresponding population Riesz scores and the remainders have the same target-specific order as in Theorem 3. Summing the two time equations and using Proposition 6 gives

$$\bar{\phi}_T - \phi(\Theta_0) = \bar{S}_T + \frac{2B_T}{T} + \frac{B_N}{N} + o_p(s_{NT}) + o_p(N^{-1} + T^{-1}).$$

Similarly,

$$\bar{\phi}_N - \phi(\Theta_0) = \bar{S}_N + \frac{B_T}{T} + \frac{2B_N}{N} + o_p(s_{NT}) + o_p(N^{-1} + T^{-1}).$$

Therefore,

$$\begin{aligned}\widehat{\phi}^{\text{corr}} - \phi(\Theta_0) &= 3 \left(\frac{B_T}{T} + \frac{B_N}{N} \right) - \left(\frac{2B_T}{T} + \frac{B_N}{N} \right) \\ &\quad - \left(\frac{B_T}{T} + \frac{2B_N}{N} \right) + \sum_{i,t} Z_{it}^{\text{corr}} + o_p(s_{NT}) \\ &= \sum_{i,t} Z_{it}^{\text{corr}} + o_p(s_{NT}^{\text{corr}}),\end{aligned}$$

where the final equality uses (A.43). The vector consisting of the full-panel and split-panel scores is a finite collection of conditionally NED random-field sums. By Lemma 1 and Proposition 1, the conditional random-field central limit theorem applies jointly. The conclusions hold for almost every independent balanced partition sequence.

A.6 Spatial HAC variance estimation

This subsection completes the proof of feasible inference. It first shows that cross-time covariance is zero because the time-level Riesz sums are martingale differences. It then shows that the spatial weighting rule captures all nonnegligible same-time covariance. Finally, it replaces the population Riesz scores by the fitted Riesz weights and residuals.

Proposition 7 (Consistency of the spatial variance estimator). *Under the assumptions of Theorem 4, the distance cutoff $h_N = \lceil c_h \log(NT) \rceil$ with sufficiently large c_h makes the omitted conditional spatial covariance negligible. Together with the recovery conclusions in Theorem 2, the target expansion in Theorem 3, and split preservation in Proposition 6, this implies consistency of (4.8) and (4.9).*

Proof. The conditional covariance inequality for NED fields bounds the covariance between two population Riesz-score contributions at spatial distance a by

$$C\{\psi_S(\lfloor a/3 \rfloor) + \bar{\alpha}(\lfloor a/3 \rfloor)^{\eta/(8+\eta)}\}$$

times their conditional $2 + \eta/4$ norms. Spatial shell growth is polynomial by Assumption 3, whereas both dependence terms decay exponentially. Summing over distances larger than h_N gives $o(s_{NT}^2)$ for sufficiently large c_h .

For retained pairs, the fourth-moment bounds from the localization lemma and $v_N(h_N) \leq C\{\log(NT)\}^{d_s}$ make the centered spatial quadratic form $o_p(s_{NT}^2)$ under Assumption 10. Finally, the Riesz and residual recovery rates imply that replacing the population Riesz score by the feasible residual Riesz score changes the quadratic form by $o_p(s_{NT}^2)$ through Cauchy–Schwarz and $\|\mathbf{W}_N\|_{\text{op}} \leq Cv_N(h_N)$. $\square$

For either the uncorrected or corrected estimator, write Z_{it}^* for the corresponding population Riesz score and $\widehat{Z}_{it}^*$ for its feasible version. The star is only a proof abbreviation that allows the

same calculation to cover both estimators. Define

$$\tilde{s}_{\text{sp}}^{2,*} = \sum_t \sum_{i,k} w_{ik,N} Z_{it}^* Z_{kt}^*.$$

Proof of Theorem 4. The split-panel calculation above gives the corrected influence expansion. It remains to establish consistency of the feasible spatial variance.

Conditional sequential exogeneity gives

$$\mathbb{E}_0 \left(\sum_i Z_{it}^* \middle| \mathcal{F}_{t-1}, \mathcal{C} \right) = 0.$$

Hence, for $s < t$,

$$\begin{aligned} \mathbb{E}_0 \left[\left(\sum_i Z_{it}^* \right) \left(\sum_k Z_{ks}^* \right) \middle| \mathcal{C} \right] &= \mathbb{E}_0 \left[\left(\sum_k Z_{ks}^* \right) \mathbb{E}_0 \left(\sum_i Z_{it}^* \middle| \mathcal{F}_{t-1}, \mathcal{C} \right) \middle| \mathcal{C} \right] \\ &= 0. \end{aligned}$$

Therefore,

$$(s_{NT}^*)^2 = \sum_t \sum_{i,k} \mathbb{E}_0(Z_{it}^* Z_{kt}^* \mid \mathcal{C}).$$

The covariance-tail argument in the proof of Proposition 7 implies

$$|\mathbb{E}_0(\tilde{s}_{\text{sp}}^{2,*} \mid \mathcal{C}) - (s_{NT}^*)^2| = o\{(s_{NT}^*)^2\}.$$

For the centered statistic, expand

$$\text{Var}_0(\tilde{s}_{\text{sp}}^{2,*} \mid \mathcal{C}) = \sum_t \sum_{i,k,j,\ell} w_{ik,N} w_{j\ell,N} \text{Cov}_0(Z_{it}^* Z_{kt}^*, Z_{jt}^* Z_{\ell t}^* \mid \mathcal{C}).$$

Every unit has at most $v_N(h_N)$ partners with nonzero weight. Splitting the quadruples according to their minimum spatial separation and applying (A.2) gives

$$\text{Var}_0(\tilde{s}_{\text{sp}}^{2,*} \mid \mathcal{C}) \leq C v_N(h_N)^3 \sum_{i,t} \mathbb{E}_0(|Z_{it}^*|^4 \mid \mathcal{C}) + o\{(s_{NT}^*)^4\}.$$

Because h_N is logarithmic, $v_N(h_N)^3$ contributes only a fixed power of $\log(NT)$. The fourth moments of local and two-way Riesz scores contain the additional polynomial factors T^{-1} or $(NT)^{-1}$ established in Proposition 1; these polynomial gains dominate every fixed logarithmic power under Assumption 10. Thus, the exponent $d_s + 6$ in the main growth condition governs the leading recovery and bias terms, while the larger logarithmic power appearing in this intermediate variance bound is asymptotically harmless. The fourth-moment argument in the proof of Proposition 7 therefore

makes the right side $o\{(s_{NT}^*)^4\}$. Conditional Chebyshev yields

$$\frac{\tilde{s}_{\text{sp}}^{2,*}}{(s_{NT}^*)^2} \rightarrow_p 1.$$

Let $e_{it}^* = \hat{Z}_{it}^* - Z_{it}^*$. In vector notation,

$$\tilde{s}_{\text{sp}}^{2,*} - s_{\text{sp}}^{2,*} = \sum_t [2(Z_t^*)' \mathbf{W}_N e_t^* + (e_t^*)' \mathbf{W}_N e_t^*].$$

Cauchy–Schwarz and positive semidefiniteness give

$$|\tilde{s}_{\text{sp}}^{2,*} - s_{\text{sp}}^{2,*}| \leq \|\mathbf{W}_N\|_{\text{op}} \left[2 \left\{ \sum_{i,t} (Z_{it}^*)^2 \sum_{i,t} (e_{it}^*)^2 \right\}^{1/2} + \sum_{i,t} (e_{it}^*)^2 \right].$$

It remains to verify the feasible-score bound for the corrected statistic, which contains residuals from three distinct fitted objects. Write

$$\hat{Z}_{it}^{\text{corr}} = 3\hat{\Psi}_{it}\hat{u}_{it} - \hat{\Psi}_{it}^T \hat{u}_{it}^T - \hat{\Psi}_{it}^N \hat{u}_{it}^N, \quad Z_{it}^{\text{corr}} = u_{it} (3\Psi_{0,it} - \Psi_{0,it}^T - \Psi_{0,it}^N).$$

For a time split h ,

$$\hat{u}_{it}^{T,h} - u_{it} = -\mathcal{Y}_{it}(\hat{\Theta}_{T,h} - \Theta_0), \quad t \in \mathcal{I}_{T,h},$$

and the analogous identity holds for each unit split. By [Proposition 6](#), every split preserves the supplied ranks and the Gram lower bounds. Applying the recovery argument of [Theorem 2](#) to the $N \times T_h$ and $N_g \times T$ panels therefore gives, uniformly over the two halves,

$$\sum_{t \in \mathcal{I}_{T,h}} \sum_i (\hat{u}_{it}^{T,h} - u_{it})^2 = O_p\{NT_h \zeta_{N,T_h}\}, \quad \sum_t \sum_{i \in \mathcal{I}_{N,g}} (\hat{u}_{it}^{N,g} - u_{it})^2 = O_p\{N_g T \zeta_{N_g,T}\}.$$

Balanced halves imply $\zeta_{N,T_h} \asymp \zeta_{NT}$ and $\zeta_{N_g,T} \asymp \zeta_{NT}$. The split Riesz recovery in [\(A.42\)](#), together with the full-panel Riesz rate, then yields

$$\|\mathbf{W}_N\|_{\text{op}} \sum_{i,t} (\hat{Z}_{it}^{\text{corr}} - Z_{it}^{\text{corr}})^2 = o_p\{(s_{NT}^{\text{corr}})^2\}.$$

Substituting this bound for $\sum (e_{it}^*)^2$ in the preceding Cauchy–Schwarz display makes $\tilde{s}_{\text{corr}}^2 - \tilde{s}_{\text{sp}}^{2,\text{corr}} = o_p\{(s_{NT}^{\text{corr}})^2\}$. For the uncorrected statistic the same calculation uses only the full-panel recovery rates. Studentization follows from the corresponding influence expansion, the population limit, and conditional Slutsky. $\square$

A.7 Rank selection

The rank theorem separates candidate coverage from final model comparison. Candidate coverage depends only on whether the candidate set contains the thresholded rank vector of one sufficiently accurate pilot. The nuclear path is useful for screening and warm starts, but no particular path value or grid resolution is used in the consistency argument. Conditional on candidate coverage, the post-refit information criterion rejects underfitted and overfitted rank vectors.

Proposition 8 (Pilot thresholding and candidate coverage). *Suppose the candidate set contains the thresholded rank vector of a pilot $\hat{\Theta}^{\text{pil}}$ satisfying*

$$\max_M \|\widehat{\mathbf{M}}^{\text{pil}} - \mathbf{M}_0\|_{\text{op}} = o_p(\tau_{NT}), \quad \tau_{NT} = o(\sqrt{NT}).$$

Then

$$\mathbb{P}_0(\mathbf{r}_0 \in \mathcal{R}_{\text{cand}} \mid \mathcal{C}) \rightarrow_p 1.$$

Under the maintained assumptions, the rank-cap pilot in (4.12) satisfies

$$\max_M \|\widehat{\mathbf{M}}^{\text{max}} - \mathbf{M}_0\|_{\text{op}} = O_p\{\sqrt{NT\zeta_{NT}}\}$$

when its normalized objective gap is $o_p(\zeta_{NT})$. Hence it supplies candidate coverage whenever

$$\sqrt{NT\zeta_{NT}} = o(\tau_{NT}) \quad \text{and} \quad \tau_{NT} = o(\sqrt{NT}).$$

Proof. For every coefficient matrix, Weyl's singular-value perturbation inequality gives

$$\left| \sigma_s(\widehat{\mathbf{M}}^{\text{pil}}) - \sigma_s(\mathbf{M}_0) \right| \leq \|\widehat{\mathbf{M}}^{\text{pil}} - \mathbf{M}_0\|_{\text{op}}.$$

If $s > r_{M,0}$, then $\sigma_s(\mathbf{M}_0) = 0$, so the corresponding estimated singular value is $o_p(\tau_{NT})$. If $s \leq r_{M,0}$, [Assumption 6](#) gives $\sigma_s(\mathbf{M}_0) \geq c\sqrt{NT}$, whereas $\tau_{NT} = o(\sqrt{NT})$ and the pilot error is $o_p(\tau_{NT})$. Thus, thresholding the pilot at τ_{NT} recovers $r_{M,0}$ for every coefficient matrix with conditional probability approaching one. Since the thresholded pilot rank is included in $\mathcal{R}_{\text{cand}}$, candidate coverage follows.

It remains to verify the stated rate for the baseline rank-cap pilot. The truth is feasible for (4.12). Let

$$\widehat{\Delta}^{\text{max}} = \widehat{\Theta}^{\text{max}} - \Theta_0.$$

Every matrix difference has rank at most $\bar{r}_M + r_{M,0}$. The same least-squares basic inequality, score bound, and conditional prediction lower bound used in [Theorem 2](#) therefore give

$$\|\widehat{\Delta}^{\text{max}}\| = O_p\{\sqrt{NT\zeta_{NT}}\}.$$

Consequently,

$$\max_M \|\widehat{\mathbf{M}}^{\text{max}} - \mathbf{M}_0\|_{\text{op}} \leq \|\widehat{\Delta}^{\text{max}}\| = O_p\{\sqrt{NT\zeta_{NT}}\}.$$

The final threshold conditions then imply the generic pilot requirement. $\square$

The same perturbation argument also proves the screening statement in [Remark 2](#); no additional approximate-low-rank inference claim is used in the rank-consistency proof.

Proof of Theorem 5. By [Proposition 8](#), it is enough to compare the incorrect vectors in $\mathcal{R}_{\text{cand}}$ with $\mathbf{r}_0$ on the event $\{\mathbf{r}_0 \in \mathcal{R}_{\text{cand}}\}$.

If $\mathbf{r}$ is underfitted, at least one fitted coefficient matrix omits a true singular direction of order $\sqrt{NT}$. Hence

$$\inf_{\boldsymbol{\Theta} \in \mathcal{M}(\mathbf{r})} \|\boldsymbol{\Theta} - \boldsymbol{\Theta}_0\|^2 \geq cNT.$$

Conditional prediction identification converts this into a residual-loss increase bounded away from zero after division by NT . The largest possible reduction in the dimension penalty is of order

$$\kappa_{NT} \frac{N+T}{NT} = o(1)$$

by (6.4). Therefore, an underfitted candidate cannot minimize (4.15).

If $\mathbf{r}$ is overfitted, the least-squares basic inequality and [Proposition 2](#) show that its best normalized residual improvement relative to $\mathbf{r}_0$ is $O_p(\zeta_{NT})$. Adding one rank to one $N \times T$ coefficient matrix raises $d(\mathbf{r})$ by order $N+T$, so the penalty increase is of order

$$\kappa_{NT} \frac{N+T}{NT},$$

which dominates ζ_{NT} by (6.4). The rank caps and number of coefficient matrices are fixed, so $\mathcal{R}_{\text{max}}$, and hence $\mathcal{R}_{\text{cand}}$, contains only finitely many distinct vectors. Numerical objective errors are smaller than the same separation by assumption. Therefore,

$$\mathbb{P}_0(\hat{\mathbf{r}} = \mathbf{r}_0 \mid \mathcal{C}) \rightarrow_p 1.$$

On this event, the selected full- and split-panel estimators use the same rank restrictions as the supplied-rank estimators, which transfers all supplied-rank limits. $\square$

B Additional Monte Carlo Results

B.1 Calibration and coefficient summaries

Table 11: DGP calibration and coefficient summaries

Placeholder. Placeholder for c_h , c_ξ , realized pooled R^2 , coefficient means and ranges, stability summaries, and conditional variance summaries for every DGP and panel size.

B.2 Entrywise and fixed-time targets

Table 12: Additional entrywise results

Placeholder. Placeholder for individual autoregressive and slope entries across target locations, including leverage diagnostics and sensitivity to the selected target time.

Table 13: Fixed-time group means and contrasts

Placeholder. Placeholder for group means and contrasts using deterministic unit groups, with diagonal and spatial standard errors.

B.3 Two-way averages and bias correction

Table 14: Full-panel means before and after bias correction

Placeholder. Placeholder for plug-in bias, time-split and unit-split bias estimates, corrected bias, RMSE, standard-error accuracy, and coverage.

Table 15: Split-panel diagnostics

Placeholder. Placeholder for retained ranks, Gram-matrix eigenvalues, target support, objective tolerances, and failure rates in the four split-panel fits.

B.4 Rank, spatial-weight, and computation sensitivity

Table 16: Rank-cap and information-criterion sensitivity

Placeholder. Placeholder for alternative rank caps, the baseline grid $(\gamma, \epsilon) = (0.8, 0.01)$, the denser grid $\gamma = \sqrt{0.8}$, singular-value threshold constants 1/2, 1, and 2, information-criterion penalty multipliers, one-coordinate local completion, selected-to-cap margins, gaps to neighboring ranks, distinct path ranks, candidate counts, target sensitivity under one-rank changes, and screened-versus-exhaustive agreement.

Table 17: Spatial HAC sensitivity

Placeholder. Placeholder for spatial distance cutoffs, omitted-covariance diagnostics, standard-error ratios, and coverage under DGPs 2–4.

Table 18: Runtime comparison with the buffered-fold procedure

Placeholder. Placeholder for nonlinear-fit counts, full-panel-equivalent data passes, Riesz solve time, memory use, and total runtime.

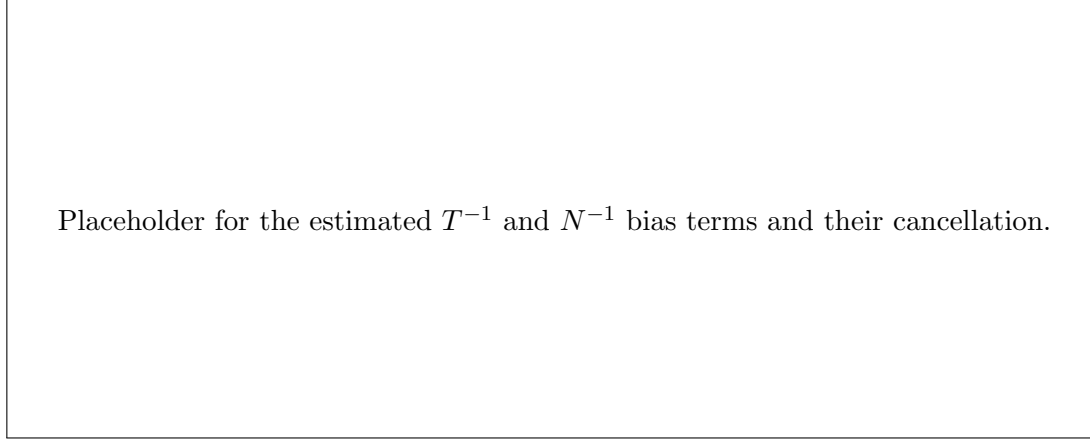


Figure 4: Bias decomposition across panel sizes

C Empirical Appendix

C.1 Data construction

Placeholder. Document sources, sample filters, seasonal adjustment, transformations, lag construction, geographic matching, missing-value treatment, and the definitions of all predetermined controls for the unemployment and housing panels.

Table 19: Empirical sample construction

Placeholder. Placeholder for raw and final sample sizes, time ranges, exclusions, coordinate matches, and control availability.

C.2 Rank selection and numerical diagnostics

Table 20: Empirical rank search and information criteria

Placeholder. Placeholder for $\lambda_{\max}$, the path endpoints, grid resolution, distinct thresholded path ranks, candidate vectors after duplicate removal, warm-start diagnostics, post-refit loss, dimension penalties, the smallest and second-smallest information criteria, selected ranks, denser-grid agreement, threshold and penalty sensitivity, rank-cap sensitivity, cap hits, and target sensitivity under the best one-coordinate neighboring ranks.

Table 21: Full- and split-panel numerical diagnostics

Placeholder. Placeholder for objective paths, first-order residuals, Riesz solver convergence, Gram-matrix eigenvalues, envelope checks, and split-panel stability.

C.3 Unemployment robustness

Table 22: Unemployment specifications and controls

Placeholder. Placeholder for baseline and control-augmented specifications, alternative sample windows, group definitions, selected ranks, estimates, and spatial standard errors.

Table 23: Unemployment target details

Placeholder. Placeholder for full means, fixed-time means, high and low group means, contrasts, selected entries, and the two-way correction terms.

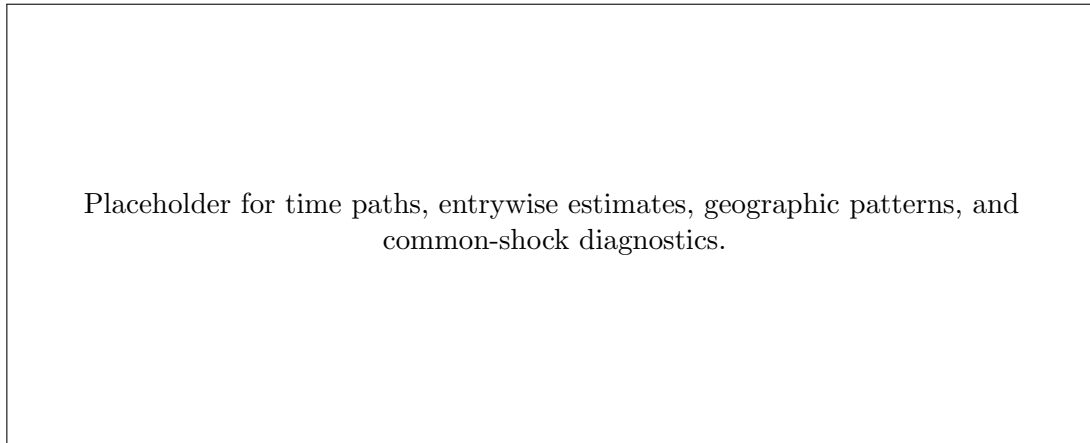


Figure 5: Unemployment coefficient paths and maps

C.4 Housing robustness

Table 24: Housing specifications and controls

Placeholder. Placeholder for lag choices, control-augmented specifications, alternative samples, selected ranks, estimates, and spatial standard errors.

Table 25: Housing target details

Placeholder. Placeholder for lag means, lag-sum persistence, price-tier means, contrasts, selected entries, and the two-way correction terms.

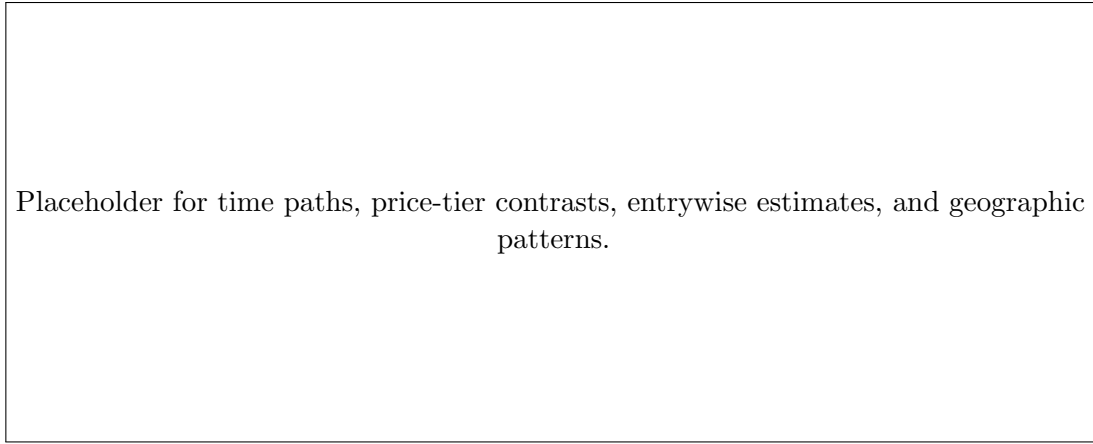


Figure 6: Housing coefficient paths and maps

C.5 Variance and dependence diagnostics

Table 26: Empirical spatial-weight and cluster sensitivity

Placeholder. Placeholder for alternative geographic distance cutoffs, known-cluster calculations, residual spatial covariance, and conditional variance diagnostics.

Table 27: Residual adequacy and homogeneous benchmarks

Placeholder. Placeholder for fit measures, residual serial correlation, residual spatial correlation, remaining singular-value shares, and comparisons with homogeneous dynamic-panel benchmarks.
